9702 Physics · Paper 4

Structured questions by topic · 2024 · 2025 · 2026

Name: _____

Class: _____

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12 Motion in a circle

7 questions · Paper 4 · 2024–2026

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Syllabus requirements · 12 Motion in a circle

考纲要求（9702 · 2025 – 2027 syllabus）

12.1

12.1.1 define the radian and express angular displacement in radians

12.1.2 understand and use the concept of angular speed

12.1.3 recall and use ω = 2π / T and v = rω

12.2

12.2.1 understand that a force of constant magnitude that is always perpendicular to the direction of motion causes centripetal acceleration

12.2.2 understand that centripetal acceleration causes circular motion with a constant angular speed

12.2.3 recall and use a = rω2 and a = v2 / r

12.2.4 recall and use F = mrω2 and F = mv2 / r

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1 **(a)** Define the radian.

.....

..... [1]

(b) A circular metal disc spins horizontally about a vertical axis, as shown in Fig. 1.1.

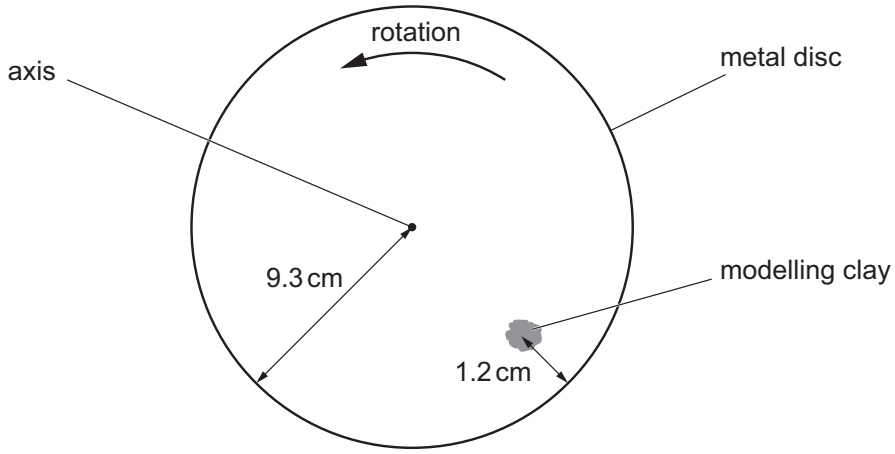


Fig. 1.1 (not to scale)

A piece of modelling clay is attached to the disc.

For the instant when the piece of modelling clay is in the position shown, draw on Fig. 1.1:

(i) an arrow, labelled V, showing the direction of the velocity of the modelling clay [1]

(ii) an arrow, labelled A, showing the direction of the acceleration of the modelling clay. [1]

(c) The metal disc in Fig. 1.1 has a radius of 9.3 cm.
The centre of gravity of the modelling clay is 1.2 cm from the rim of the disc and moves with a speed of 0.68 m s^{-1}.

(i) Calculate the angular speed ω of the disc.

ω = ... rad s^{-1} [2]

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(ii) Calculate the acceleration a of the centre of gravity of the modelling clay.

a = ... $m\,s^{-2}$ [2]

(d) A second piece of modelling clay is attached to the disc in the position shown in Fig. 1.2.

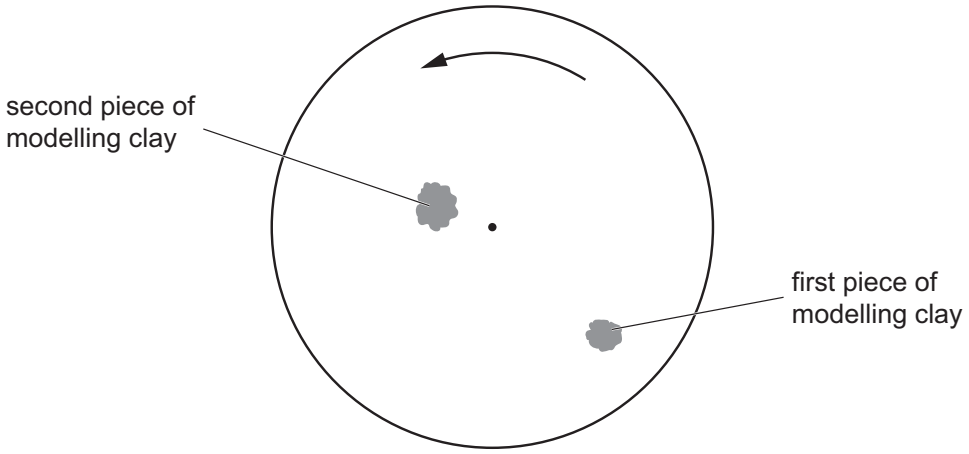


Fig. 1.2

The second piece of modelling clay has a larger mass than the first piece.

By placing **one** tick (✓) in each row, complete Table 1.1 to show how the quantities indicated compare for the two pieces of modelling clay.

Table 1.1

quantity	less for second piece than first piece	same for both pieces	greater for second piece than first piece
angular speed			
linear speed			
acceleration			

[3]

[Total: 10]

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1 A steel ball is placed on the inside surface of a hollow circular cone. The ball moves in a horizontal circle at constant speed, as shown in Fig. 1.1.

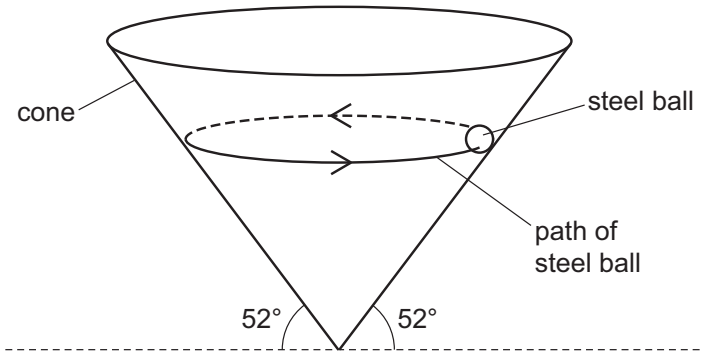


Fig. 1.1

The angle of the side of the cone to the horizontal is 52°. There is no friction between the ball and the cone.

(a) Fig. 1.2 shows a cross-section through the cone and the steel ball.

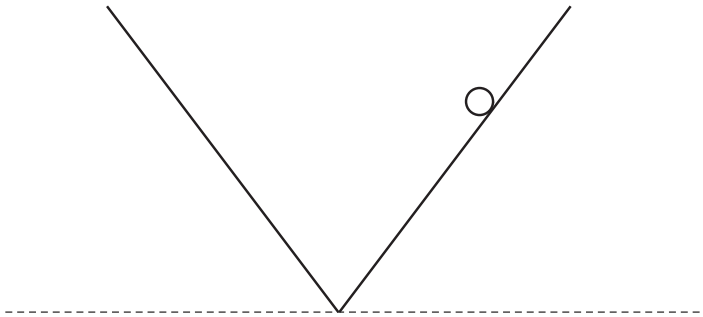


Fig. 1.2

On Fig. 1.2, draw labelled arrows to show the **two** forces acting on the ball. [1]

(b) Describe how the forces acting on the ball cause its acceleration to be centripetal.

.....

.....

.....

..... [2]

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(c) The ball moves in a circle of radius 0.15 m.

Show that the speed of the ball is 1.4 m s^{-1}.

[3]

(d) Calculate the angular speed ω of the ball.

ω = ... rad s^{-1} [2]

(e) The speed of the ball is increased.

Explain why the radius of the circular path of the ball increases.

.....

.....

..... [1]

[Total: 9]

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[Turn over

1 **(a)** Define the radian.

.....

..... [1]

(b) The rear wheel and the pedals of a bicycle are connected by a chain that passes around two cogs (toothed wheels), as shown in Fig. 1.1.

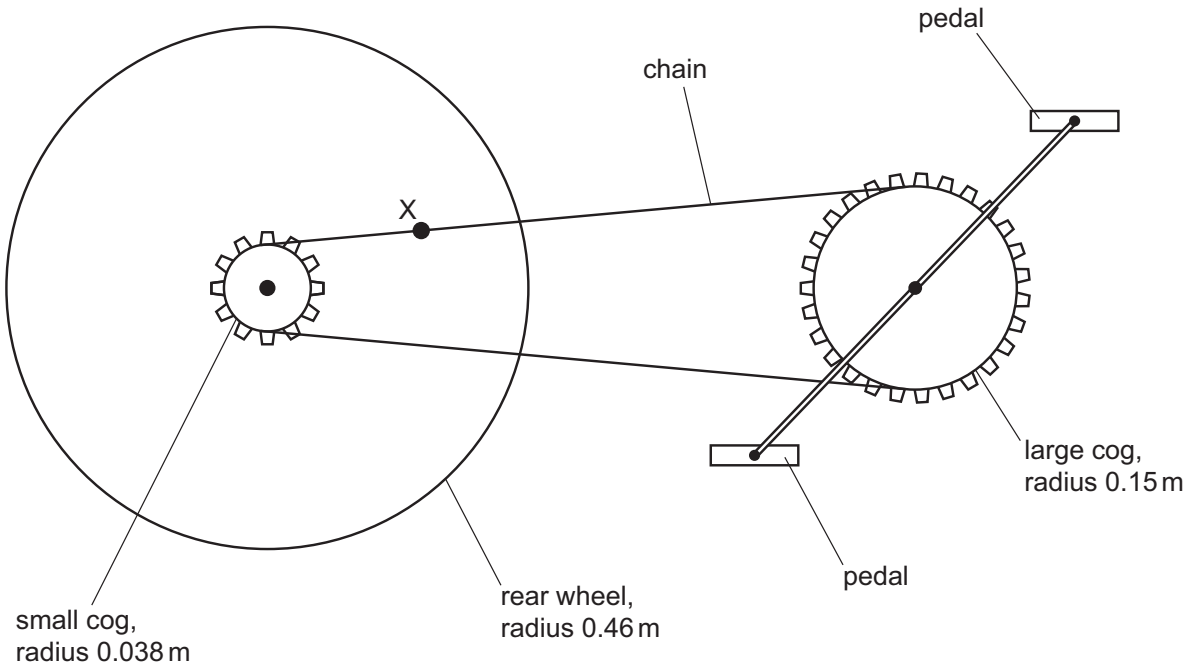


Fig. 1.1 (not to scale)

The small cog has a radius of 0.038 m and is fixed to the rear wheel so that it rotates with it. The large cog has a radius of 0.15 m and is fixed to the pedals so that it rotates with them. The rear wheel has a radius of 0.46 m.

The bicycle is being pedalled so that it moves in a straight line at a constant speed of 17 m s^{-1}.

(i) Calculate the angular speed of the rear wheel.

angular speed = ... rad s^{-1} [2]

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(ii) Calculate the period of rotation of the small cog.

period = ... s [2]

(iii) Show that the distance moved by point X on the chain during one full rotation of the small cog is 0.24 m.

[1]

(iv) Use the information in **(b)(iii)** to determine the angle through which the large cog rotates during one full rotation of the small cog.

angle = ... rad [2]

(c) The chain of the bicycle in **(b)** is moved onto a smaller cog fixed to the rear wheel. The speed of the bicycle does not change.

Explain, without calculation, the effect of this change on the angular speed of the pedals.

.....

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..... [2]

[Total: 10]

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1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

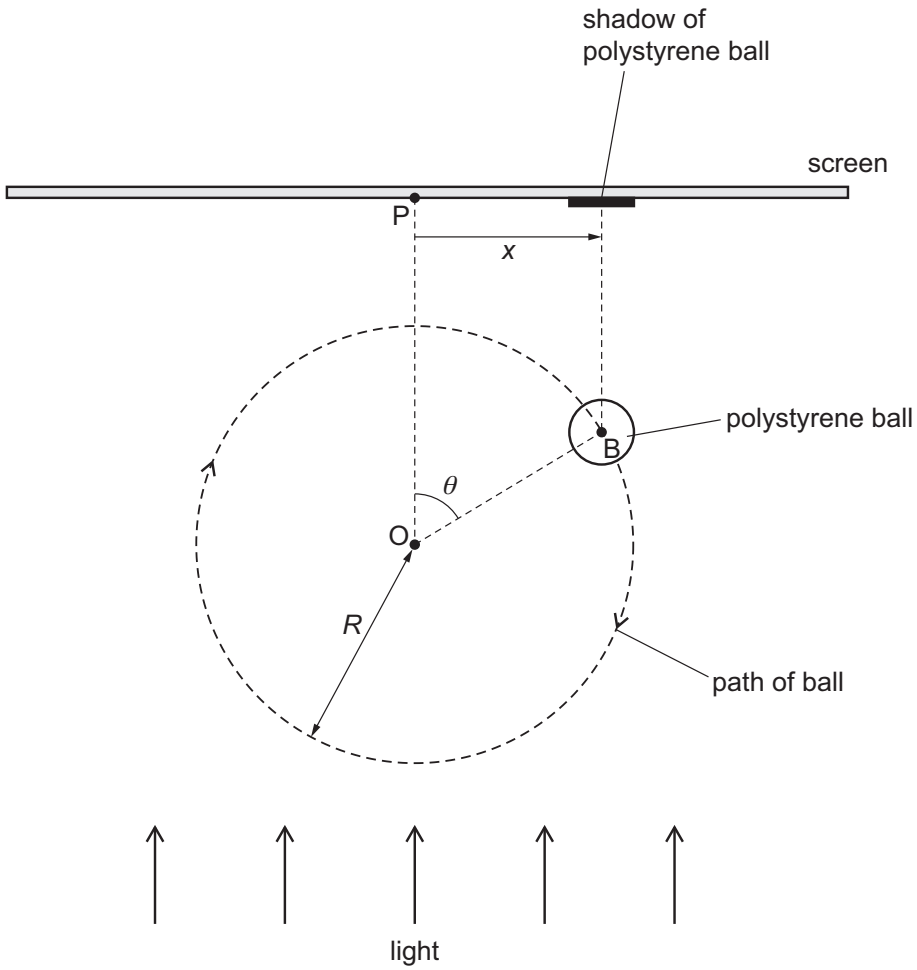


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

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(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time t = 0.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

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(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... m s^{-2} [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

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1 The Earth may be considered as a uniform sphere of radius 6.37×10^6 m.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Fig. 1.1.

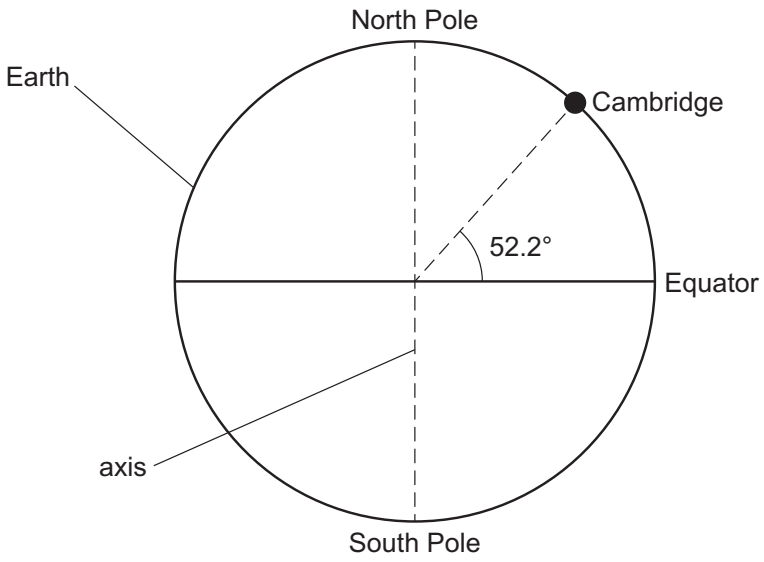


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) (i) Show that the radius of the circle around which Cambridge moves is 3.90×10^6 m.

[1]

(ii) Calculate the speed at which Cambridge moves around the circle.

speed = ... $m\,s^{-1}$ [3]

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(b) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = ... N [2]

(ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

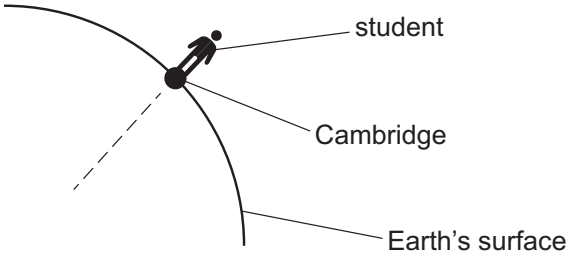


Fig. 1.2 (not to scale)

[1]

(iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in **(b)(ii)**.

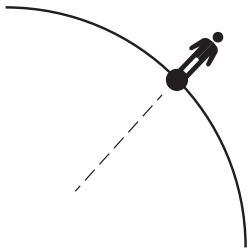


Fig. 1.3 (not to scale)

[2]

[Total: 9]

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1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

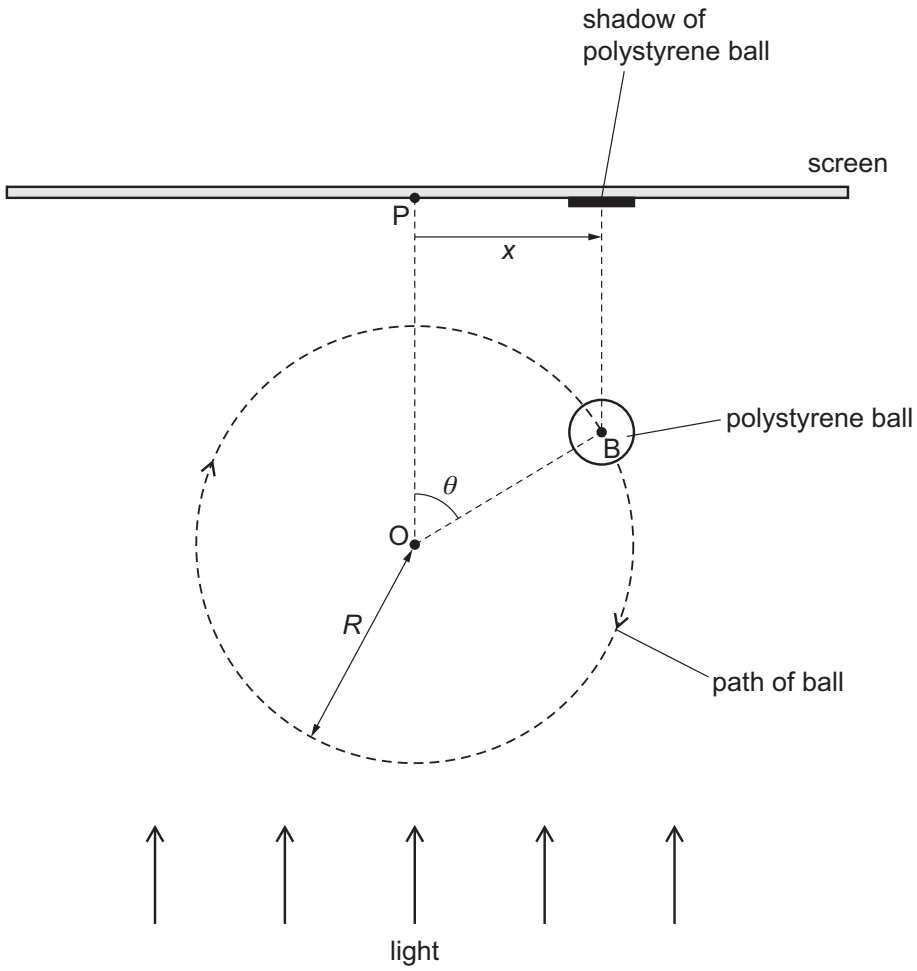


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

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(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

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(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... m s^{-2} [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

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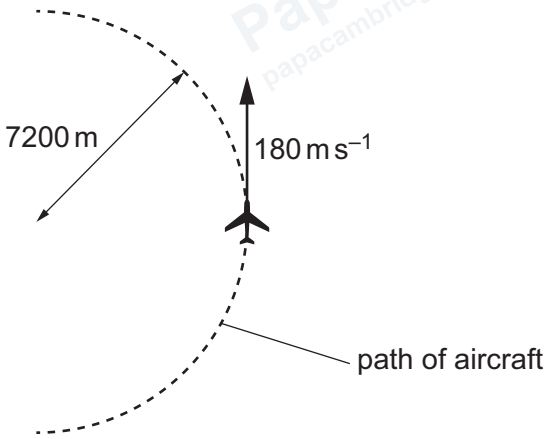
1 **(a)** With reference to velocity and acceleration, describe uniform circular motion.

.....

.....

..... [2]

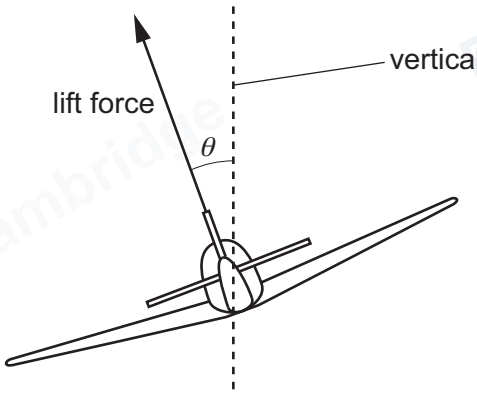
(b) An aircraft is travelling in a horizontal circle of radius 7200 m at a speed of 180 m s^{-1}, as shown in the view from above in Figure 1.1.



not to scale

Figure 1.1

In order to move in a circle, the aircraft has to tilt its wings so that the lift force from the wings is not vertical, as shown in the view from behind in Figure 1.2.



not to scale

Figure 1.2

(i) With reference to the forces acting on the aircraft, explain why the circular motion of the aircraft requires the lift force to be in the direction shown rather than vertically upwards.

.....

.....

..... [2]

(ii) On Figure 1.2, draw an arrow from the aircraft to represent the direction of the resultant force on the aircraft. [1]

(iii) Calculate the centripetal acceleration of the aircraft.

acceleration = ... $m\,s^{-2}$ [2]

(iv) Determine the angle θ at which the aircraft is tilted. Explain your reasoning.

θ = ... ° [3]

[Total: 10]

13 Gravitational fields

15 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 13 Gravitational fields

考纲要求（9702 · 2025 – 2027 syllabus）

13.1

13.1.1 understand that a gravitational field is an example of a field of force and define gravitational field as force per unit mass

13.1.2 represent a gravitational field by means of field lines

13.2

13.2.1 understand that, for a point outside a uniform sphere, the mass of the sphere may be considered to be a point mass at its centre

13.2.2 recall and use Newton’s law of gravitation F = Gm1m2 / r2 for the force between two point masses

13.2.3 analyse circular orbits in gravitational fields by relating the gravitational force to the centripetal acceleration it causes

13.2.4 understand that a satellite in a geostationary orbit remains at the same point above the Earth’s surface, with an orbital period of 24 hours, orbiting from west to east, directly above the Equator

13.3

13.3.1 derive, from Newton’s law of gravitation and the definition of gravitational field, the equation g = GM / r 2 for the gravitational field strength due to a point mass

13.3.2 recall and use g = GM / r 2

13.3.3 understand why g is approximately constant for small changes in height near the Earth’s surface

13.4

13.4.1 define gravitational potential at a point as the work done per unit mass in bringing a small test mass from infinity to the point

13.4.2 use ϕ = –GM / r for the gravitational potential in the field due to a point mass

13.4.3 understand how the concept of gravitational potential leads to the gravitational potential energy of two point masses and use EP = –GMm / r

1 **(a)** Explain why the gravitational potential near to a point mass is negative.

.....

.....

.....

..... [2]

(b) A planet may be assumed to be a uniform sphere. It has gravitational potential ϕ at distance r from the centre of the planet.

The variation with $\frac{1}{r}$ of ϕ is shown in Fig. 1.1.

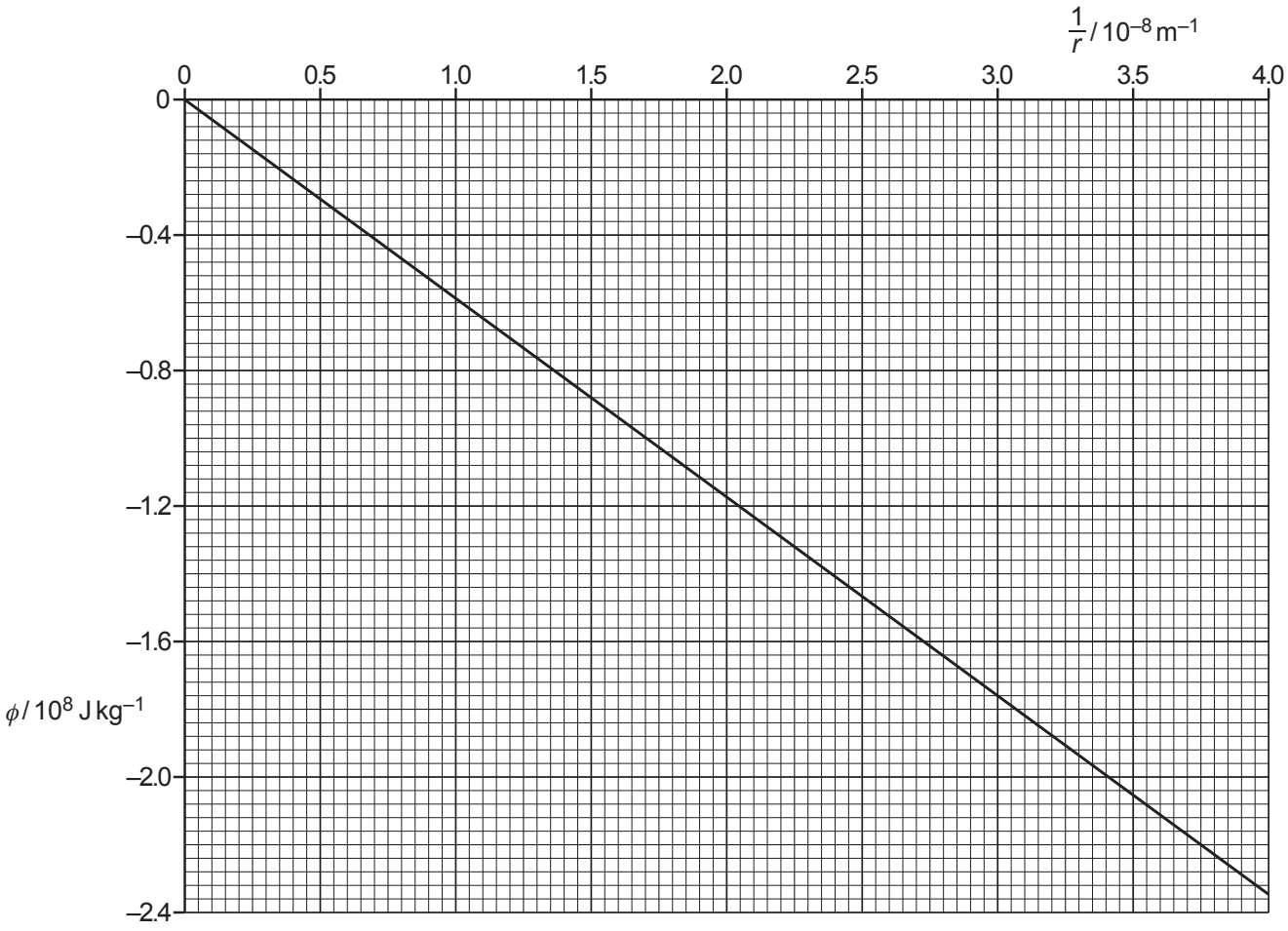


Fig. 1.1

(i) Show that the mass of the planet is 8.8×10^{25} kg.

[2]

(ii) The period of rotation of the planet is 0.72 Earth days.

A satellite in orbit around the planet remains above the same point on the surface of the planet.

Use the mass of the planet in **(b)(i)** to determine the radius R of the orbit of the satellite.

R = ... m [3]

(iii) The speed of the satellite in **(b)(ii)** is 8400 m s^{-1}. The mass of the satellite is 1200 kg.

Determine the additional energy required to move the satellite from its orbit to infinity.

energy required = ... J [3]

[Total: 10]

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1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A satellite X, of mass M, orbits a planet at a constant distance $4R$ from the centre of the planet, as shown in Fig. 1.1.

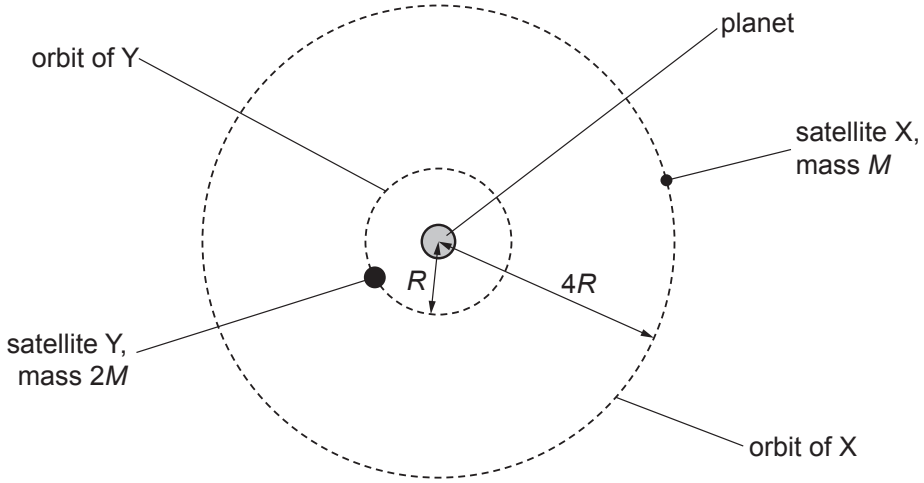


Fig. 1.1 (not to scale)

A second satellite Y, of mass $2M$, orbits the planet with orbital radius R.

The gravitational potential at X due to the planet is $-\Phi$. The planet is a uniform sphere.

(i) Explain why the gravitational potential at X is negative.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ, for the gravitational potential at Y due to the planet.

gravitational potential = ... [2]

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(iii) Complete Table 1.1 by giving expressions, in terms of some or all of M, R and Φ, for the quantities indicated for each of the satellites X and Y.

Table 1.1

	satellite X	satellite Y
gravitational field strength at satellite due to planet		
gravitational potential energy of satellite		

[4]

[Total: 10]

* 0000800000004 *

1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) A planet may be considered as a uniform sphere.

A satellite is in circular orbit of period T around the planet at a height h above the surface. The height of the orbit can be adjusted by use of the satellite's rocket engines.

Fig. 1.1 shows the variation with h of $T^{\frac{2}{3}}$.

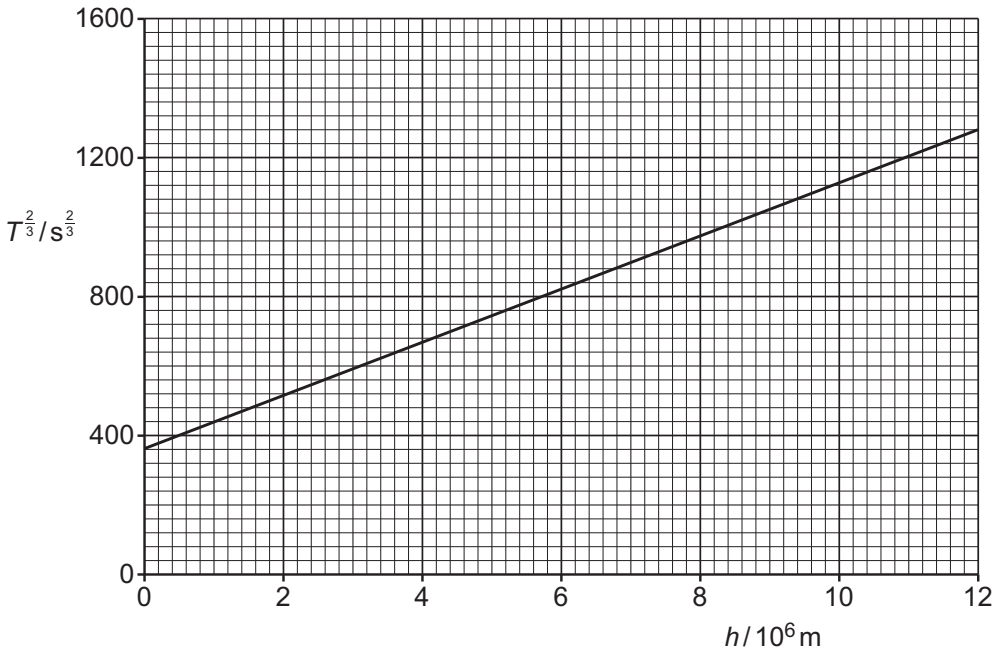


Fig. 1.1

(i) By reference to forces, explain why the orbit of the satellite is circular.

.....

.....

..... [2]

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(ii) Use Newton's law of gravitation to show that h and T are related by

$$(h + B)^3 = \frac{GA}{4\pi^2} T^2$$

where G is the gravitational constant and A and B are constants that depend on the properties of the planet.

[3]

(iii) Use the gradient and intercept of the line in Fig. 1.1 to determine values for A and B. Give units with your answers.

A = ... unit .....

B = ... unit .....

[5]

[Total: 12]

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2 The Sun may be considered as a uniform sphere with a mass of 1.99×10^{30} kg and a surface temperature of 5780 K.

A probe with a mass of 2.63 kg moves in a straight line towards the Sun.
When it is at a distance x from the centre of the Sun, the probe measures the gravitational field strength g due to the Sun and the radiant flux intensity F of radiation from the Sun.

(a) Define gravitational field.

.....

..... [1]

(b) For the position of the probe where $x = 1.47 \times 10^{11}$ m:

(i) calculate g

g = ... N kg^{-1} [2]

(ii) determine the gravitational potential energy E_P of the probe.

E_P = ... J [2]

(c) **(i)** Show that, for any particular value of x, the numerical values of g and F are related by

$$g = \frac{4\pi GM}{L} F$$

where M is the mass of the Sun, L is the luminosity of the Sun and G is the gravitational constant.

[3]

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(ii) Fig. 2.1 shows the variation of g with F.

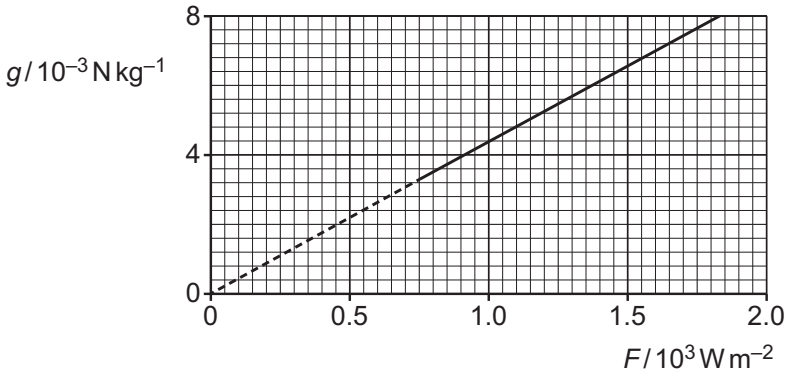


Fig. 2.1

Determine a value for the luminosity L of the Sun. Give a unit with your answer.

L = ... unit ..... [2]

(iii) Use your answer in **(c)(ii)** to determine the radius r of the Sun.

r = ... m [2]

[Total: 12]

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2 **(a)** The magnitude of the gravitational potential on the surface of a planet of radius R is ϕ. The planet can be considered to be an isolated sphere.

On Fig. 2.1, sketch the variation of the gravitational potential with distance x from the centre of the planet for values of x between R and $4R$.

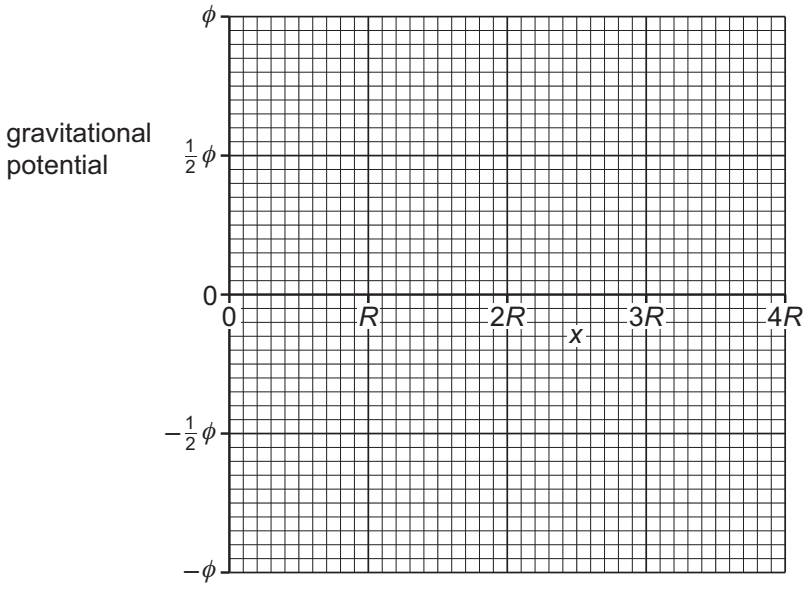


Fig. 2.1

[3]

(b) A satellite is in a geostationary orbit above the Earth. At time t = 0, the magnitude of the gravitational potential due to the Earth at the location of the satellite is ϕ.

On Fig. 2.2, sketch the variation of the gravitational potential due to the Earth at the location of the satellite for values of t between t = 0 and t = 24 hours.

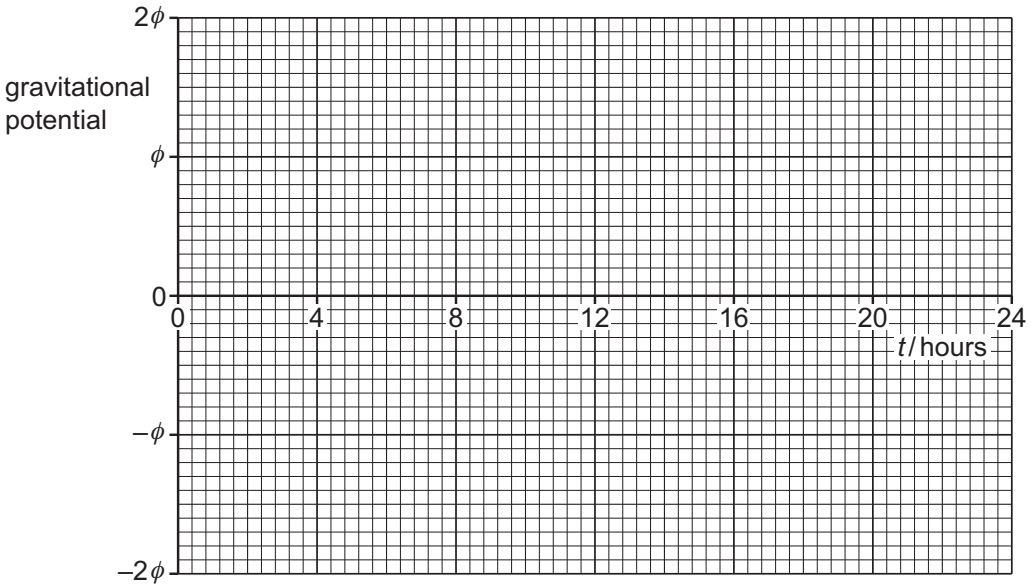


Fig. 2.2

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(c) The electric potential difference (p.d.) between two parallel plates is V, as shown in Fig. 2.3.

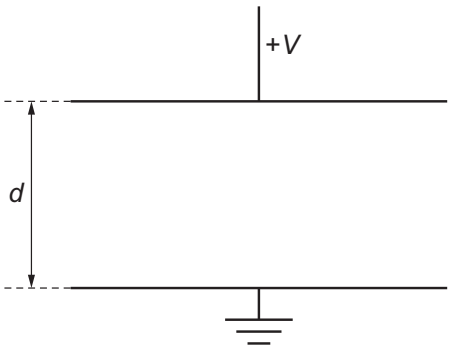


Fig. 2.3

The distance between the plates is d. The region between the plates is a vacuum.

On Fig. 2.4, sketch the variation of the electric potential with distance from the positive plate.

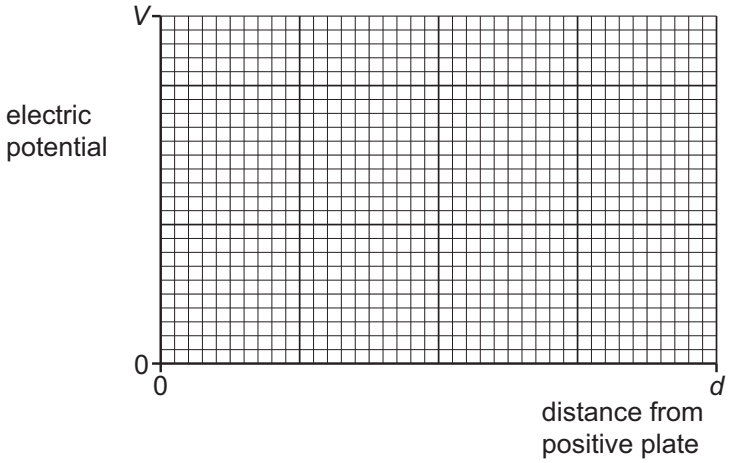


Fig. 2.4

[2]

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1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

(i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

(ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

ϕ = ... unit ..... [2]

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(c) The satellite in **(b)** is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....

..... [1]

(ii) State **one** other feature of this orbit.

.....

..... [1]

[Total: 9]

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1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

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[3]

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ϕ = ... unit ..... [2]

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(c) The satellite in **(b)** is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....

..... [1]

(ii) State **one** other feature of this orbit.

.....

..... [1]

[Total: 9]

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1 **(a)** Define gravitational field.

.....

..... [1]

(b) The gravitational field strength g at a distance x from the centre of a uniform spherical planet of mass M is given by the expression

$$g = \frac{GM}{x^2}$$

where G is the gravitational constant and distance x is greater than the radius of the planet.

(i) Describe the pattern of the field lines outside the planet that represent the gravitational field due to the planet.

.....

.....

..... [2]

(ii) Explain why, for small changes in vertical height near the surface of the planet, g may be assumed to be constant.

.....

.....

..... [2]

(c) Assume that the Earth is a uniform sphere. For the Earth, the product GM is equal to $3.99 \times 10^{14}\,m^3\,s^{-2}$.

(i) Determine a value, to three significant figures, for the radius R of the Earth.

R = ... m [2]

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(ii) Calculate the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(d) Explain why the gravitational potential energy of two point masses is always negative.

.....

.....

..... [2]

[Total: 11]

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3 **(a)** Define gravitational field at a point.

.....

..... [1]

(b) Fig. 3.1 shows an isolated point mass of mass *M*.

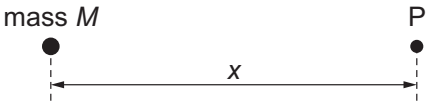


Fig. 3.1

Point P is at distance *x* from the point mass.

(i) By considering the force exerted by the point mass on a test mass of mass *m* placed at P, derive an equation for the gravitational field strength *g* at P, in terms of *M* and *x*. Identify any other symbols you use.

[2]

(ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

(iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]

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(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.3.

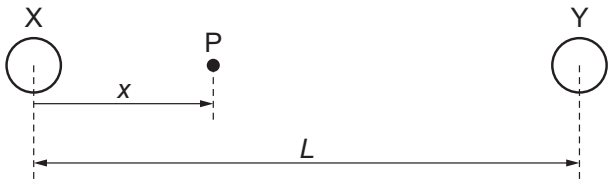


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.
The gravitational field strength at the surface of each sphere is g_0.

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

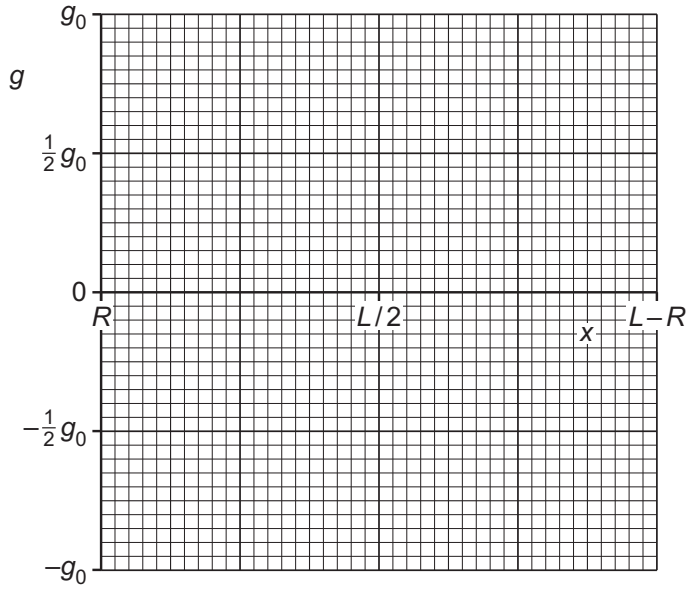


Fig. 3.4

[3]

[Total: 9]

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3 **(a)** Define gravitational field at a point.

.....

..... [1]

(b) Fig. 3.1 shows an isolated point mass of mass *M*.

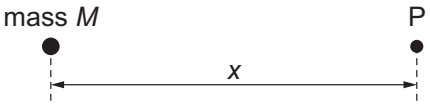


Fig. 3.1

Point P is at distance *x* from the point mass.

(i) By considering the force exerted by the point mass on a test mass of mass *m* placed at P, derive an equation for the gravitational field strength *g* at P, in terms of *M* and *x*. Identify any other symbols you use.

[2]

(ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

(iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]

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(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.3.

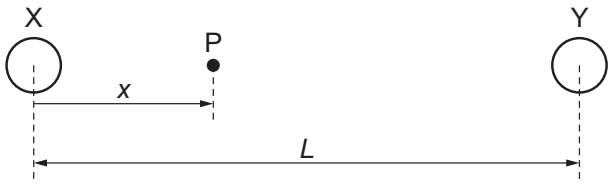


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.
The gravitational field strength at the surface of each sphere is g_0.

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

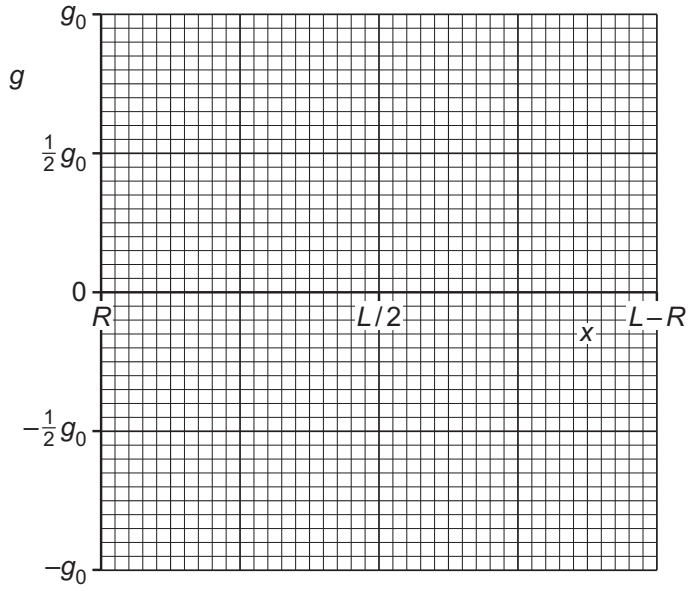


Fig. 3.4

[3]

[Total: 9]

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1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) A binary star consists of star A, of mass 4.0×10^{30} kg, and star B, of mass 2.0×10^{30} kg, separated by a distance of 3.3×10^{12} m. The stars are both in circular orbit around their common centre of gravity X, as shown in Fig. 1.1.

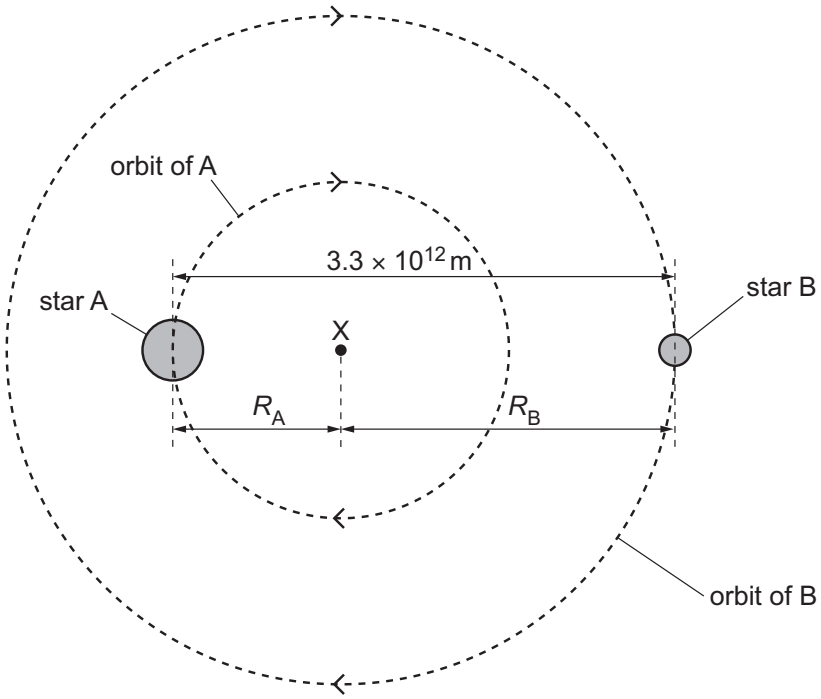


Fig. 1.1

The radius R_B of the orbit of star B is double the radius R_A of the orbit of star A.

(i) Use Newton's law of gravitation to calculate the magnitude of the gravitational force exerted by each star on the other.

force = ... N [2]

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(ii) Calculate the centripetal acceleration of star A.

acceleration = ... $m\,s^{-2}$ [1]

(iii) Use your answer in **(b)(ii)** to determine the period of the orbit of star A.

period = ... s [3]

(iv) By placing a tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for star B compare with the same quantities for star A.

Table 1.1

	B less than A	B equal to A	B greater than A
centripetal acceleration			
linear speed			
period			

[3]

[Total: 11]

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1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

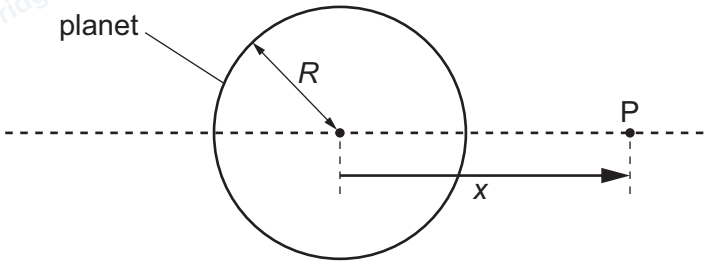


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

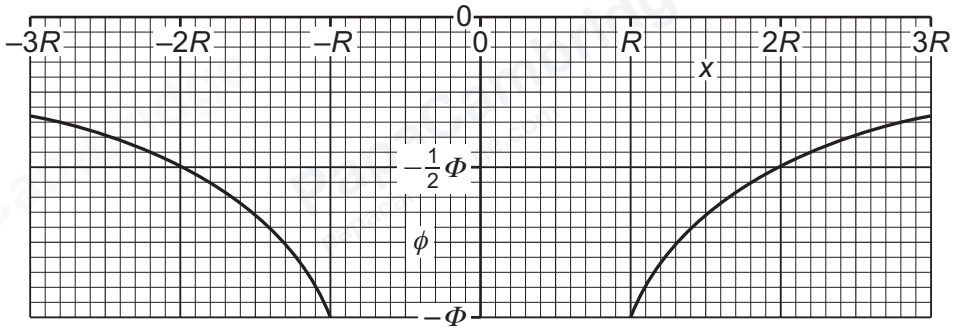


Figure 1.2

(i) Explain why ϕ is negative for all values of x.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

$M =$ ..... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

$g_0 =$ ..... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

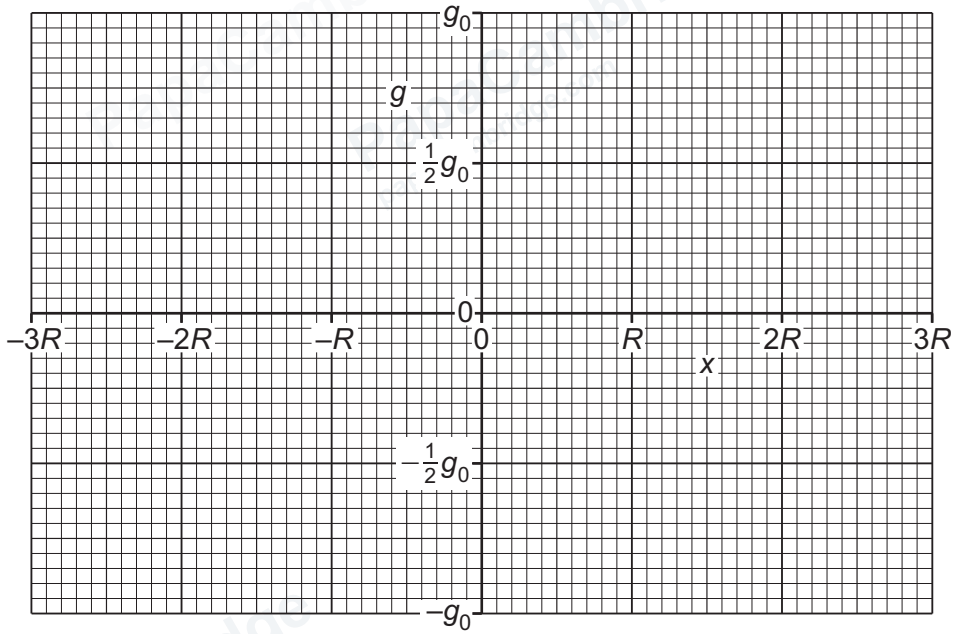


Figure 1.3

[3]

[Total: 10]

2 **(a)** **(i)** Define gravitational field at a point.

.....

..... [1]

(ii) Figure 2.1 shows a horizontal section through a small piece of ground at the Earth's surface and the air at a small distance above it.

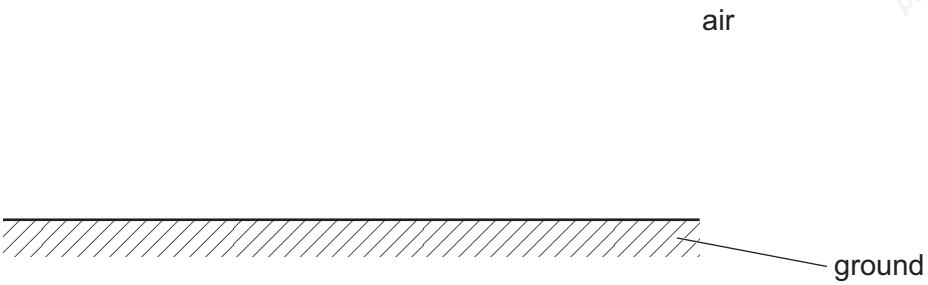


Figure 2.1

On Figure 2.1, draw four field lines to represent the gravitational field of the Earth in the air above the ground. [2]

(b) The Earth may be considered to be an isolated uniform sphere of radius 6.37×10^6 m.

(i) By considering the gravitational field strength at the Earth's surface, calculate a value for the mass of the Earth. Give your answer to three significant figures.

mass = ... kg [2]

(ii) Determine, to three significant figures, the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(iii) A satellite of mass 96.0 kg is in circular orbit at a height of 1.93×10^6 m above the Earth's surface.

Determine the gravitational potential energy of the satellite.

gravitational potential energy = ... J [2]

[Total: 9]

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

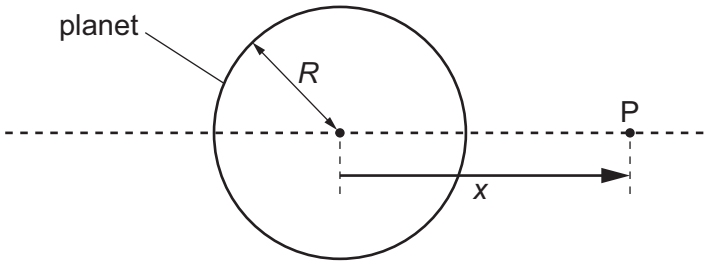


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

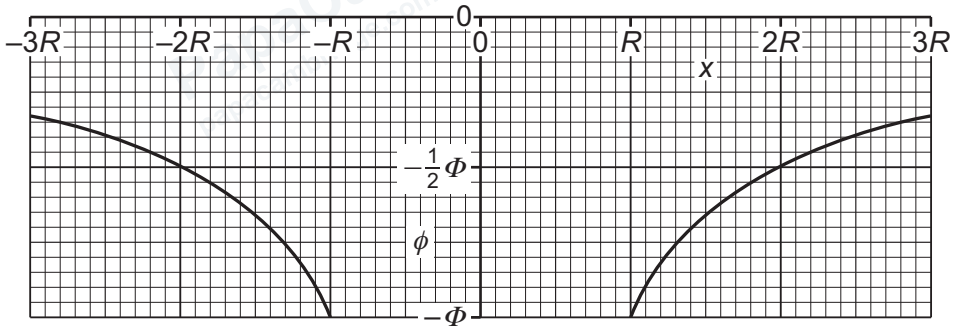


Figure 1.2

(i) Explain why ϕ is negative for all values of x.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

$M =$ ..... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

$g_0 =$ ..... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

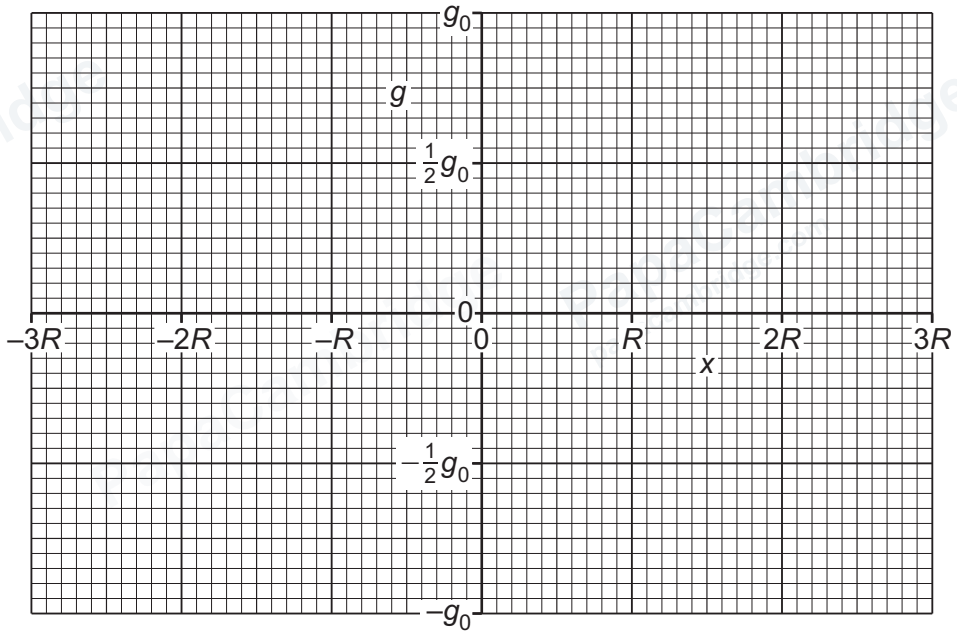


Figure 1.3

[3]

[Total: 10]

2 A satellite is in geostationary orbit around the Earth. The orbit is circular.

(a) (i) State what is meant by a geostationary orbit.

.....

.....

..... [1]

(ii) State **two** properties of the orbit.

1 .....

.....

2 .....

..... [2]

(b) The orbit has radius R and period T.

(i) Determine an expression, in terms of R and T, for the speed v of the satellite in its orbit.

v = ... [1]

(ii) Use Newton's law of gravitation to show that the mass M of the Earth is given by

$$M = \frac{4\pi^2 R^3}{GT^2}$$

where G is the gravitational constant.

[2]

(c) The Earth may be considered to be a sphere of radius 6370 km.
Geostationary satellites orbit at a height of 35 900 km above the surface of the Earth.

Determine a value, to three significant figures, for the mean density of the Earth.

density = ... $kg\,m^{-3}$ [4]

[Total: 10]

14 Temperature

9 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 14 Temperature

考纲要求（9702 · 2025 – 2027 syllabus）

14.1

14.1.1 understand that (thermal) energy is transferred from a region of higher temperature to a region of lower temperature

14.1.2 understand that regions of equal temperature are in thermal equilibrium

14.2

14.2.1 understand that a physical property that varies with temperature may be used for the measurement of temperature and state examples of such properties, including the density of a liquid, volume of a gas at constant pressure, resistance of a metal, e.m.f. of a thermocouple

14.2.2 understand that the scale of thermodynamic temperature does not depend on the property of any particular substance

14.2.3 convert temperatures between kelvin and degrees Celsius and recall that $T / K = \theta / °C + 273.15$

14.2.4 understand that the lowest possible temperature is zero kelvin on the thermodynamic temperature scale and that this is known as absolute zero

14.3

14.3.1 define and use specific heat capacity

14.3.2 define and use specific latent heat and distinguish between specific latent heat of fusion and specific latent heat of vaporisation

2 **(a)** **(i)** State the magnitude and unit of absolute zero on the thermodynamic temperature scale.

..... [1]

(ii) Explain why temperature measured using a laboratory liquid-in-glass thermometer does **not** give a measurement of thermodynamic temperature.

.....

..... [1]

(b) Fig. 2.1 shows a simplified diagram of a type of thermometer called a platinum resistance thermometer.

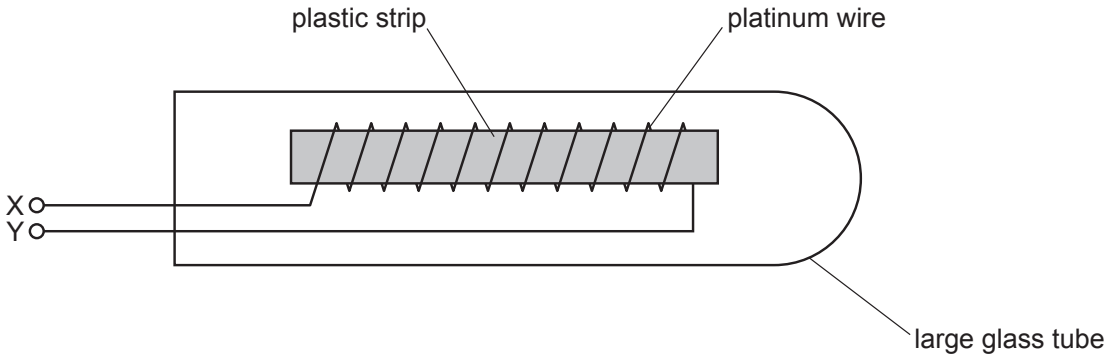


Fig. 2.1

The glass tube is immersed in the environment for which the temperature is to be determined. The resistance between the terminals X and Y is measured.

Fig. 2.2 shows the variation of the resistivity ρ of platinum with thermodynamic temperature T.

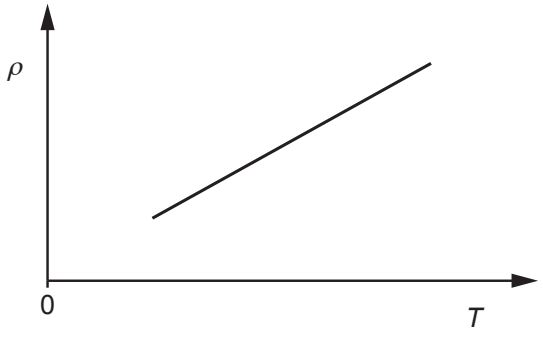


Fig. 2.2

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(i) Explain how Fig. 2.2 shows that platinum is a suitable metal for use in a resistance thermometer.

.....

.....

..... [2]

(ii) Suggest a reason why a platinum resistance thermometer is **not** suitable for measuring a rapidly changing temperature.

.....

.....

..... [1]

(iii) Suggest a type of thermometer that is suitable for measuring a rapidly changing temperature.

..... [1]

(c) A negative temperature coefficient thermistor may be used as a type of resistance thermometer.

State **one** way in which the variation with temperature of the resistance of a thermistor differs from that of a platinum wire.

.....

..... [1]

[Total: 7]

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2 **(a)** Define specific heat capacity.

.....

.....

..... [2]

(b) Two solid blocks X and Y are made from different metals. The blocks have different initial temperatures. Block Y is initially at room temperature.

The blocks are placed in direct thermal contact with each other at time $t = 0$. Fig. 2.1 shows the variation with t of the temperatures of the two blocks.

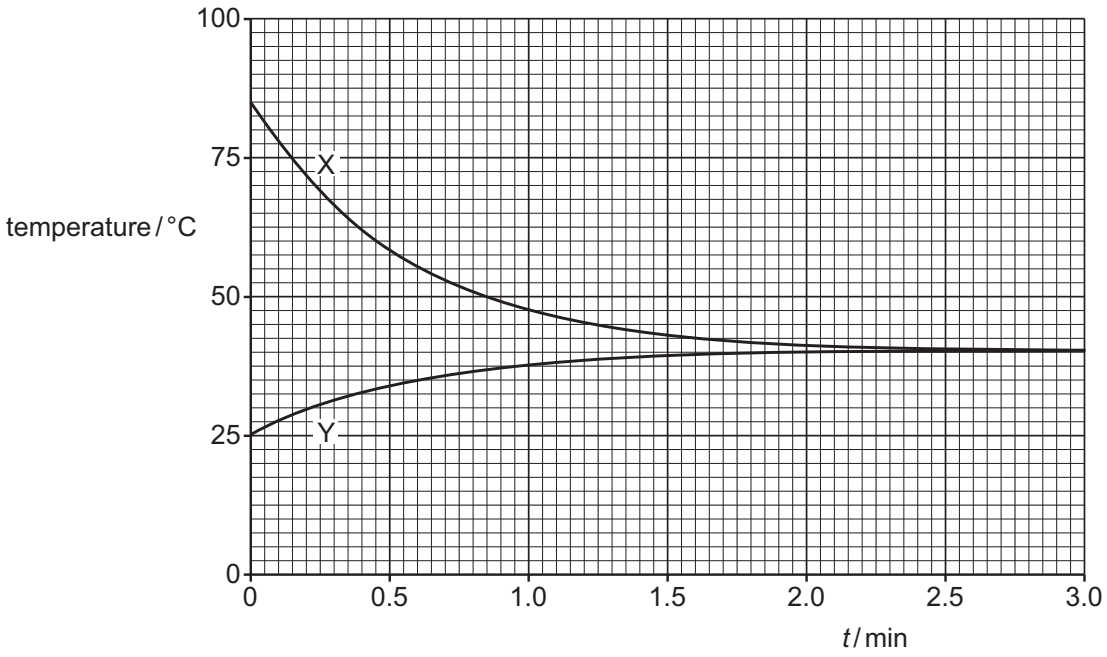


Fig. 2.1

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(i) State **three** conclusions that may be drawn from Fig. 2.1. The conclusions may be qualitative or quantitative.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(ii) The ratio $\frac{\text{mass of block Y}}{\text{mass of block X}}$ is equal to 1.3.

The metal in block Y has a specific heat capacity of 901 $J\,kg^{-1}\,K^{-1}$.

Determine the specific heat capacity of the metal in block X.

specific heat capacity = ... $J\,kg^{-1}\,K^{-1}$ [3]

[Total: 8]

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3 **(a)** Define specific latent heat.

.....

.....

..... [2]

(b) A dish containing $7.2 \times 10^{-5}\,m^3$ of a substance rests on a laboratory bench. The substance is initially a liquid of density $710\,kg\,m^{-3}$. Atmospheric pressure is $1.0 \times 10^5\,Pa$.

The liquid is heated at its boiling point so that it completely vaporises. The increase in the internal energy of the substance during this process is 17.6 kJ. The final volume of the vapour is $0.017\,m^3$.

(i) Show that the magnitude of the work done on the substance when it vaporises is 1.7 kJ.

[2]

(ii) Use the information in **(b)(i)** to calculate the thermal energy Q, in kJ, supplied to the substance to cause it to vaporise.

Q = ... kJ [2]

(iii) Use your answer in **(b)(ii)** to determine a value for the specific latent heat of vaporisation L_V, in $kJ\,kg^{-1}$, of the substance.

L_V = ... $kJ\,kg^{-1}$ [2]

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(c) The substance in **(b)** has a specific latent heat of fusion L_F.

Suggest and explain whether L_F is likely to be less than, the same as, or greater than the answer in **(b)(iii)**.

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..... [3]

[Total: 11]

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3 **(a)** Define specific latent heat.

.....

.....

..... [2]

(b) Explain why, for a substance, the specific latent heat of vaporisation is usually greater than the specific latent heat of fusion.

.....

.....

.....

.....

..... [3]

(c) An ice cube of mass 37.0 g at temperature 0.0 °C is placed in a beaker containing water of mass 208 g at temperature 26.4 °C.

When all the ice has melted, and all the water in the beaker has reached thermal equilibrium, the final temperature of all the water is 10.3 °C.

The specific heat capacity of water is 4.18 $J\,g^{-1}\,°C^{-1}$.

The beaker has negligible specific heat capacity and is perfectly insulated from the surroundings.

Determine a value, to three significant figures, for the specific latent heat of fusion of water.

specific latent heat of fusion = ... $J\,g^{-1}$ [4]

[Total: 9]

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3 **(a)** Define specific heat capacity.

.....

.....

..... [2]

(b) A block of aluminium has a volume of $3.612 \times 10^{-3}\,m^3$ at a temperature of 0 °C.

Aluminium has a density of $2.700 \times 10^{3}\,kg\,m^{-3}$ at 0 °C.
It has a density of $2.620 \times 10^{3}\,kg\,m^{-3}$ at 500 °C.

The block is heated so that its temperature increases from 0 °C to 500 °C at an atmospheric pressure of $1.01 \times 10^{5}\,Pa$.
The increase in internal energy of the block is 4.38 MJ.

(i) Calculate the mass of the block.

mass = ... kg [2]

(ii) Show that the volume of the block at a temperature of 500 °C is $3.722 \times 10^{-3}\,m^3$.

[1]

(iii) Use the information in **(b)(ii)** to determine the magnitude of the work done on the block when its temperature is raised from 0 °C to 500 °C.

work done = ... J [2]

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(iv) Explain whether the work done on the block is positive or negative.

.....

.....

..... [2]

(v) Use the first law of thermodynamics to determine, to three significant figures, a value for the specific heat capacity of aluminium. Explain your reasoning. Give a unit with your answer.

specific heat capacity = ... unit ..... [3]

(c) Without further calculation, suggest with a reason how doubling the pressure in **(b)** is likely to affect the answer in **(b)(v)**.

.....

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..... [1]

[Total: 13]

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3 **(a)** Define specific latent heat.

.....

.....

..... [2]

(b) Explain why, for a substance, the specific latent heat of vaporisation is usually greater than the specific latent heat of fusion.

.....

.....

.....

.....

..... [3]

(c) An ice cube of mass 37.0 g at temperature 0.0 °C is placed in a beaker containing water of mass 208 g at temperature 26.4 °C.

When all the ice has melted, and all the water in the beaker has reached thermal equilibrium, the final temperature of all the water is 10.3 °C.

The specific heat capacity of water is 4.18 $J\,g^{-1}\,°C^{-1}$.

The beaker has negligible specific heat capacity and is perfectly insulated from the surroundings.

Determine a value, to three significant figures, for the specific latent heat of fusion of water.

specific latent heat of fusion = ... $J\,g^{-1}$ [4]

[Total: 9]

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2 **(a)** State what is meant by two objects being in thermal equilibrium.

.....

.....

..... [2]

(b) A mass X of ice at 0 °C is placed in a beaker containing a mass M of water at Celsius temperature t. The beaker is perfectly insulated and has negligible heat capacity. After some time, the ice that was added reaches thermal equilibrium with the original water in the beaker.

The specific latent heat of fusion of water is L. The specific heat capacity of water is c. The final Celsius temperature of the system is θ.

Give expressions, in terms of some or all of X, M, t, θ, L and c, for the thermal energy:

(i) E_1, gained by the ice as it melts to become water at 0 °C

E_1 = ... [1]

(ii) E_2, lost by the water as its Celsius temperature decreases from t to θ

E_2 = ... [1]

(iii) E_3, gained by the melted ice as its Celsius temperature increases from 0 °C to θ.

E_3 = ... [1]

(c) Use your answers in **(b)** to show that the final Celsius temperature θ of the system is given by

$$\theta = \frac{Mct - XL}{c(M + X)}.$$

[2]

[Total: 7]

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3 **(a)** State what is meant by two objects being in thermal equilibrium.

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.....

..... [2]

(b) Fig. 3.1 shows a type of thermometer called a constant volume gas thermometer.

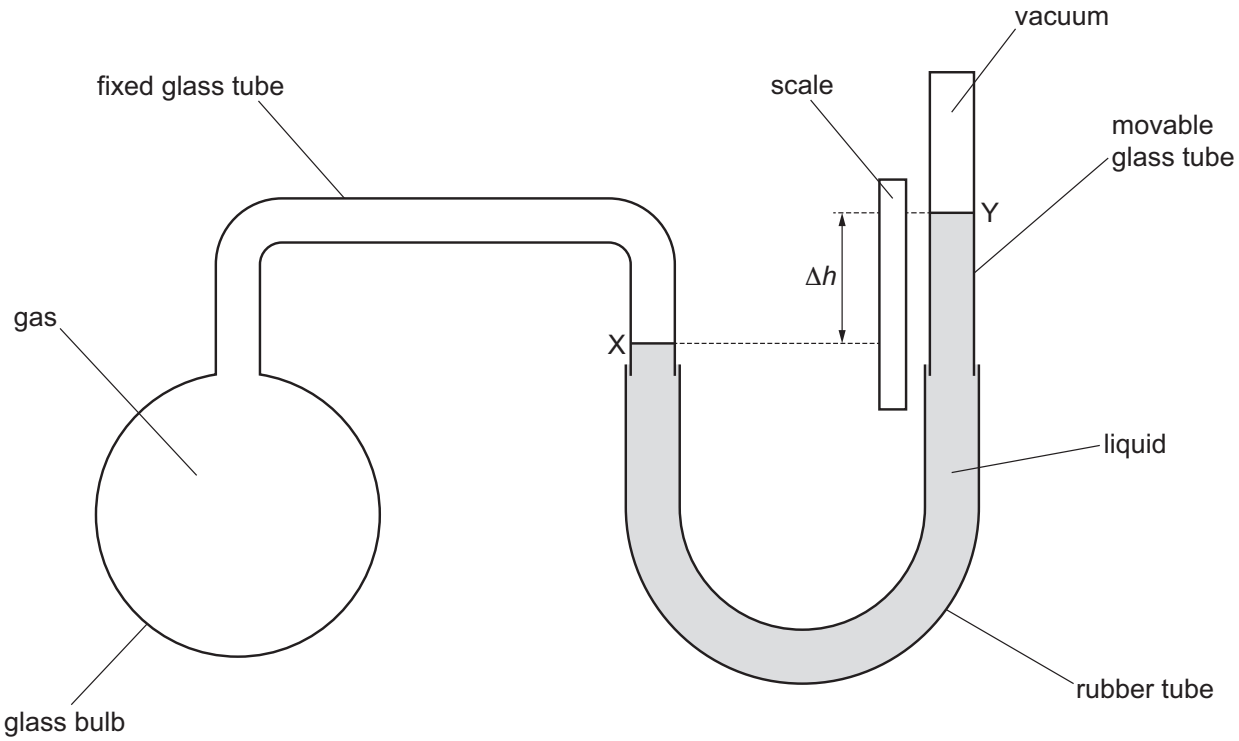


Fig. 3.1 (not to scale)

The thermometer is used to determine the thermodynamic temperature T of the gas in the glass bulb.

The glass bulb is immersed in the environment for which the temperature is to be measured. The height of the movable glass tube is then adjusted so that the level of the liquid on the left-hand side aligns with the reference line X marked on the fixed glass tube. The reference line Y is marked on the side of the movable glass tube. The level of the liquid at Y is higher than at X as a result of the pressure of the gas in the glass bulb.

The difference in height Δh between the liquid levels at X and Y is then measured using the scale. The thermodynamic temperature T of the gas is directly proportional to the pressure of the gas. This pressure is directly proportional to Δh.

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(i) The value of Δh can be used to calculate the pressure of the gas. In order to do this, the gravitational field strength is used, along with a property of the liquid.

State the property of the liquid that is used to calculate the pressure.

..... [1]

(ii) Before the measurement of Δh can be made, the glass bulb needs to reach thermal equilibrium with the environment for which the temperature is to be measured.

State **two** disadvantages of using a constant volume gas thermometer to measure temperature.

1 .....

.....

2 .....

.....
[2]

(iii) Suggest **one** situation in which a constant volume gas thermometer would be an appropriate type of thermometer to choose for measuring temperature.

.....

.....

..... [1]

(iv) Level X aligns with 2.31 cm on the scale. At 0 °C, level Y aligns with 8.69 cm.

At temperature θ, level Y aligns with 7.83 cm on the scale.

Determine a value for θ in °C.

θ = ... °C [3]

[Total: 9]

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3 **(a)** State what is meant by two objects being in thermal equilibrium.

.....

.....

..... [2]

(b) An insulated container with negligible specific heat capacity contains a mass of 63 g of water at an initial temperature of 78.0 °C.

A thermistor with a mass of 8.5 g is at an initial temperature of 25.0 °C. The thermistor is connected into a circuit containing an ammeter that is accurately calibrated to give readings of temperature in °C.

The thermistor is immersed in the water. When thermal equilibrium is reached, the ammeter indicates a temperature of 76.8 °C.

(i) Explain why the thermistor does not measure the temperature as 78.0 °C.

.....

.....

..... [2]

(ii) The specific heat capacity of water is $4.2\,J\,g^{-1}\,K^{-1}$.

Calculate a value for the specific heat capacity, in $J\,g^{-1}\,K^{-1}$, of the material of the thermistor.

specific heat capacity = ... $J\,g^{-1}\,K^{-1}$ [3]

(c) Suggest, with a reason, how the initial temperature of the water in **3(b)** could be measured more accurately than using the thermistor in **3(b)**.

.....

.....

.....

..... [2]

[Total: 9]

15 Ideal gases

14 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 15 Ideal gases

考纲要求（9702 · 2025 – 2027 syllabus）

15.1

15.1.1 understand that amount of substance is an SI base quantity with the base unit mol

15.1.2 use molar quantities where one mole of any substance is the amount containing a number of particles of that substance equal to the Avogadro constant NA

15.2

15.2.1 understand that a gas obeying $pV \propto T$, where T is the thermodynamic temperature, is known as an ideal gas

15.2.2 recall and use the equation of state for an ideal gas expressed as $pV = nRT$, where n = amount of substance (number of moles) and as $pV = NkT$, where N = number of molecules

15.2.3 recall that the Boltzmann constant k is given by $k = R / N_A$

15.3

15.3.1 state the basic assumptions of the kinetic theory of gases

15.3.2 explain how molecular movement causes the pressure exerted by a gas and hence deduce the relationship $p = \frac{1}{3}\rho\langle c^2\rangle$

15.3.3 understand that the root-mean-square speed $c_{r.m.s.}$ is given by $\sqrt{\langle c^2\rangle}$

15.3.4 compare $pV = \frac{1}{3}Nm\langle c^2\rangle$ with $pV = NkT$ to deduce that the average translational kinetic energy of a molecule is $(3/2)kT$, and recall and use this expression

3 **(a)** **(i)** State what is meant by an ideal gas.

.....

.....

..... [2]

(ii) Use one of the basic assumptions of the kinetic theory to explain what can be deduced about the potential energy associated with the random motion of molecules in an ideal gas.

.....

.....

..... [2]

(b) A sample of 0.26 m^3 of an ideal gas is at pressure 2.0×10^5 Pa and temperature 290 K.

Determine:

(i) the number N of molecules of the gas

N = ... [2]

(ii) the average translational kinetic energy E_K of one molecule of the gas

E_K = ... J [2]

(iii) the internal energy of the gas. Explain your reasoning.

internal energy = ... J [2]

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(c) The volume V of the gas in **(b)** is now varied, keeping its pressure constant.

On Fig. 3.1, sketch the variation with V of the internal energy U of the gas.



Fig. 3.1

[2]

[Total: 12]

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2 **(a)** With reference to thermal energy, state what is meant by two objects being in thermal equilibrium.

.....

.....

..... [1]

(b) Two cylinders X and Y each contain a sample of an ideal gas. The samples are in thermal equilibrium with each other.

X has a volume of 0.0260 m^3 and contains 0.740 mol of gas at a pressure of 1.20×10^5 Pa. Y has a volume of 0.0430 m^3 and contains gas at a pressure of 2.90×10^5 Pa. Data for the two cylinders are shown in Fig. 2.1.



Fig. 2.1

(i) Show that the temperature of the gas in X is 234 °C.

[3]

(ii) Determine the number N of molecules of the gas in Y. Explain your reasoning.

N = ... [3]

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(iii) The gas in X consists of molecules that each have a mass that is four times the mass of a molecule of the gas in Y.

Explain how the root-mean-square (r.m.s.) speed of the molecules in X compares with the r.m.s. speed of the molecules in Y.

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..... [3]

[Total: 10]

* 0000800000008 *

3 **(a)** **(i)** State what is meant by the Avogadro constant.

.....

.....

..... [1]

(ii) State the relationship between the Avogadro constant N_A, the molar gas constant R and the Boltzmann constant k.

[1]

(b) Two samples X and Y of ideal gases are both at thermodynamic temperature T.

Sample X has volume V and consists of N molecules, each of mass m.
Sample Y has volume $2V$ and consists of $2N$ molecules, each of mass $2m$.

(i) Complete Table 3.1 by giving expressions, in terms of some or all of N, m, T, V and the constants in **(a)(ii)**, for the quantities indicated.

Table 3.1

	sample X	sample Y
pressure		
amount of substance		
mean-square speed of molecules		
internal energy		

[4]

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(ii) The temperature of sample X is now varied.

On Fig. 3.1, sketch the variation with thermodynamic temperature of the root-mean-square (r.m.s.) speed of the molecules of the gas.

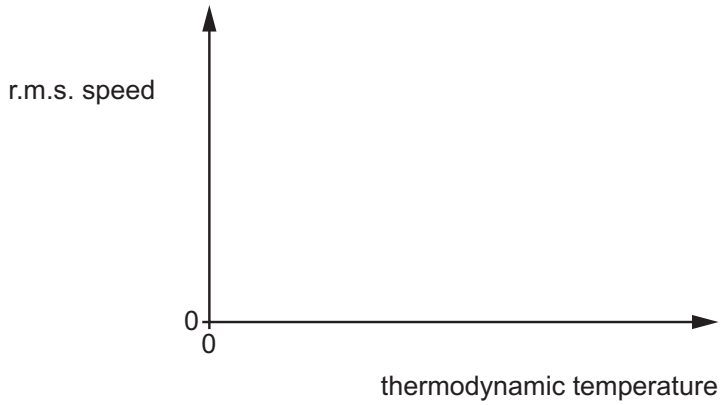


Fig. 3.1

[2]

[Total: 8]

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4 **(a)** State **three** of the basic assumptions of the kinetic theory of gases.

1 .....

.....

2 .....

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3 .....

.....

[3]

(b) Explain how molecular movement causes the pressure exerted by a gas.

.....

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..... [3]

(c) Fig. 4.1 shows the variation with thermodynamic temperature T of the mean-square speeds $\langle c^2 \rangle$ for two gases X and Y.

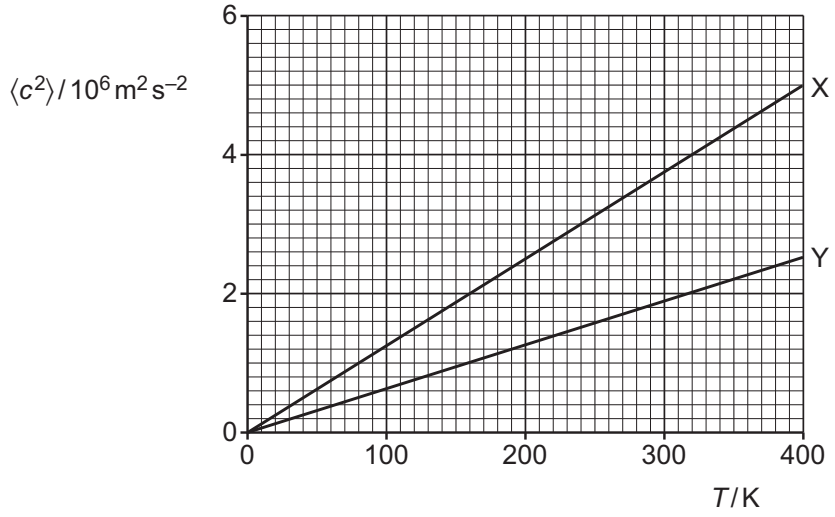


Fig. 4.1

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Fig. 4.2 shows the variation with T of the product pV for samples of the two gases, where p is the pressure of the gas and V is the volume of the gas.

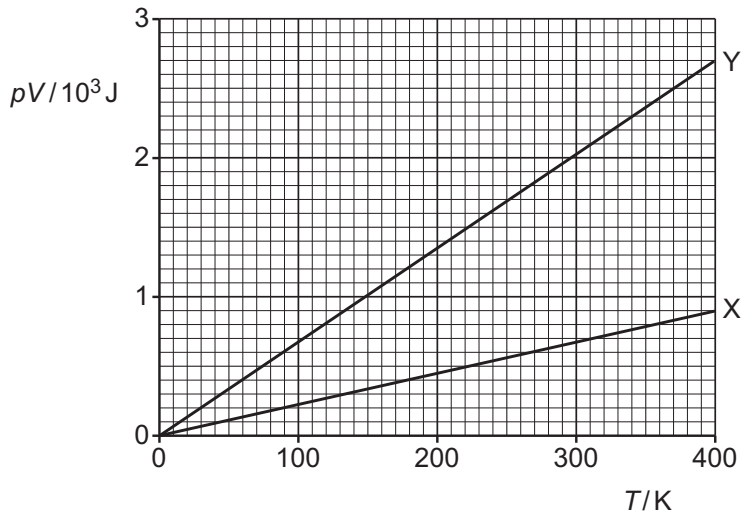


Fig. 4.2

State **three** conclusions about the gases and their samples that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

1 .....

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2 .....

.....

3 .....

.....

[3]

[Total: 9]

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3 **(a)** Two metal cuboids P and Q are in thermal contact with each other.

(i) P and Q are in thermal equilibrium.

State what is meant by the term thermal equilibrium.

.....

.....

..... [2]

(ii) Data for P and Q are given in Table 3.1.

Table 3.1

	P	Q
specific heat capacity / J kg^{-1} K^{-1}	390	910
mass / kg	0.54	0.37

P and Q are initially both at the same temperature.

P is supplied with 24 kJ of thermal energy. After some time, P and Q are once again both at the same temperature as each other.

P and Q are perfectly insulated from the surroundings.

Determine the change in temperature ΔT of Q.

ΔT = ... K [3]

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(b) Nitrogen may be assumed to be an ideal gas. A fixed amount of nitrogen gas is contained at a constant pressure of 1.6×10^5 Pa.

The variation of the volume V of the gas with the temperature θ of the gas is shown in Fig. 3.1.

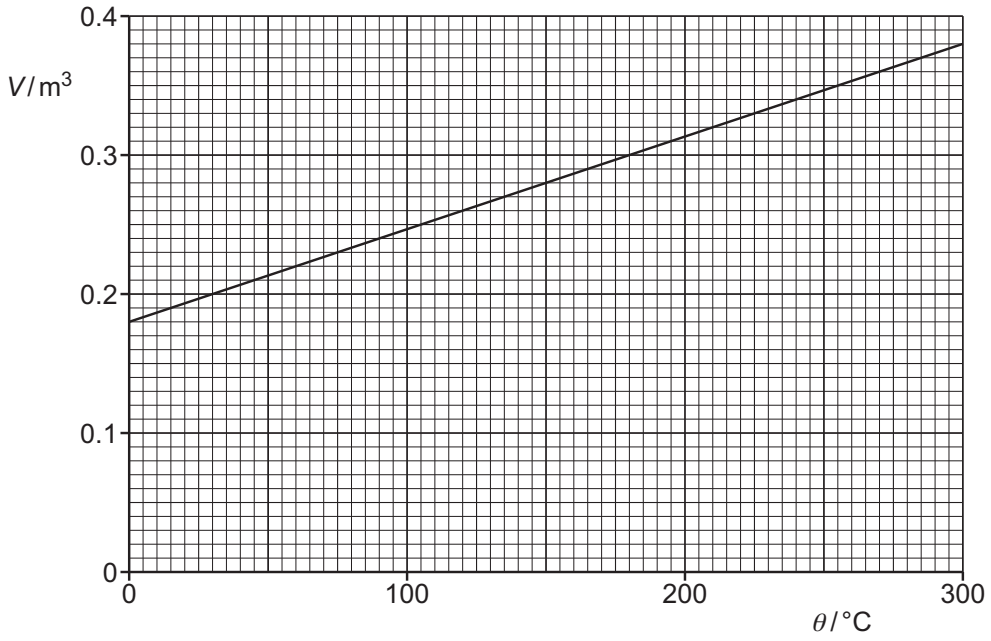


Fig. 3.1

(i) The temperature of the nitrogen gas is increased from 0 °C to 210 °C.
Determine the work done on the gas.

work done = ... J [3]

(ii) Determine the number N of molecules of nitrogen gas.

N = ... [2]

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(iii) The mass of a nitrogen molecule is 4.7×10^{-26} kg.

Calculate the root-mean-square (r.m.s.) speed of a nitrogen molecule at 210 °C.

r.m.s. speed = ... $m\,s^{-1}$ [2]

[Total: 12]

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4 **(a)** The equation of state for an ideal gas may be written as

$$pVA = NBT$$

where p is the pressure of the gas, V is the volume of the gas, A is the Avogadro constant, B is another constant and N is the number of molecules of the gas.

(i) State the meaning, in the equation, of the symbol T.

..... [1]

(ii) Identify the constant B.

..... [1]

(b) The product pV for an ideal gas is also given by

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle.$$

(i) State the meanings, in this equation, of the symbols m and $\langle c^2\rangle$.

m: .....

$\langle c^2\rangle$: .....
[2]

(ii) Use the equations in **(a)** and **(b)** to derive an expression, in terms of A, B and T, for the mean kinetic energy E_K of a molecule of the gas.

E_K = ... [2]

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(c) On Fig. 4.1, sketch the variation with T of the root-mean-square (r.m.s.) speed of the molecules of an ideal gas.

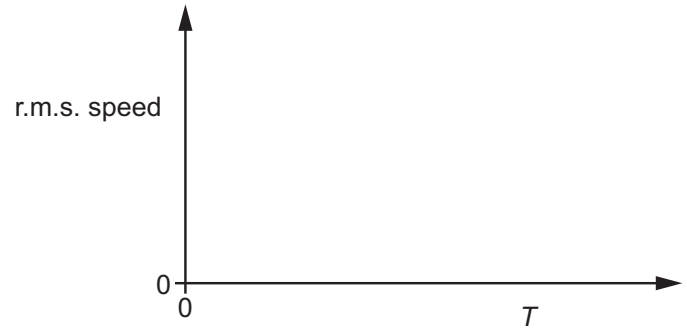


Fig. 4.1

[2]

[Total: 8]

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3 **(a)** State what is meant by an ideal gas.

.....

.....

..... [2]

(b) An ideal gas at a pressure of 1.6×10^5 Pa has a density of $1.9\,\text{kg}\,\text{m}^{-3}$.

(i) Show that the root-mean-square (r.m.s.) speed of molecules of this gas is approximately $500\,\text{m}\,\text{s}^{-1}$.

[3]

(ii) One molecule of the gas has a mass of 4.7×10^{-26} kg.

Determine the thermodynamic temperature of the gas.

temperature = ... K [2]

(c) Calculate the internal energy U of 6.0 mol of the gas in **(b)**. Explain your reasoning.

U = ... J [3]

[Total: 10]

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4 **(a)** State the value of absolute zero on:

(i) the Celsius temperature scale

temperature = ... °C [1]

(ii) the thermodynamic temperature scale. Give a unit with your answer.

temperature = ... unit ..... [1]

(b) A sample contains a fixed amount of gas. The gas has pressure p, volume V and thermodynamic temperature T.

Fig. 4.1 shows the variation of pV with kT for the sample, where k is the Boltzmann constant.

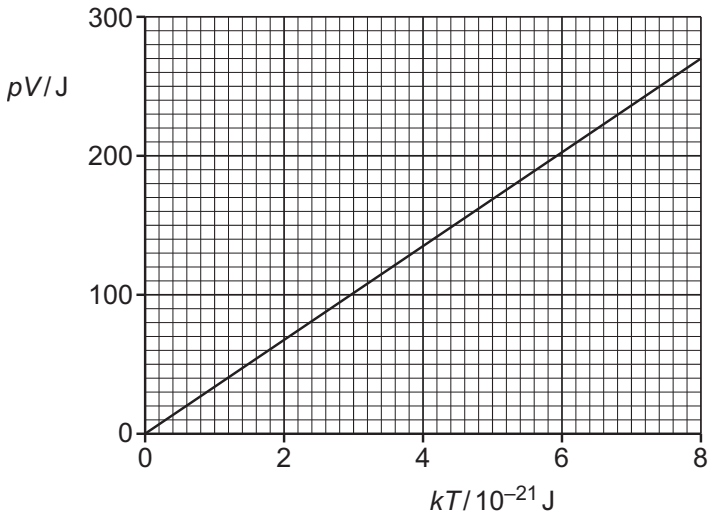


Fig. 4.1

(i) State what is indicated about the nature of the gas from the variation shown in Fig. 4.1.

..... [1]

(ii) Determine the number N of molecules of the gas in the sample.

N = ... [2]

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(iii) Use your answer in **(b)(ii)** to determine the amount n of gas in the sample.

n = ... mol [1]

(c) The root-mean-square (r.m.s.) speed of the molecules of the gas is 1900 m s^{-1} when pV is equal to 270 J.

Determine the mass, in u, of one molecule of the gas, where u is the unified atomic mass unit.

mass = ... u [4]

[Total: 10]

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2 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) One of the basic assumptions of the kinetic theory of gases is that there are no forces exerted between the molecules of the gas except during collisions.

State **two** other basic assumptions of the kinetic theory of gases.

1 .....

.....

2 .....

.....
[2]

(c) Hydrogen gas consists of molecules that each have a mass of 3.34×10^{-27} kg. Hydrogen may be considered to be an ideal gas.

A spherical balloon contains 0.0160 mol of hydrogen gas at a temperature of 282 K. At this temperature, the volume of gas in the balloon is 1.87×10^{-4} m^3.

(i) Determine the pressure of the gas.

pressure = ... Pa [2]

(ii) Estimate the average separation of the hydrogen molecules in the gas.

average separation = ... m [2]

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(d) (i) Use your answer in **(c)(ii)** to calculate the average gravitational force between adjacent molecules in hydrogen gas.

average force = ... N [2]

(ii) By considering the weight of a molecule, suggest with a reason whether your answer in **(d)(i)** is consistent with the assumption of the kinetic theory of gases that there are no forces exerted between molecules.

.....

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..... [1]

[Total: 11]

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4 **(a)** State the value of absolute zero on:

(i) the Celsius temperature scale

temperature = ... °C [1]

(ii) the thermodynamic temperature scale. Give a unit with your answer.

temperature = ... unit ..... [1]

(b) A sample contains a fixed amount of gas. The gas has pressure p, volume V and thermodynamic temperature T.

Fig. 4.1 shows the variation of pV with kT for the sample, where k is the Boltzmann constant.

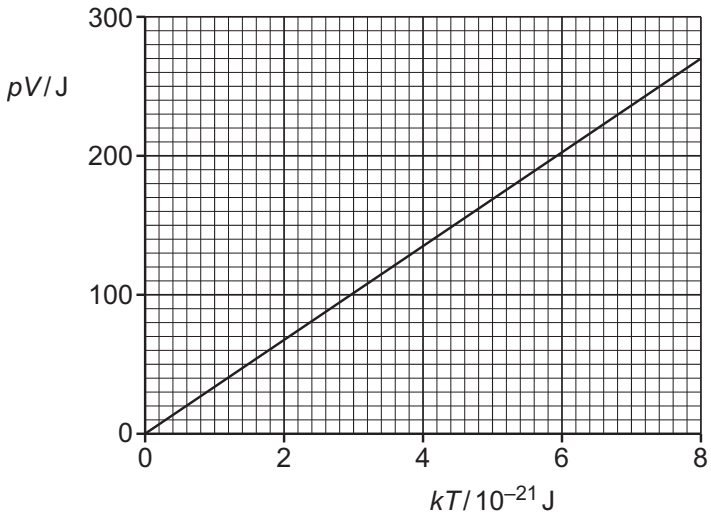


Fig. 4.1

(i) State what is indicated about the nature of the gas from the variation shown in Fig. 4.1.

..... [1]

(ii) Determine the number N of molecules of the gas in the sample.

N = ... [2]

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(iii) Use your answer in **(b)(ii)** to determine the amount n of gas in the sample.

n = ... mol [1]

(c) The root-mean-square (r.m.s.) speed of the molecules of the gas is 1900 m s^{-1} when pV is equal to 270 J.

Determine the mass, in u, of one molecule of the gas, where u is the unified atomic mass unit.

mass = ... u [4]

[Total: 10]

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2 **(a)** The equation of state for an ideal gas may be expressed as

$$pV = NkT.$$

(i) State the meaning of each of the symbols in this equation.

p: .....

V: .....

N: .....

k: .....

T: .....

[3]

(ii) Using the equation of state, derive an expression for the average translational kinetic energy E_K of a particle in the gas in terms of some or all of N, k and T.

E_K = ... [2]

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(b) A molecule of hydrogen gas consists of two hydrogen atoms, each of nucleon number 1. A molecule of oxygen gas consists of two oxygen atoms, each of nucleon number 16.

Assume that hydrogen and oxygen both behave as ideal gases.

A sample of hydrogen gas is at the same temperature as a sample of oxygen gas.

For the two samples, determine the ratio

$$\frac{\text{root-mean-square (r.m.s.) speed of hydrogen molecules}}{\text{root-mean-square (r.m.s.) speed of oxygen molecules}}.$$

ratio = ... [2]

[Total: 7]

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2 **(a)** State what is meant by an ideal gas.

.....

.....

..... [2]

(b) Ideal gases may be used to measure the temperature of the environment in a laboratory experiment. In some situations, however, it is preferable to use a thermocouple thermometer.

(i) Explain how a thermometer that uses an ideal gas can be advantageous compared with a thermometer that uses a liquid in glass.

.....

.....

..... [1]

(ii) Describe a laboratory situation in which a thermocouple thermometer is preferable to a thermometer that uses an ideal gas.

.....

.....

..... [1]

(c) A very small sample of an ideal gas consists of five molecules, each of mass 3.34×10^{-27} kg. At a particular instant, the speeds of the molecules are as shown in Figure 2.1.

$357\,m\,s^{-1}$	$424\,m\,s^{-1}$	$509\,m\,s^{-1}$	$286\,m\,s^{-1}$	$435\,m\,s^{-1}$

Figure 2.1

Determine, to three significant figures, the root-mean-square (r.m.s.) speed c_0 of the molecules.

c_0 = ... $m\,s^{-1}$ [2]

(d) A larger sample of the gas in **2(c)** consists of 5.0×10^{23} molecules. These molecules have the same r.m.s. speed as the molecules of the gas in **2(c)**.

(i) Calculate the amount of gas.

amount of gas = ... mol [2]

(ii) Determine the thermodynamic temperature T_0 of the gas.

T_0 = ... K [2]

(iii) The temperature of the gas is now gradually increased.

On Figure 2.2, sketch the variation of the r.m.s. speed of the molecules of the gas with thermodynamic temperature.

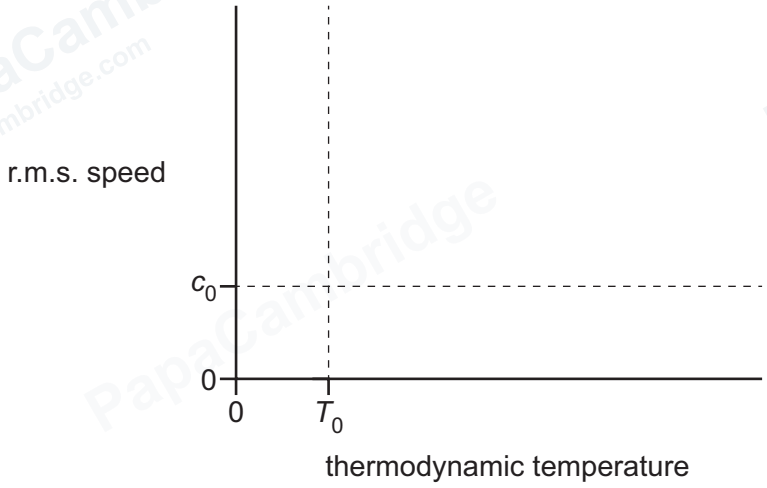


Figure 2.2

[3]

[Total: 13]

4 **(a)** State the **two** assumptions of the kinetic theory of gases that relate to collisions of particles with each other and with the walls of the container of the gas.

1 .....

.....

2 .....

.....

[2]

(b) A single particle of an ideal gas has mass *m*. The particle is moving at speed *u* in a cubic box of side length *L*, as shown in Figure 4.1.

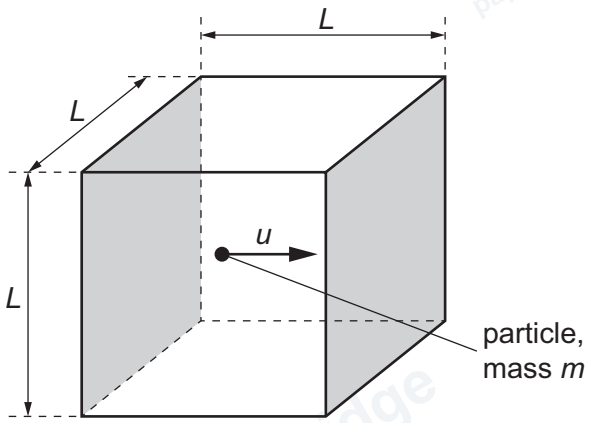


Figure 4.1

The particle is moving in a direction that is perpendicular to the two shaded faces, continually rebounding from one face and then the other.

Determine an expression, in terms of some or all of *m*, *u* and *L*, for:

(i) the magnitude of the change in momentum of the particle during each collision with a shaded face

change in momentum = ... [1]

(ii) the time taken between consecutive collisions of the particle with **one** of the shaded faces

time = ... [1]

(iii) the mean force exerted on the particle by one of the shaded faces

mean force = ... [1]

(iv) the mean pressure p exerted over the area of each of the shaded faces by the particle. Explain your reasoning.

p = ... [2]

(c) An ideal gas consists of N particles moving in many different directions with a mean-square speed of $\langle c^2 \rangle$.

Explain how the equation in **4(b)(iv)** is modified to arrive at the more general equation

$$pV = \tfrac{1}{3} Nm \langle c^2 \rangle$$

where V is the volume of the gas.

.....

.....

.....

.....

..... [3]

[Total: 10]

2 **(a)** State what is meant by an ideal gas.

.....

.....

..... [2]

(b) Ideal gases may be used to measure the temperature of the environment in a laboratory experiment. In some situations, however, it is preferable to use a thermocouple thermometer.

(i) Explain how a thermometer that uses an ideal gas can be advantageous compared with a thermometer that uses a liquid in glass.

.....

.....

..... [1]

(ii) Describe a laboratory situation in which a thermocouple thermometer is preferable to a thermometer that uses an ideal gas.

.....

.....

..... [1]

(c) A very small sample of an ideal gas consists of five molecules, each of mass 3.34×10^{-27} kg. At a particular instant, the speeds of the molecules are as shown in Figure 2.1.

$357\,m\,s^{-1}$	$424\,m\,s^{-1}$	$509\,m\,s^{-1}$	$286\,m\,s^{-1}$	$435\,m\,s^{-1}$

Figure 2.1

Determine, to three significant figures, the root-mean-square (r.m.s.) speed c_0 of the molecules.

c_0 = ... $m\,s^{-1}$ [2]

(d) A larger sample of the gas in **2(c)** consists of 5.0×10^{23} molecules. These molecules have the same r.m.s. speed as the molecules of the gas in **2(c)**.

(i) Calculate the amount of gas.

amount of gas = ... mol [2]

(ii) Determine the thermodynamic temperature T_0 of the gas.

T_0 = ... K [2]

(iii) The temperature of the gas is now gradually increased.

On Figure 2.2, sketch the variation of the r.m.s. speed of the molecules of the gas with thermodynamic temperature.

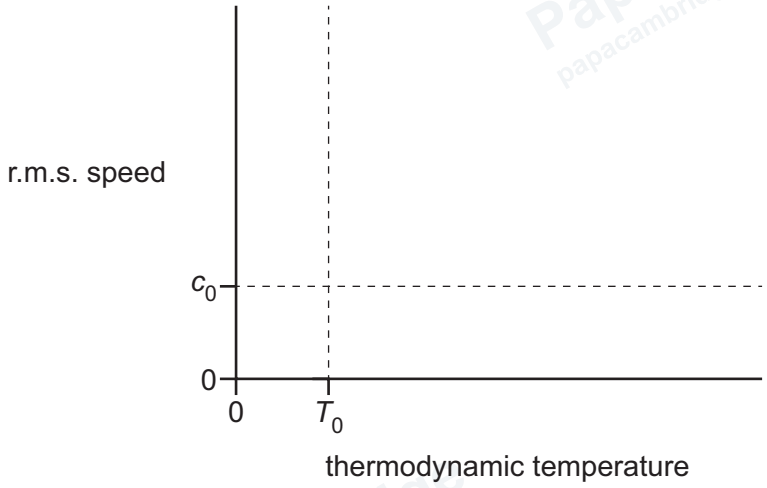


Figure 2.2

[3]

[Total: 13]

16 Thermodynamics

12 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 16 Thermodynamics

考纲要求（9702 · 2025 – 2027 syllabus）

16.1

16.1.1 understand that internal energy is determined by the state of the system and that it can be expressed as the sum of a random distribution of kinetic and potential energies associated with the molecules of a system

16.1.2 relate a rise in temperature of an object to an increase in its internal energy

16.2

16.2.1 recall and use W = pΔV for the work done when the volume of a gas changes at constant pressure and understand the difference between the work done by the gas and the work done on the gas

16.2.2 recall and use the first law of thermodynamics ΔU = q + W expressed in terms of the increase in internal energy, the heating of the system (energy transferred to the system by heating) and the work done on the system

2 **(a)** By referring to both kinetic energy and potential energy, explain what is meant by the internal energy of an ideal gas.

.....

.....

.....

..... [2]

(b) A fixed mass of an ideal gas at a temperature of 20 °C is sealed in a cylinder by a piston, as shown in Fig. 2.1.

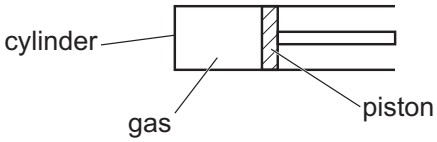


Fig. 2.1

The initial volume of the gas is $1.24 \times 10^{-4}\,m^3$.
Thermal energy is supplied to the gas and its volume increases by $5.20 \times 10^{-5}\,m^3$.

(i) The piston is freely moving so that the gas is always at atmospheric pressure.

Atmospheric pressure is $1.01 \times 10^5\,Pa$.

Calculate the work done by the gas.

work done by gas = ... J [2]

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(ii) Calculate the final thermodynamic temperature T of the gas.

T = ... K [2]

(iii) The mass of the gas is 16 g. For this expansion, there is a net transfer of 960 J of thermal energy to the gas.

Calculate the specific heat capacity c of the gas at this pressure.

c = ... $J\,kg^{-1}\,K^{-1}$ [2]

(c) The gas in **(b)** is allowed to return to its starting temperature. The piston is now fixed in position.

Thermal energy is supplied to increase the temperature to the same final temperature as in **(b)**.

Use the first law of thermodynamics to suggest and explain how the specific heat capacity of the gas for this situation compares with the value in **(b)(iii)**.

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..... [3]

[Total: 11]

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3 **(a)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

(b) With reference to molecular kinetic and potential energies, describe and explain how the internal energy of the system changes when:

(i) a gas is heated at constant volume so that its temperature increases

.....

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..... [3]

(ii) a wire is stretched within its elastic limit at constant temperature.

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..... [3]

[Total: 8]

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4 **(a)** **(i)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

(ii) Explain why the internal energy of an ideal gas is directly proportional to the thermodynamic temperature of the gas.

.....

.....

.....

..... [2]

(b) A sample of an ideal gas at thermodynamic temperature T has internal energy U.

The gas is compressed so that its temperature increases to $3T$.
During this compression, work W is done on the gas.

The gas is then cooled at constant volume so that its temperature decreases to $2T$.

Complete Table 4.1 to show, in terms of some or all of W, T and U, the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas for each of the two processes.

Table 4.1

	work done on gas	thermal energy supplied to gas	increase in internal energy of gas
compression	$+W$		
cooling			

[4]

[Total: 8]

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4 **(a)** **(i)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

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.....

.....

.....

..... [2]

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Table 4.1

	work done on gas	thermal energy supplied to gas	increase in internal energy of gas
compression	$+W$		
cooling			

[4]

[Total: 8]

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2 **(a)** State **two** ways in which the first law of thermodynamics describes that the internal energy of a system may be changed.

1 .....

.....

2 .....

.....

[2]

(b) **(i)** Use the first law of thermodynamics to explain why a bicycle pump gets hot when it is used to pump up a tyre quickly.

.....

.....

.....

.....

..... [3]

(ii) With reference to molecular energies, explain why the temperature of water remains at 100 °C when it vaporises in a kettle, even though it is being heated.

.....

.....

.....

.....

..... [3]

[Total: 8]

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4 A cylinder contains a fixed mass of an ideal gas at pressure $2Y$ and volume $6X$.

The gas undergoes a sequence of changes from its initial state A, through states B, C and D, then finally back to its initial state A, as shown in Fig. 4.1.

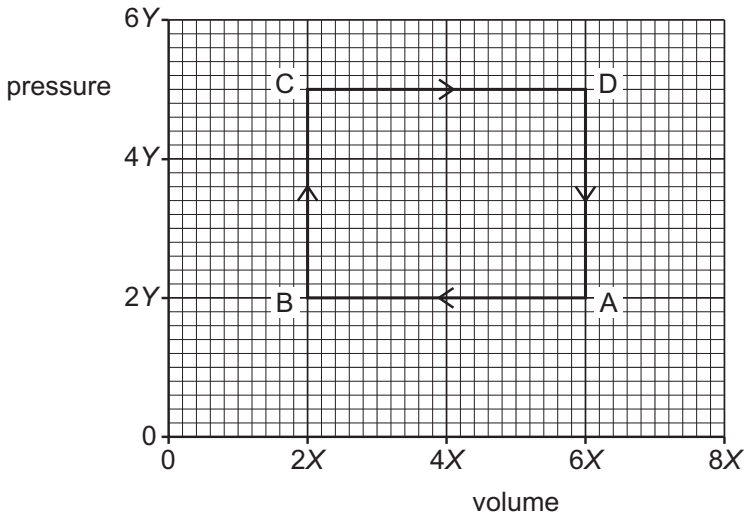


Fig. 4.1

Fig. 4.2 shows the variation with time of the internal energy of the gas.

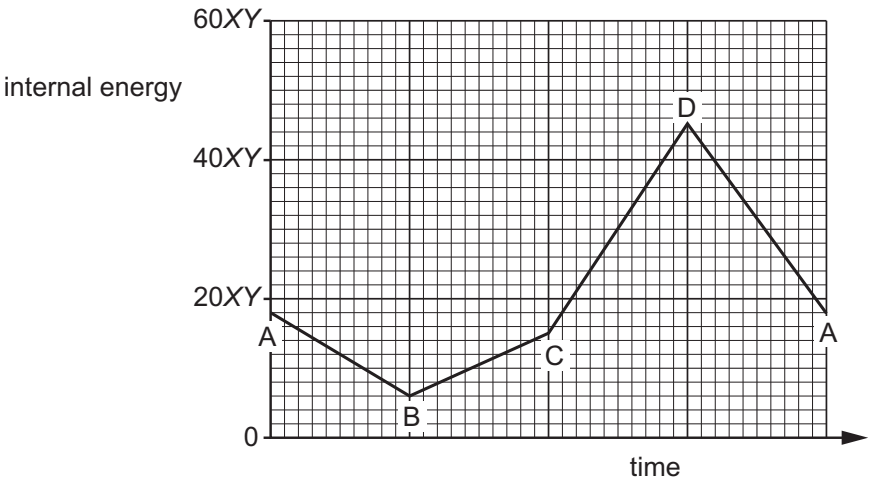


Fig. 4.2

(a) State the first law of thermodynamics.

.....

.....

..... [2]

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(b) **(i)** Use Fig. 4.1 and Fig. 4.2 to determine the general expression for the internal energy U of the gas when it has pressure p and volume V.

U = ... [1]

(ii) An ideal gas at thermodynamic temperature T contains N molecules.

Use your answer in **(b)(i)** and the equation of state for an ideal gas to deduce an expression for U in terms of N and T. Identify any other symbols you use.

U = ... [2]

(c) Determine expressions, in terms of X and Y, for the work W done on the gas during:

(i) change AB

W = ... [1]

(ii) change CD.

W = ... [1]

(d) Use your answers in **(c)** and the first law of thermodynamics to determine an expression, in terms of X and Y, for the net thermal energy Q supplied to the gas during one full cycle ABCDA. Explain your reasoning.

Q = ... [3]

[Total: 10]

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2 **(a)** State **two** ways in which the first law of thermodynamics describes that the internal energy of a system may be changed.

1 .....

.....

2 .....

.....

[2]

(b) **(i)** Use the first law of thermodynamics to explain why a bicycle pump gets hot when it is used to pump up a tyre quickly.

.....

.....

.....

.....

..... [3]

(ii) With reference to molecular energies, explain why the temperature of water remains at 100 °C when it vaporises in a kettle, even though it is being heated.

.....

.....

.....

.....

..... [3]

[Total: 8]

3 **(a)** With reference to molecular kinetic energy and molecular potential energy, explain what is meant by the internal energy of an ideal gas.

.....

.....

..... [2]

(b) A sample of an ideal gas is initially in state A, at a pressure of 2.0×10^5 Pa and with a volume of $0.016\,m^3$, as shown in Fig. 3.1.

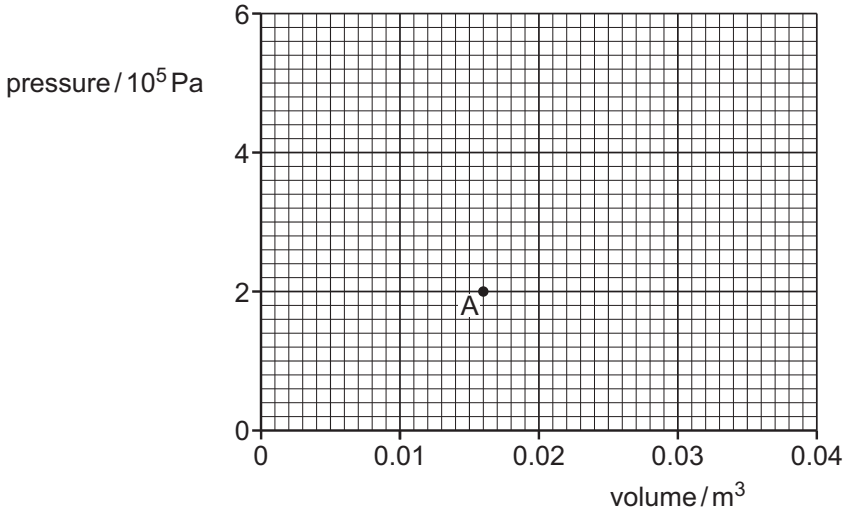


Fig. 3.1

In state A, the temperature of the gas is 400 K.

The gas undergoes two successive changes X and Y.

In change X, it is heated at constant volume to a pressure of 4.0×10^5 Pa. At the end of change X, the gas is in state B.

In change Y, it is then allowed to expand at constant temperature back to its original pressure. At the end of change Y, the gas is in state C.

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(i) Determine the internal energy of the gas in state A.

internal energy = ... J [2]

(ii) Determine the temperature of the gas in state B.

temperature = ... K [1]

(iii) Determine the volume of the gas in state C.

volume = ... m^3 [1]

(iv) On Fig. 3.1, draw two lines, one to represent change X and one to represent change Y. Label your lines X and Y respectively. [3]

[Total: 9]

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3 **(a)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

(b) An ideal gas containing 8.7×10^{23} molecules is at a temperature of 300 K and a pressure of 2.0×10^{5} Pa.

(i) Show that the volume of the gas is $0.018\,m^3$.

[2]

(ii) Determine the internal energy of the gas. Explain your reasoning.

internal energy = ... J [3]

(c) The gas in **3(b)** undergoes a sequence of changes AB, BC and CA in its pressure p and volume V, as shown in Figure 3.1.

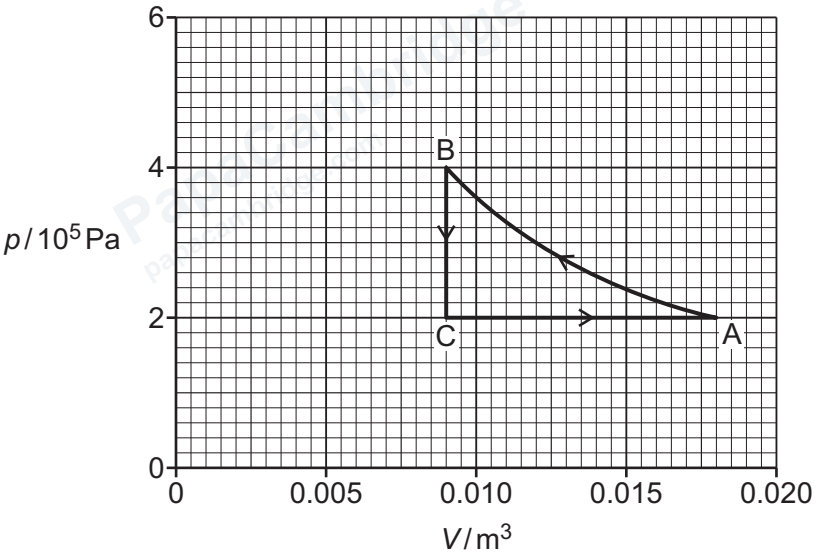


Figure 3.1

Determine **three** conclusions that can be drawn from Figure 3.1 about the individual and overall effects of these changes on the energy of the gas. The conclusions may be qualitative or quantitative, and may refer to any of work done, transfer of thermal energy and change in internal energy. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 10]

3 **(a)** Define specific heat capacity.

.....

.....

..... [2]

(b) A sample of an ideal gas contains 6.00×10^{23} molecules, each of mass 6.64×10^{-27} kg. The gas is at a pressure of 1.01×10^{5} Pa. It has an initial volume of $0.0243\,m^3$, an initial internal energy of 3676 J and an initial temperature of 296 K.

The gas is now cooled at constant pressure so that its volume becomes 80% of the initial volume.

(i) Show that the magnitude of the work done on the gas is 491 J.

[2]

(ii) Explain whether the work done on the gas is positive or negative.

.....

..... [1]

(iii) Explain why the increase in internal energy of the gas is –735 J.

.....

.....

.....

..... [2]

(iv) Determine the thermal energy Q transferred to the gas.

Q = ... J [2]

(v) Calculate a value for the specific heat capacity c of the gas for this change.

c = ... $J\,kg^{-1}\,K^{-1}$ [2]

(c) The gas in **3(b)** is now heated at constant volume so that its temperature returns to 296 K.

Use the first law of thermodynamics to explain, without calculation, whether the specific heat capacity for this change is less than, the same as, or greater than the answer in **3(b)(v)**.

.....

.....

.....

.....

..... [3]

[Total: 14]

3 **(a)** State what is meant by the internal energy of a system.

.....

.....

..... [2]

(b) An ideal gas containing 8.7×10^{23} molecules is at a temperature of 300 K and a pressure of 2.0×10^{5} Pa.

(i) Show that the volume of the gas is $0.018\,m^3$.

[2]

(ii) Determine the internal energy of the gas. Explain your reasoning.

internal energy = ... J [3]

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(c) The gas in **3(b)** undergoes a sequence of changes AB, BC and CA in its pressure p and volume V, as shown in Figure 3.1.

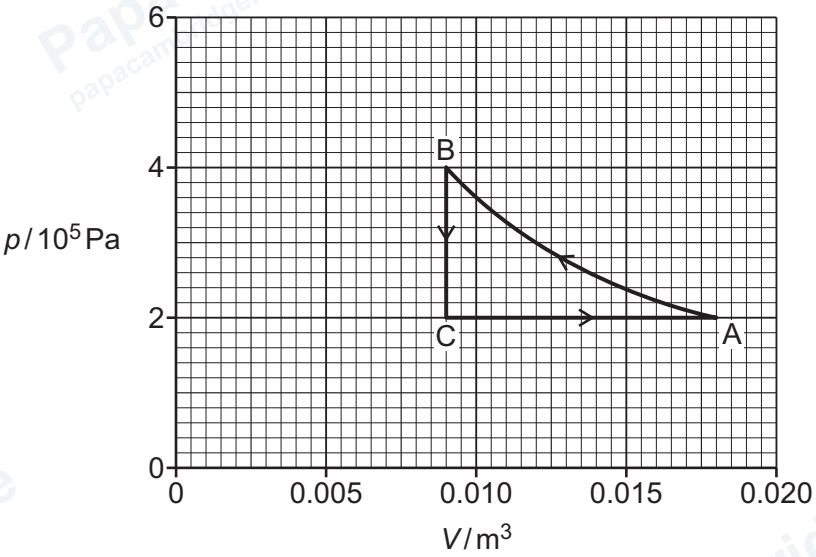


Figure 3.1

Determine **three** conclusions that can be drawn from Figure 3.1 about the individual and overall effects of these changes on the energy of the gas. The conclusions may be qualitative or quantitative, and may refer to any of work done, transfer of thermal energy and change in internal energy. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 10]

4 **(a)** State the first law of thermodynamics.

.....

.....

.....

..... [2]

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(b) A sample of an ideal gas contains 3.58×10^{24} molecules. The sample is at a temperature of 298 K and a pressure of 1.84×10^{5} Pa.

(i) Show that the volume of the gas is $0.0800\,m^3$.

[2]

(ii) Calculate the amount of gas n in the sample.

n = ... mol [1]

(iii) Determine the average translational kinetic energy E_K of one molecule of the gas.

E_K = ... J [2]

(iv) Explain why the average potential energy of one molecule of the gas is zero.

.....

..... [1]

(v) Calculate the internal energy U_A of the gas.

U_A = ... J [1]

(c) The gas in **4(b)** undergoes two changes AB and BC, as shown in Figure 4.1.

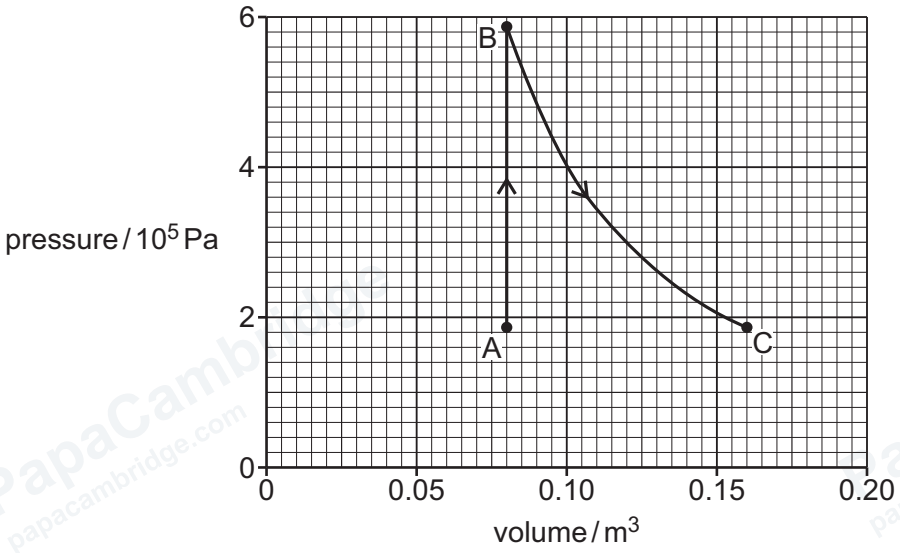


Figure 4.1

In change AB, the gas is heated at constant volume from its initial pressure of 1.84×10^5 Pa to a pressure of 5.84×10^5 Pa.

In change BC, the gas rapidly expands and returns to its original pressure with no transfer of thermal energy. The volume of the gas after this expansion is 0.160 m^3.

(i) Determine an expression, in terms of U_A, for the thermal energy Q supplied to the gas during change AB. Explain your reasoning.

Q = ... [2]

(ii) Determine an expression, in terms of U_A, for the work W done on the gas during change BC. Explain your reasoning.

W = ... [2]

[Total: 13]

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17 Oscillations

15 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 17 Oscillations

考纲要求（9702 · 2025 – 2027 syllabus）

17.1

17.1.1 understand and use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations, and express the period in terms of both frequency and angular frequency

17.1.2 recall and use ω = 2πf and ω = 2π/T

17.1.3 use a = –ω2x and recall and use, as a solution to this equation, x = x0 sin ωt

17.1.4 use the equations v = v0 cos ωt and v = ± ω ( ) x x 0

17.1.5 analyse and interpret graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion

17.2

17.2.1 describe the interchange between kinetic and potential energy during simple harmonic motion

17.2.2 recall and use $E = ½mω^2x_0^2$ for the total energy of a system undergoing simple harmonic motion

17.3

17.3.1 understand that a resistive force acting on an oscillating system causes damping

17.3.2 understand and use the terms light, critical and heavy damping and sketch displacement–time graphs illustrating these types of damping

17.3.3 understand that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency

3 A small object of mass 24 g rests on a platform. The platform is attached to an oscillator, as shown in Fig. 3.1.

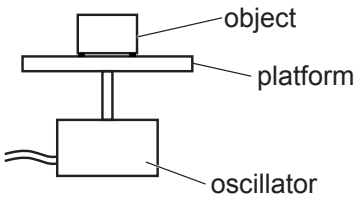


Fig. 3.1

The oscillator moves the platform up and down.

(a) The total energy of the oscillations of the object is 2.2×10^{-4} J.
In one oscillation the object travels a total distance of 14 mm.

Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [3]

(b) The frequency of the oscillator is fixed, and the amplitude of the oscillations is gradually increased.

(i) Calculate the maximum amplitude of the oscillations so the object does not lose contact with the platform.

amplitude = ... m [2]

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(ii) The amplitude of the oscillations is increased so it is greater than the value in **(b)(i)**.

State and explain the position in an oscillation where the object first loses contact with the platform.

.....

.....

.....

..... [2]

[Total: 7]

4 **(a)** State what is meant by resonance.

.....

.....

..... [2]

(b) A small ball is held in place using a stretched string. One end of the string is fixed to a wall and the other end is attached to a vibration generator, as shown in Fig. 4.1.

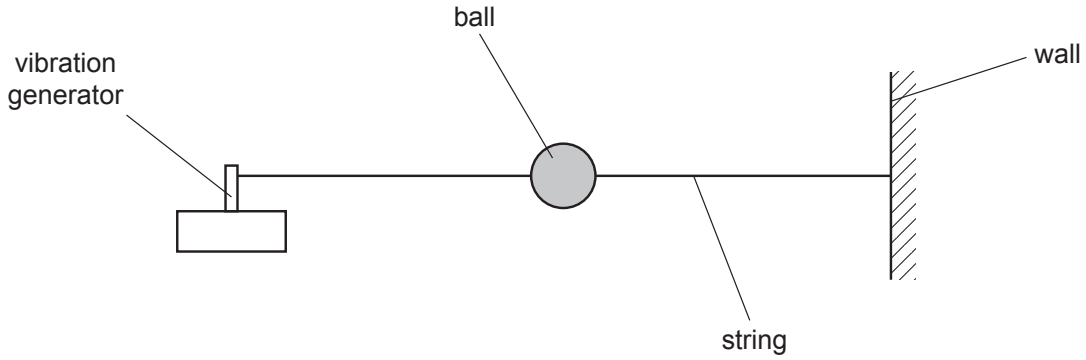


Fig. 4.1

Initially, the vibration generator is switched off.

A student displaces the ball vertically and then releases it. Fig. 4.2 shows the variation of the displacement of the ball with time after it is released.

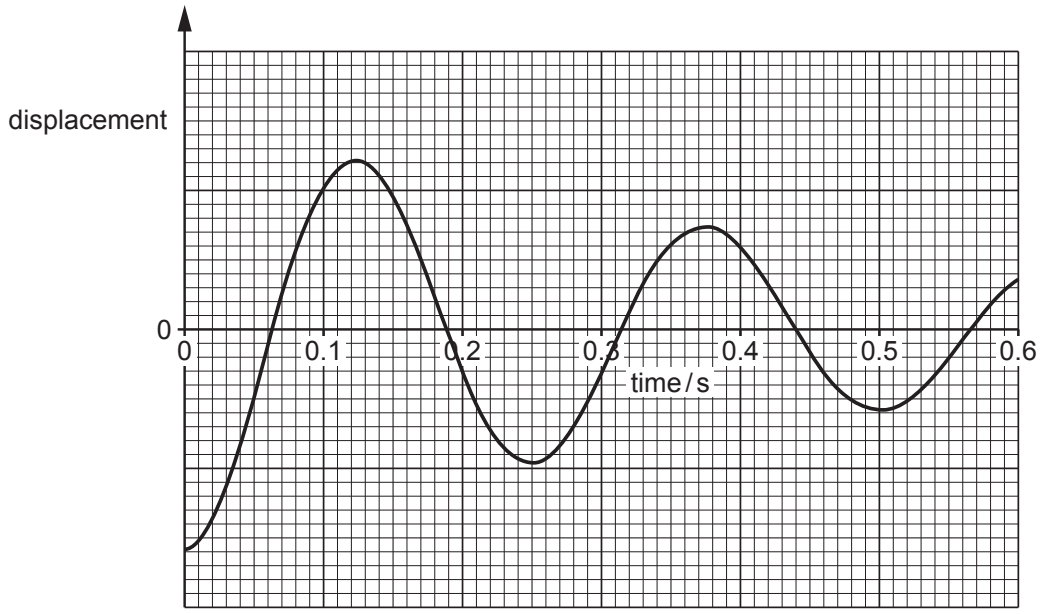


Fig. 4.2

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(i) State the name of the phenomenon illustrated by the decrease in the amplitude of the oscillations in Fig. 4.2.

..... [1]

(ii) Explain the decrease with time of the amplitude of the oscillations of the ball.

.....

.....

..... [2]

(iii) Determine the frequency of the oscillations of the ball.

frequency = ... Hz [1]

(c) The vibration generator in **(b)** is switched on and its frequency *f* of vibration is gradually increased from 0 to 10 Hz.

On Fig. 4.3, sketch the variation with *f* of the amplitude of the oscillations of the ball.

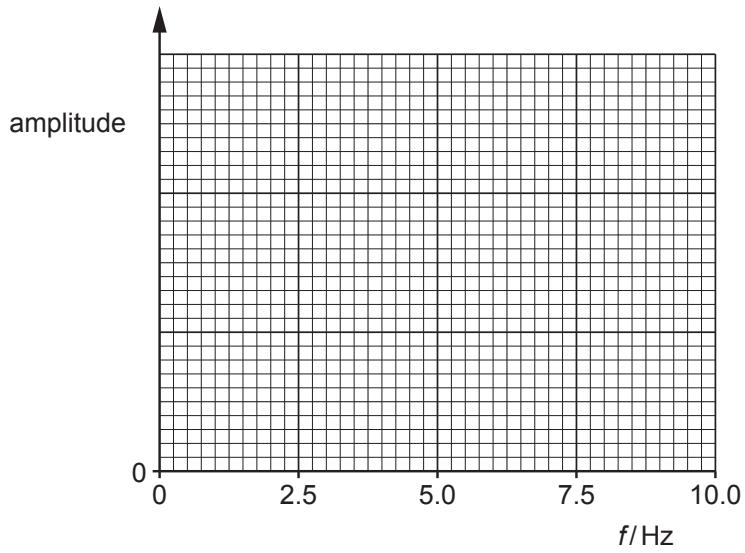


Fig. 4.3

[2]

[Total: 8]

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* 0019655342210 *

4 A block of mass *m* oscillates vertically on a spring, as shown in Fig. 4.1.

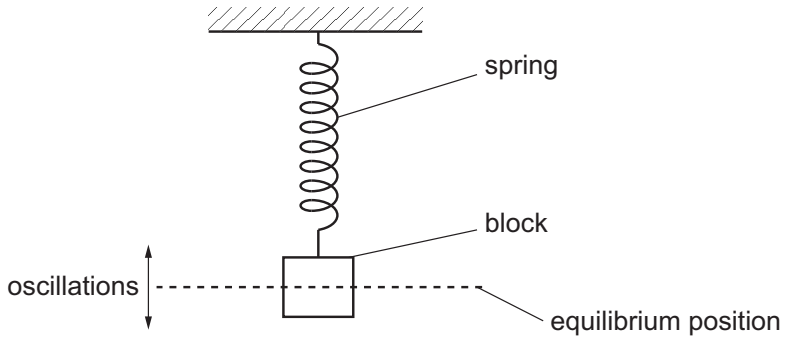


Fig. 4.1

The acceleration *a* of the block varies with displacement *x* from its equilibrium position, as shown in Fig. 4.2.

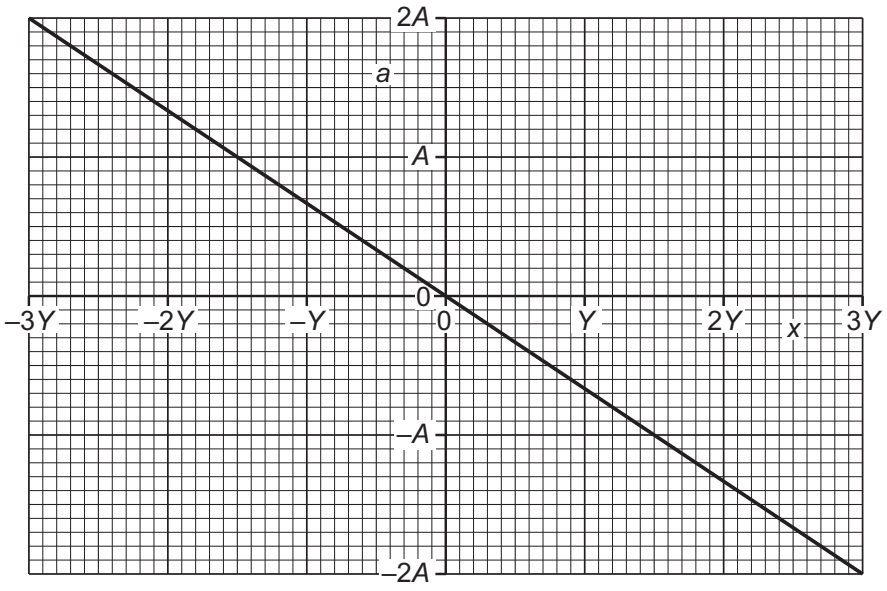


Fig. 4.2

The amplitude of the oscillations is 3*Y* and the maximum acceleration is 2*A*.

(a) Explain how Fig. 4.2 shows that the oscillations of the block are simple harmonic.

.....

.....

..... [2]

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(b) Deduce expressions, in terms of some or all of m, A and Y, for:

(i) the angular frequency ω of the oscillations

ω = ... [1]

(ii) the maximum speed v_0 of the oscillations

v_0 = ... [2]

(iii) the energy E of the oscillations.

E = ... [2]

(c) The period of the oscillations is 0.75 s and the value of $3Y$ is 1.8 cm.

Determine an expression for x in terms of time t, where x is in cm and t is in seconds.

x = ... [2]

[Total: 9]

* 0000800000010 *

4 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A block is suspended from a spring, as shown in Fig. 4.1.

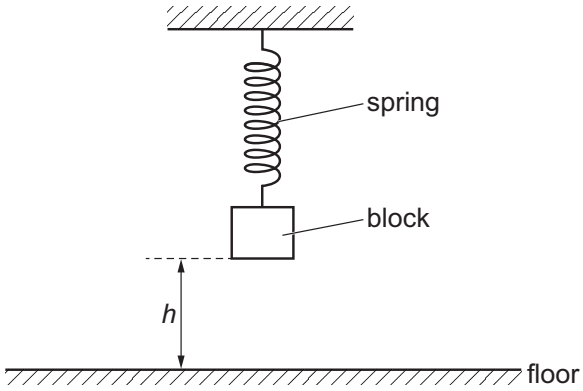


Fig. 4.1

The block is pulled down and released at time $t = 0$. It then oscillates vertically with simple harmonic motion.

Fig. 4.2 shows the variation of the velocity v of the block with height h of the base of the block above the floor.

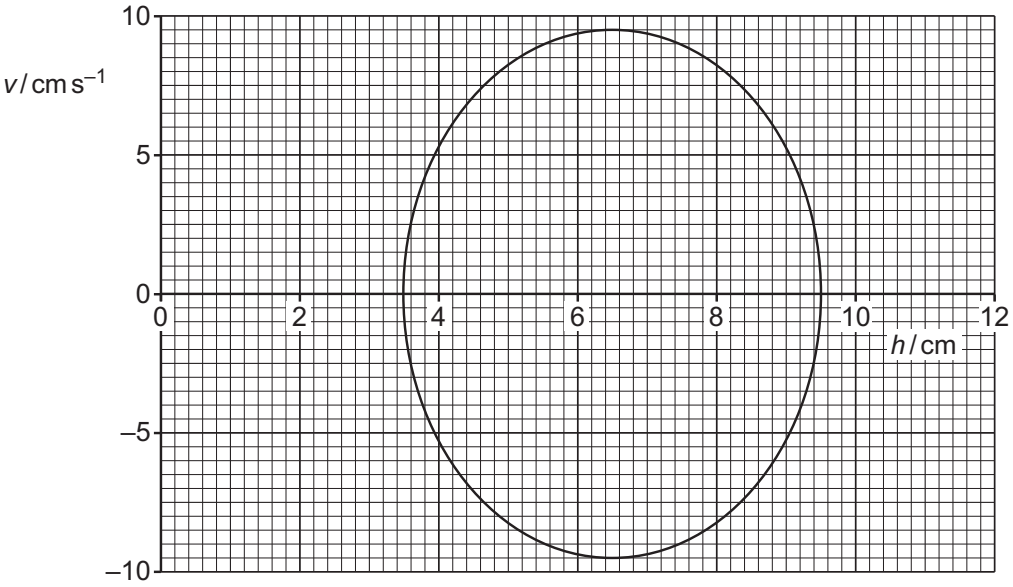


Fig. 4.2

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(i) Determine the amplitude, in cm, of the oscillations.

amplitude = ... cm [1]

(ii) Show that the angular frequency of the oscillations is 3.2 rad s^{-1}.

[2]

(iii) Calculate the period T of the oscillations.

T = ... s [2]

(iv) On Fig. 4.3, sketch the variation of h with time t from $t = 0$ to $t = 6.0$ s.

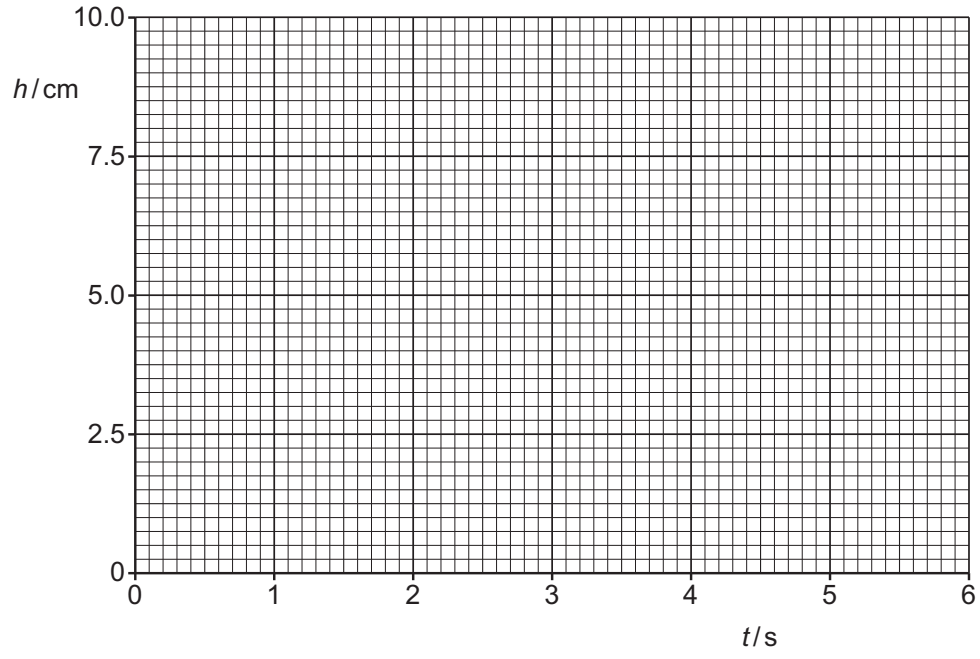


Fig. 4.3

[4]

[Total: 11]

* 0000800000012 *

5 Fig. 5.1 shows a pendulum consisting of a metal sphere suspended by a thin string.

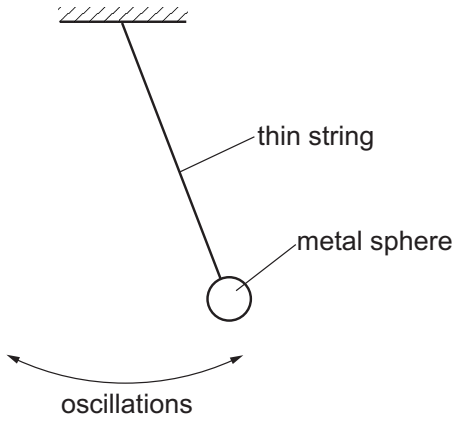


Fig. 5.1 (not to scale)

The sphere undergoes small oscillations about its equilibrium position. The oscillations may be considered to be simple harmonic.

Fig. 5.2 shows the variation with time t of the displacement x of the sphere from its equilibrium position.

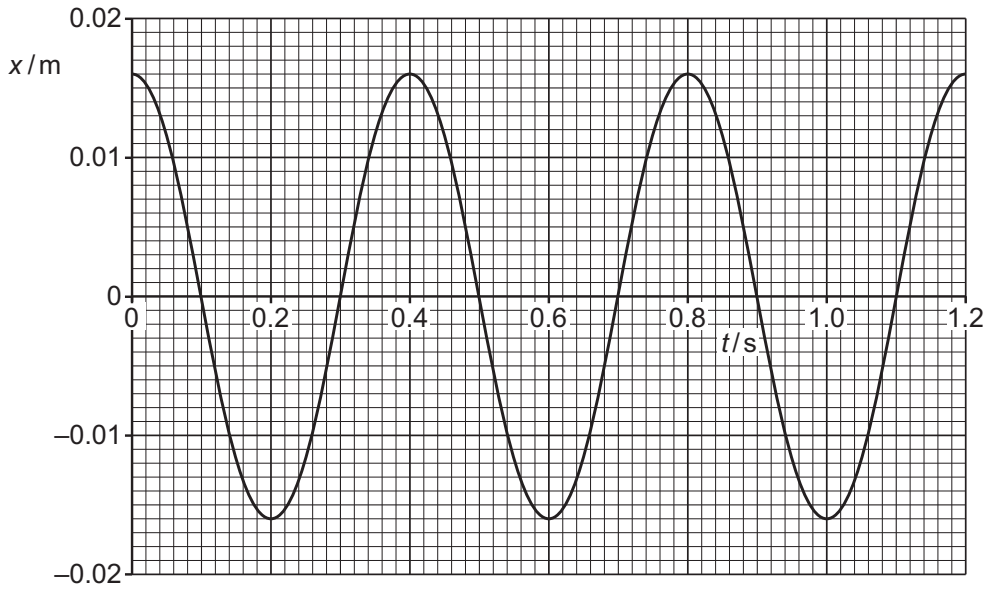


Fig. 5.2

(a) On Fig. 5.1, draw an arrow, from the centre of the sphere, to represent the direction of the resultant force acting on the sphere when it is in the position shown. [1]

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(b) The mass of the sphere is 0.15 kg.

(i) State the amplitude of the oscillations.

amplitude = ... m [1]

(ii) Determine the angular frequency of the oscillations.

angular frequency = ... rad s^{-1} [2]

(iii) Calculate the total energy of the oscillations.

total energy = ... J [2]

(c) On Fig. 5.3, sketch the variation with x of the kinetic energy E_K of the sphere.

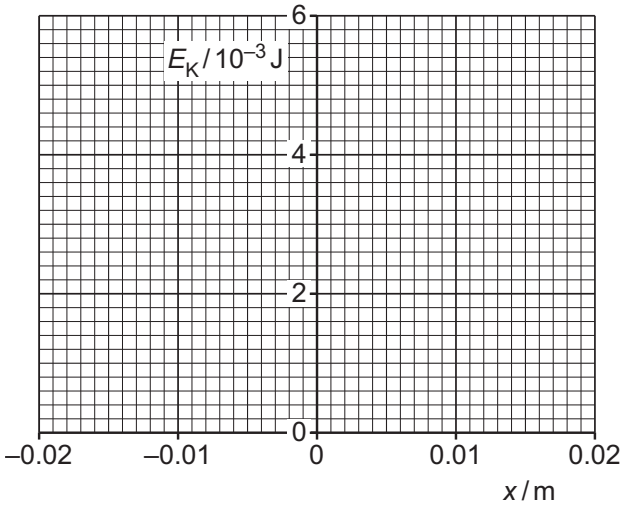


Fig. 5.3

[3]

[Total: 9]

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4 A small crystal is made to vibrate with simple harmonic motion. The variation with time t of the displacement x of one surface of the crystal from its equilibrium position is shown in Fig. 4.1.

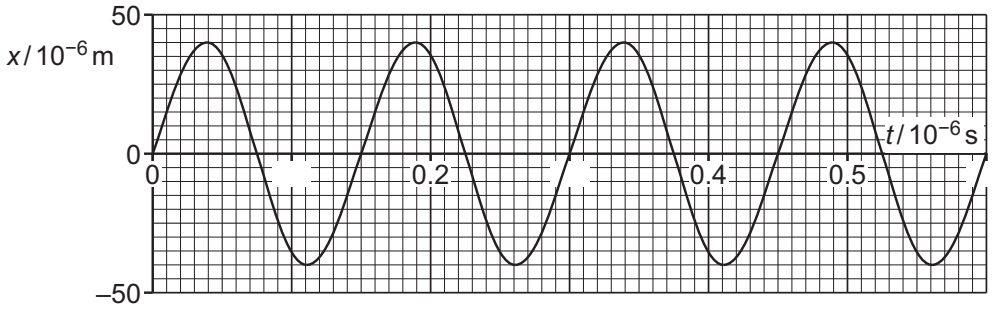


Fig. 4.1

(a) Show that the angular frequency of the vibration of the surface is 4.2×10^7 rad s^{-1}.

[2]

(b) Determine the maximum acceleration a_0 of the vibration of the surface.

a_0 = ... m s^{-2} [2]

(c) The crystal may be modelled as a single mass of 2.4×10^{-4} kg that vibrates as shown in Fig. 4.1.

Calculate the total energy E of the vibrations.

E = ... J [3]

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(d) The crystal generates ultrasound waves that are used to obtain diagnostic information about internal structures.

(i) The crystal is made from piezoelectric material.

Explain how the crystal is made to vibrate.

.....

.....

.....

..... [2]

(ii) A parallel beam of ultrasound waves is incident on a muscle-bone boundary. Data for muscle and bone are given in Table 4.1.

Table 4.1

material	density / kg m^{-3}	speed of sound / m s^{-1}
muscle	1100	1600
bone	1900	4100

Calculate the percentage of the intensity of the ultrasound beam that is transmitted at this boundary.

percentage transmitted = ... % [3]

[Total: 12]

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5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

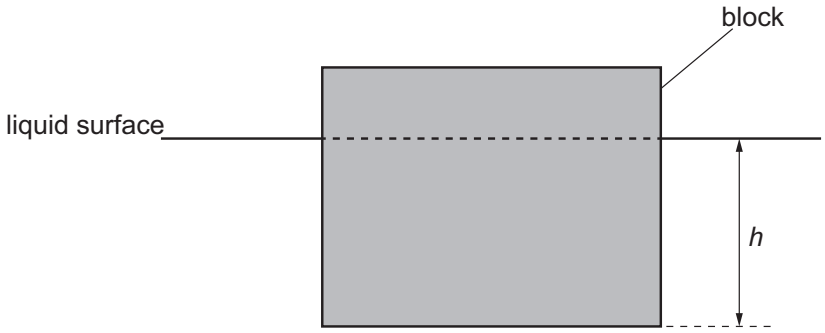


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

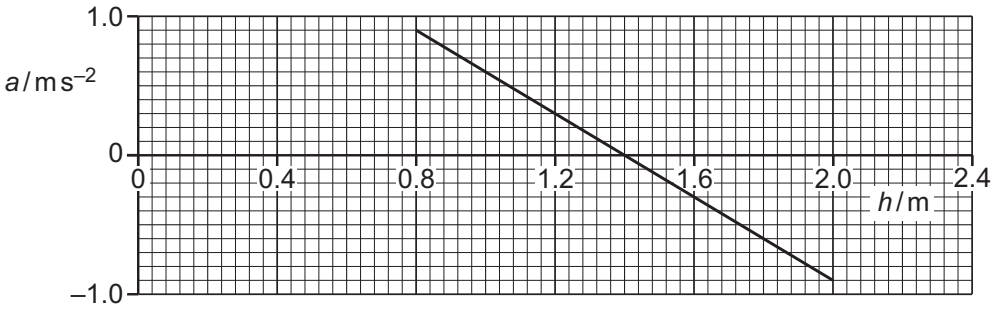


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

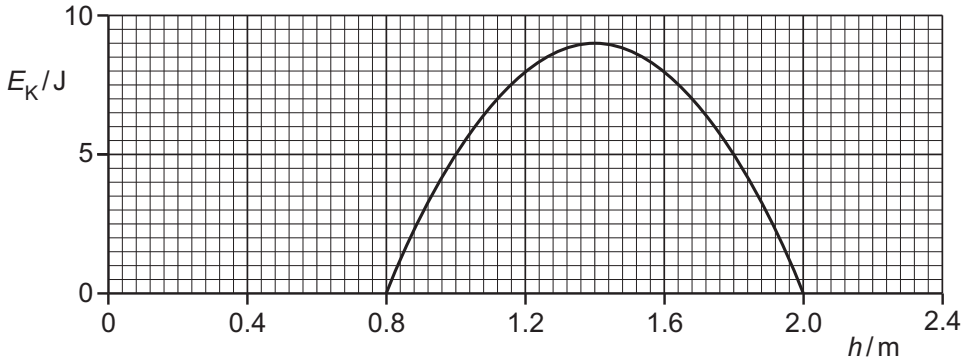


Fig. 5.3

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(a) **(i)** Determine the amplitude of the oscillations.

amplitude = ... m [1]

(ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

(b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) On Fig. 5.4, sketch the variation with h of the potential energy E_P of the oscillations.

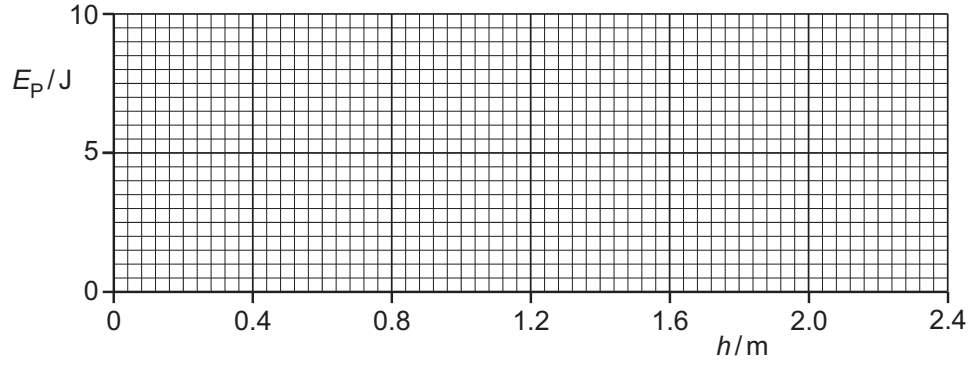


Fig. 5.4

[3]

[Total: 8]

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5 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A block is suspended by a spring. The block oscillates vertically with simple harmonic motion.

The velocity v of the block varies with time t according to

$$v = 0.56 \cos 16t$$

where v is in m s^{-1} and t is in s.

(i) Calculate the period of the oscillation.

period = ... s [1]

(ii) Determine the amplitude x_0 of the oscillation.

x_0 = ... m [2]

(iii) Use your answer in **(b)(ii)** to determine the equation for v in terms of the displacement x of the block, where v is in m s^{-1} and x is in m.

v = ... [1]

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(iv) On Fig. 5.1, sketch the variation of v with x.

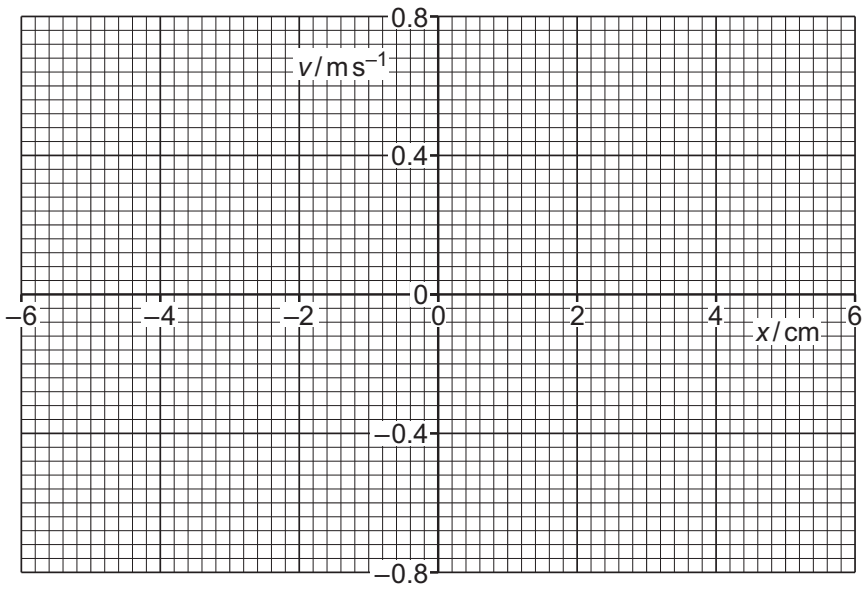


Fig. 5.1

[3]

[Total: 9]

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5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

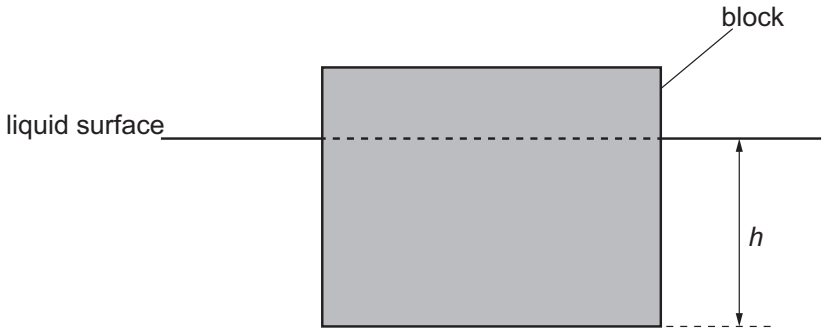


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

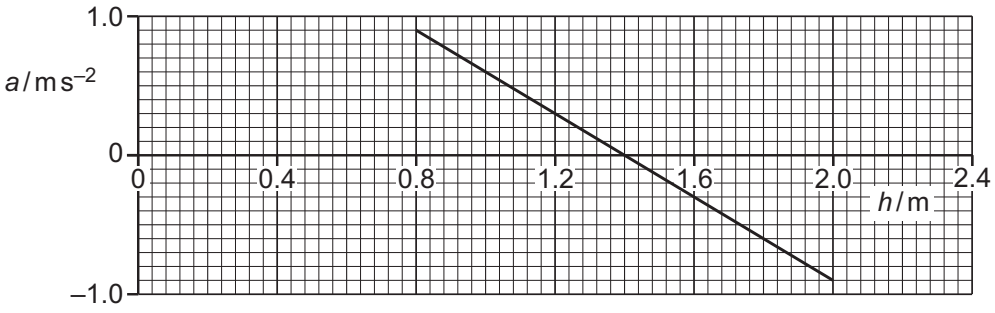


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

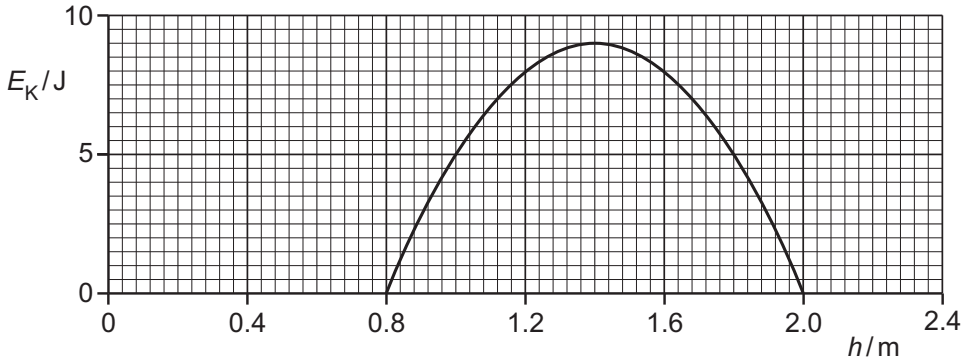


Fig. 5.3

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(a) **(i)** Determine the amplitude of the oscillations.

amplitude = ... m [1]

(ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

(b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) On Fig. 5.4, sketch the variation with h of the potential energy E_P of the oscillations.

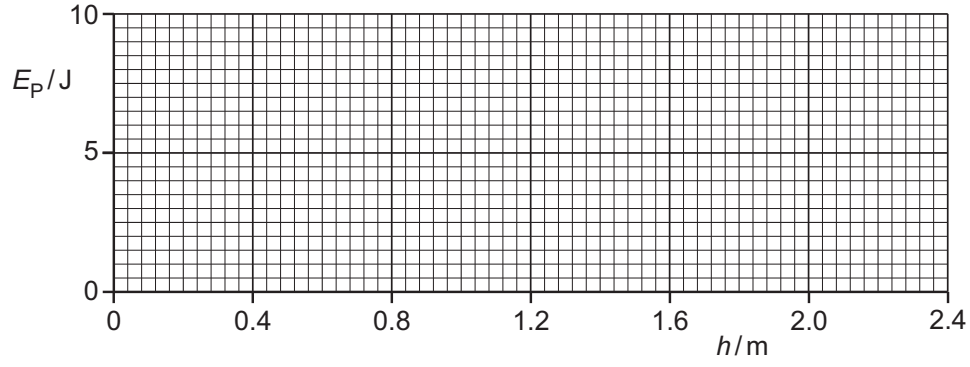


Fig. 5.4

[3]

[Total: 8]

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4 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A small sphere is suspended from a fixed point P by a string of negligible mass, as shown in Fig. 4.1.

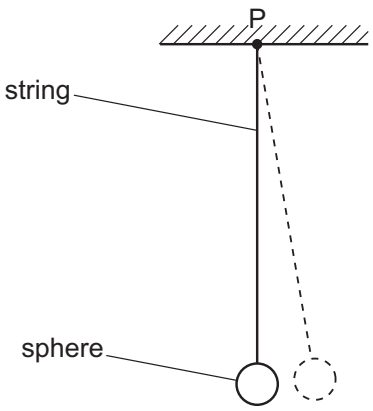


Fig. 4.1

The sphere is given a small horizontal displacement and is then released.

The variation with time of the horizontal velocity v of the sphere is shown in Fig. 4.2.

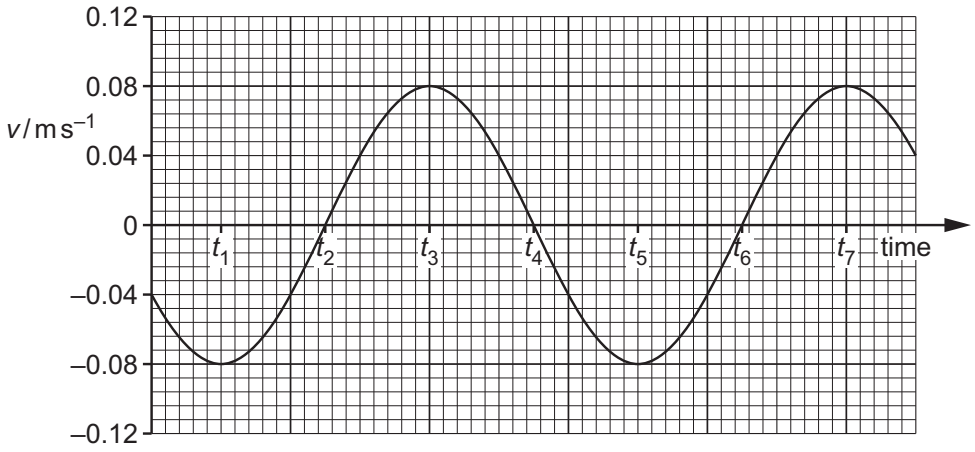


Fig. 4.2

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(i) State two times at which the sphere is passing in the same direction through the equilibrium position.

time ..... and time ..... [1]

(ii) The time interval between t_1 and t_6 is 2.2 s.

Calculate the frequency of oscillation of the sphere.

frequency = ... Hz [2]

(c) The sphere in **(b)** is undergoing simple harmonic motion.

Use your answer in **(b)(ii)** and data from Fig. 4.2 to determine the maximum displacement of the sphere from its equilibrium position.

maximum displacement = ... m [3]

[Total: 8]

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7 A bar magnet is suspended from a spring. One pole of the magnet oscillates freely in a coil of wire, as shown in Fig. 7.1.

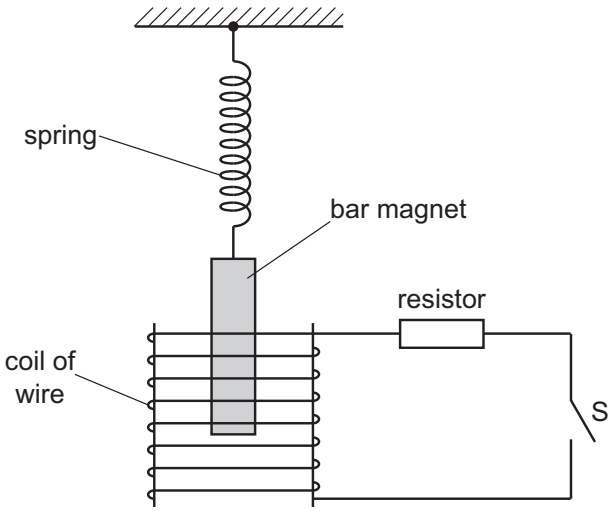


Fig. 7.1

The switch S is initially open.

(a) The switch S is now closed. As a result, the oscillations of the magnet are lightly damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) Describe what is observed to indicate that the damping is light.

.....

..... [1]

(iii) By reference to electromagnetic induction and to conservation of energy, explain why the oscillations are damped.

.....

.....

.....

..... [3]

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(b) The procedure in **(a)** is repeated after replacing the resistor with one of greater resistance.

Suggest, with a reason, the effect of this change on the oscillations.

.....

.....

..... [2]

[Total: 8]

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5 A steel ball on the end of a thin string oscillates with small oscillations, as shown in Fig. 5.1.

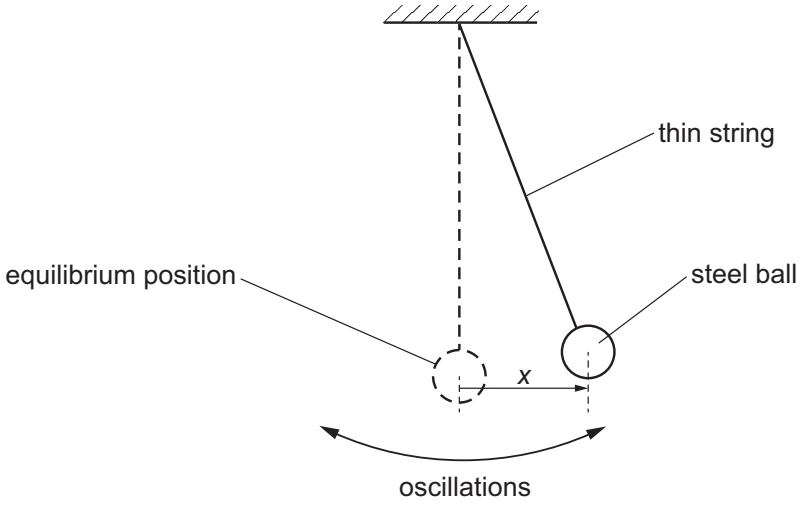


Fig. 5.1 (not to scale)

The displacement of the centre of the ball from its equilibrium position is x.

(a) Fig. 5.2 shows the variation with x of the acceleration a of the ball.

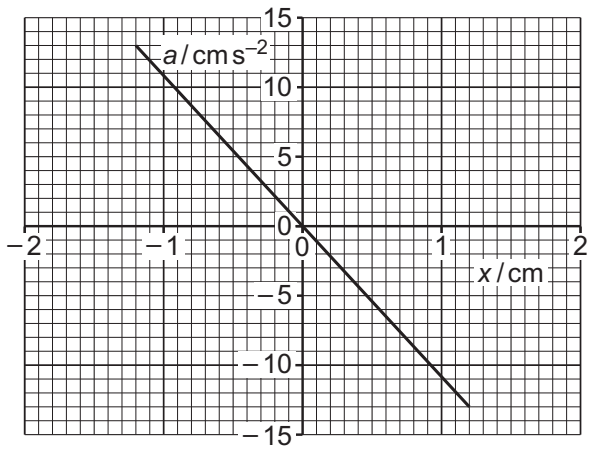


Fig. 5.2

(i) Explain how Fig. 5.2 shows that the oscillations of the ball are simple harmonic.

.....

.....

..... [2]

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(ii) Determine the period T of the oscillations.

T = ... s [3]

(b) At time $t = 0$, when the displacement of the ball has its maximum value, the ball is immersed in a trough containing thick oil so that the ball is just below the surface of the oil. This results in the subsequent motion of the ball being heavily damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) On Fig. 5.3, sketch a possible variation of the displacement x of the ball with t between $t = 0$ and $t = 2T$.

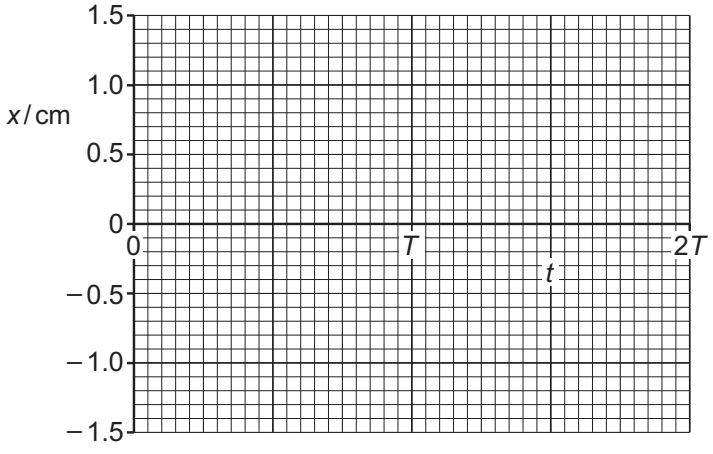


Fig. 5.3

[3]

[Total: 10]

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4 **(a)** State what is meant by the frequency of the oscillations of an oscillating object.

.....

..... [1]

(b) An object is oscillating.

Fig. 4.1 shows the variation of the acceleration a of the object with its displacement x from the equilibrium position.

Fig. 4.2 shows the variation of the kinetic energy E_K of the object with time t.

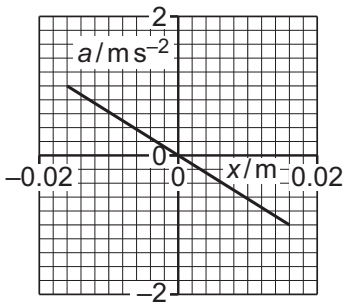


Fig. 4.1

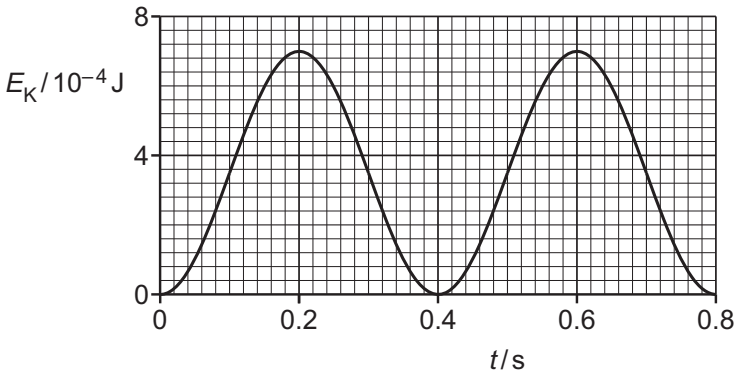


Fig. 4.2

(i) Explain how Fig. 4.2 shows that the period of the oscillations is 0.80 s.

.....

.....

..... [1]

(ii) Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [2]

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(iii) Apart from the period, frequency and angular frequency of the oscillations, determine **three** other conclusions about the object and its oscillations that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working.

1 .....

.....

2 .....

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3 .....

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[3]

(iv) Describe the interchange between kinetic energy and potential energy during the oscillations. Numerical values are **not** required.

.....

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.....

..... [3]

[Total: 10]

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5 A block of mass 27 g is suspended from a spring so that it rests in equilibrium, as shown in Figure 5.1.

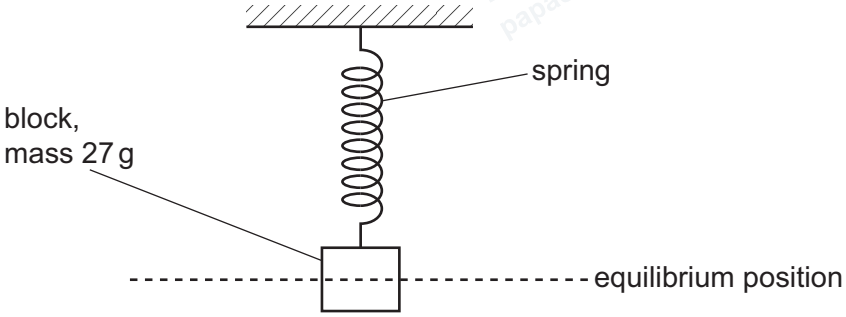


Figure 5.1

The block is displaced downwards through a small distance. The block is released and then oscillates vertically with simple harmonic motion.

The oscillations have a period of 1.7 s and an energy of 13 μJ.

(a) (i) Show that the angular frequency of the oscillations is 3.7 rad s^{-1}.

[1]

(ii) State what is meant by the amplitude of the oscillations.

.....

.....

..... [1]

(iii) Calculate the amplitude of the oscillations.

amplitude = ... m [3]

(b) On Figure 5.2, sketch the variation of the kinetic energy E_K of the block with displacement x from the equilibrium position for one complete oscillation of the block. Add a suitable scale to the x-axis.

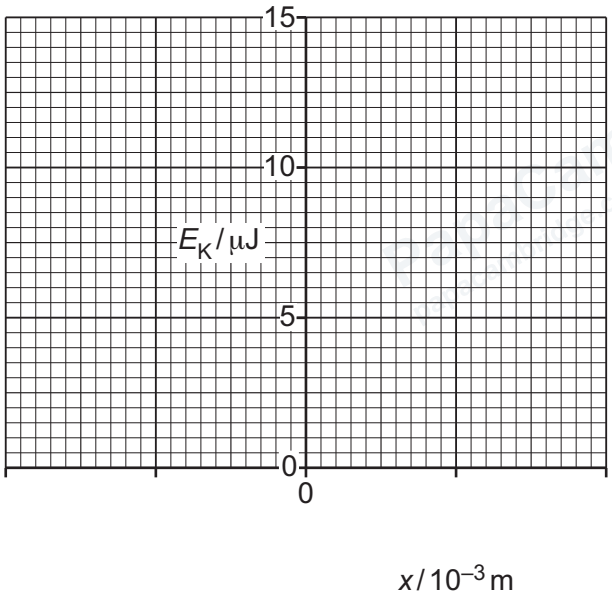


Figure 5.2

[3]

[Total: 8]

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5 A metal ball is attached to the lower end of a spring, as shown in Figure 5.1.

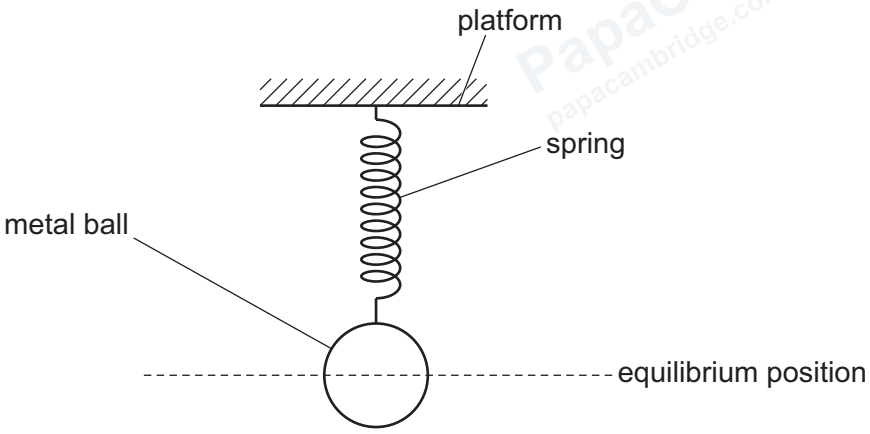


Figure 5.1

The upper end of the spring is attached to a platform that is initially stationary.

(a) The ball is pulled down by a small distance and released so that it oscillates with simple harmonic motion. The oscillations have an amplitude of 2.5 cm and a maximum speed of 4.4 cm s^{-1}.

(i) On Figure 5.2, sketch the variation of the velocity v of the ball with the displacement x from its equilibrium position as the ball undergoes one full oscillation.

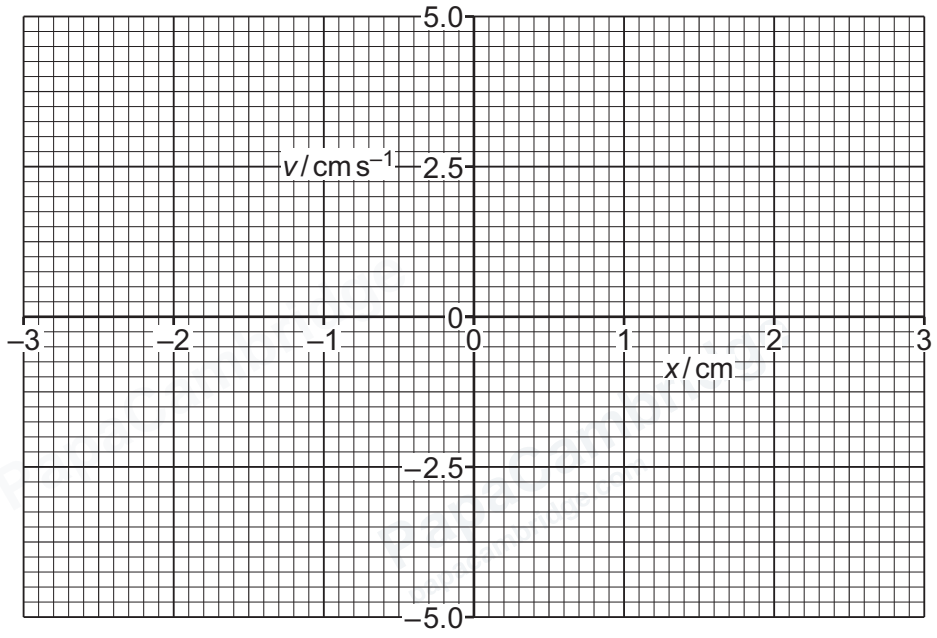


Figure 5.2

[3]

(ii) Show that the frequency of the oscillations is 0.28 Hz.

[2]

(b) The ball is returned to rest at its equilibrium position.

The platform is now moved up and down by a motor so that it oscillates vertically with simple harmonic motion of constant amplitude but variable frequency. The oscillating platform causes the ball to oscillate.

(i) State what is meant by resonance.

.....

.....

..... [2]

(ii) The frequency *f* of the oscillations of the platform slowly increases from 0 to 0.40 Hz.

On Figure 5.3, sketch the variation with *f* of the amplitude of the oscillations of the **ball**.

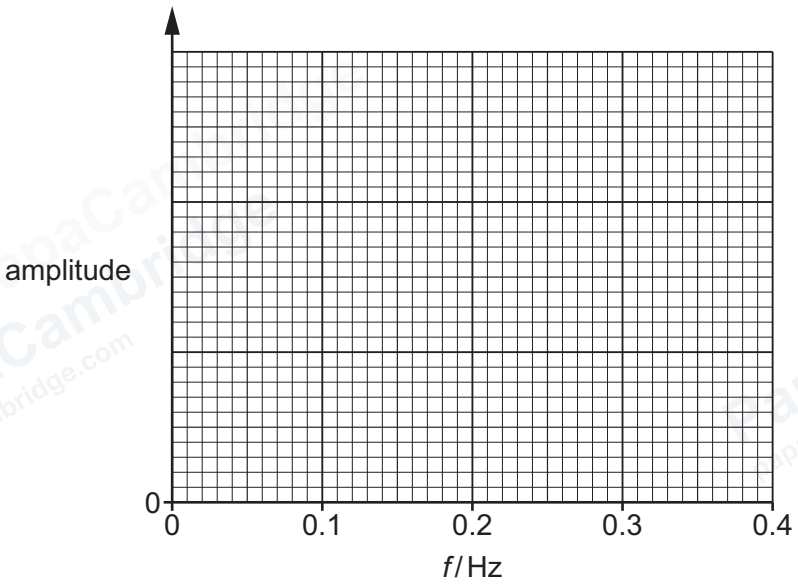


Figure 5.3

[2]

[Total: 9]

18 Electric fields

16 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 18 Electric fields

考纲要求（9702 · 2025 – 2027 syllabus）

18.1

18.1.1 understand that an electric field is an example of a field of force and define electric field as force per unit positive charge

18.1.2 recall and use F = qE for the force on a charge in an electric field

18.1.3 represent an electric field by means of field lines

18.2

18.2.1 recall and use E = ΔV / Δd to calculate the field strength of the uniform field between charged parallel plates

18.2.2 describe the effect of a uniform electric field on the motion of charged particles

18.3

18.3.1 understand that, for a point outside a spherical conductor, the charge on the sphere may be considered to be a point charge at its centre

18.3.2 recall and use Coulomb’s law F = Q1Q2 / (4πε0 r 2) for the force between two point charges in free space

18.4

18.4.1 recall and use E = Q / (4πε0 r 2) for the electric field strength due to a point charge in free space

18.5

18.5.1 define electric potential at a point as the work done per unit positive charge in bringing a small test charge from infinity to the point

18.5.2 recall and use the fact that the electric field at a point is equal to the negative of potential gradient at that point

18.5.3 use V = Q / (4πε0r) for the electric potential in the field due to a point charge

18.5.4 understand how the concept of electric potential leads to the electric potential energy of two point charges and use EP = Qq / (4πε0 r)

5 **(a)** Define electric field.

.....

.....

..... [2]

(b) Fig. 5.1 shows two parallel conducting plates that are in a vacuum. The plates are separated by a distance of 6.7 cm and have a potential difference (p.d.) of 430 V between them.

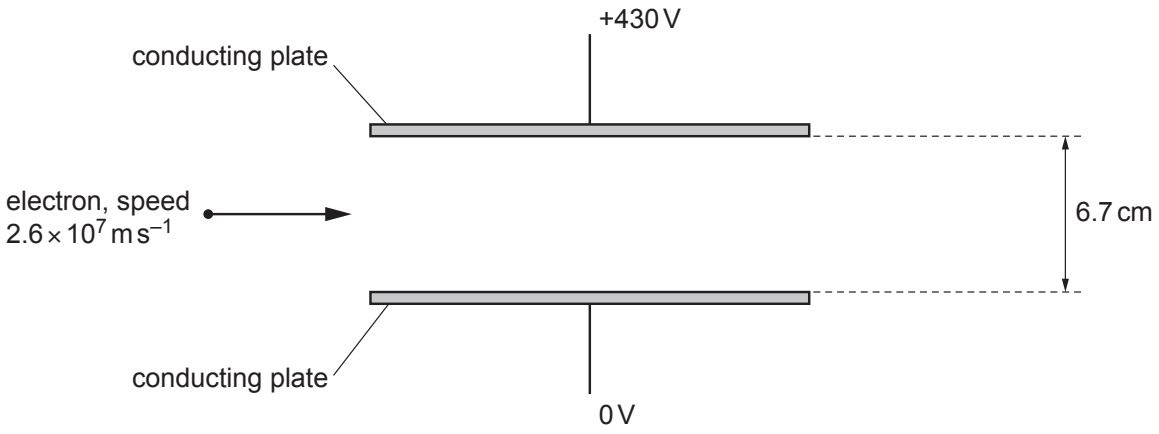


Fig. 5.1

(i) On Fig. 5.1, draw four field lines to represent the electric field between the plates. [2]

(ii) Determine the strength E of the electric field between the plates.

E = ... $N C^{-1}$ [2]

(iii) An electron travels at a speed of $2.6 \times 10^7 m s^{-1}$ towards the region between the plates, as shown in Fig. 5.1.

On Fig. 5.1, draw the path of the electron as it moves between and beyond the plates. [2]

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(c) A uniform magnetic field is now applied in the region of the electric field in Fig. 5.1, so that the electron in **(b)(iii)** travels undeviated through the region.

(i) Determine the direction of the uniform magnetic field.

..... [1]

(ii) Explain, with reference to the forces exerted by the two fields on the electron, why the path of the electron is undeviated.

.....

.....

..... [2]

(iii) Determine the flux density *B* of the uniform magnetic field. Give a unit with your answer.

B = ... unit ..... [2]

[Total: 13]

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5 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) Two isolated charged metal spheres X and Y are near to each other in a vacuum. The centres of the spheres are 1.2 m apart, as shown in Fig. 5.1.

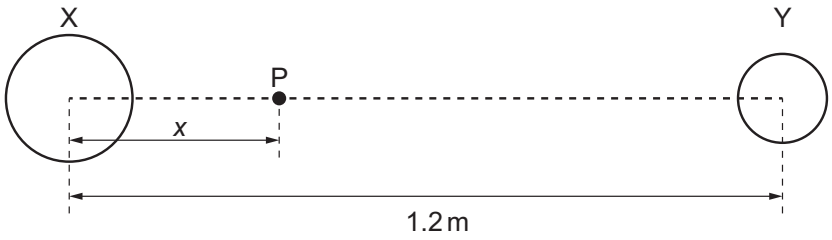


Fig. 5.1 (not to scale)

Point P is on the line joining the centres of spheres X and Y and is at a variable distance *x* from the centre of X.

Fig. 5.2 shows the variation with *x* of the total electric potential *V* due to the two spheres.

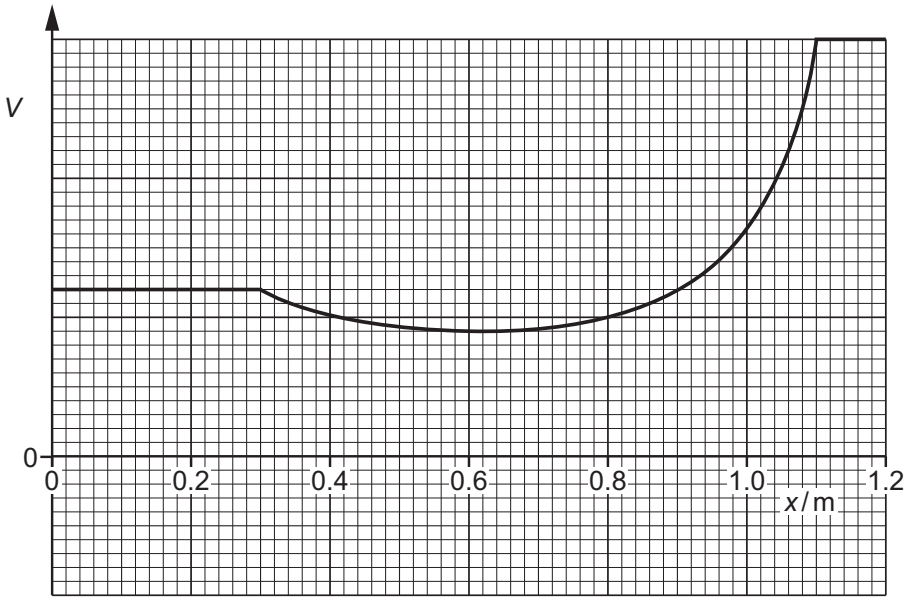


Fig. 5.2

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State **three** conclusions that may be drawn about the spheres from Fig. 5.2. The conclusions may be qualitative or quantitative.

1 .....

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2 .....

.....

3 .....

.....

[3]

(c) A proton is held at rest on the line joining the centres of the spheres in **(b)** at the position where $x = 0.60\,\text{m}$.

The proton is released.

Describe and explain, without calculation, the subsequent motion of the proton.

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..... [2]

[Total: 7]

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5 **(a)** State the relationship between electric field and electric potential.

.....

.....

..... [2]

(b) Two charged isolated insulating spheres X and Y are near to each other, as shown in Fig. 5.1.



Fig. 5.1

P is a point on the line joining the centres of the spheres.

Explain why it is **not** possible for the total electric potential and the resultant electric field to simultaneously be zero at point P.

.....

.....

.....

.....

..... [3]

(c) The magnitudes of the charges on spheres X and Y in Fig. 5.1 are Q and $2Q$ respectively. The spheres may be considered as point charges at their centres.

Point P is a distance x from the centre of sphere X.

The electric potential at point P is zero.

(i) Show that the distance y of point P from the centre of sphere Y is equal to $2x$.

[2]

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(ii) State an expression, in terms of Q, x and the permittivity of free space ε_0, for the electric field strength E_X at P due to sphere X.

E_X = ... [1]

(iii) Determine an expression, in terms of Q, x and ε_0, for the resultant electric field strength E at point P due to the two spheres.

E = ... [2]

[Total: 10]

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6 **(a)** State Coulomb's law.

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..... [2]

(b) Fig. 6.1 shows an isolated hollow conducting sphere that is positively charged.

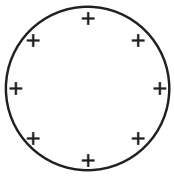


Fig. 6.1

On Fig. 6.1, draw field lines to represent the electric field outside the sphere. [3]

(c) Fig. 6.2 shows the variation of the electric field strength E with distance x from the centre of the sphere in **(b)**.

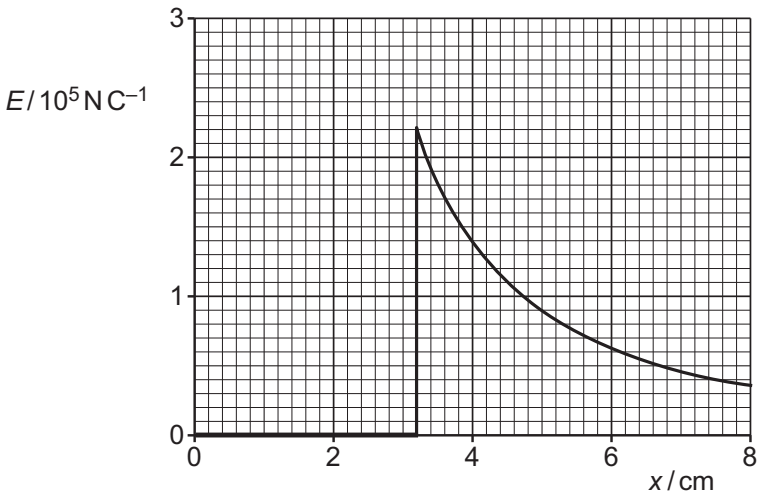


Fig. 6.2

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(i) Determine the radius, in cm, of the sphere.

radius = ... cm [1]

(ii) Calculate the charge on the sphere.

charge = ... C [3]

(iii) Suggest an explanation for the fact that the electric field inside the sphere is zero.

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..... [1]

[Total: 10]

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2 A helium atom may be modelled as a nucleus surrounded by two electrons in diametrically opposite circular orbits, each of radius 170 pm, as shown in Fig. 2.1.

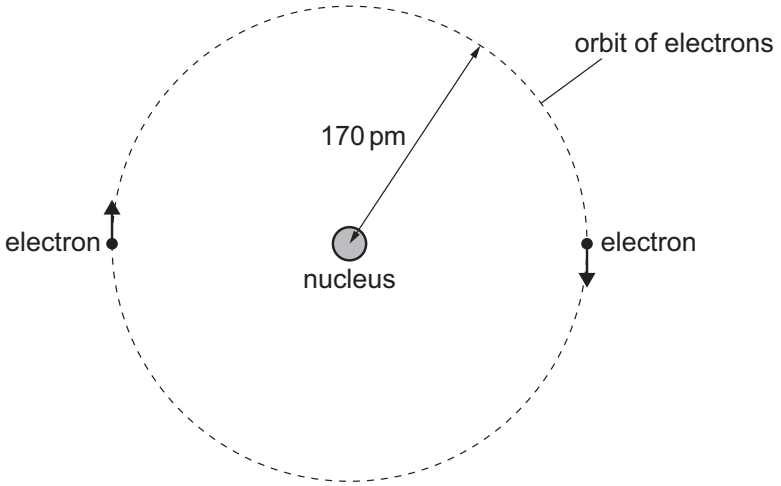


Fig. 2.1

(a) State Coulomb's law.

.....

.....

..... [2]

(b) **(i)** State the charge on the nucleus, in terms of the elementary charge *e*.

charge = ... *e* [1]

(ii) Show that the electric force between the nucleus and one of the electrons is 1.6×10^{-8} N.

[1]

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(c) Assume that the force in **(b)(ii)** is the only force on the electrons.

(i) Calculate the speed of the orbiting electrons.

speed = ... $m\,s^{-1}$ [2]

(ii) Calculate the period of the orbit of the electrons.

period = ... s [2]

(d) In practice, the orbit of each electron is affected by the presence of the other electron.

(i) For the position of one of the electrons, determine the ratio

$$\frac{\text{electric field strength due to the other electron}}{\text{electric field strength due to the nucleus}}.$$

ratio = ... [2]

(ii) Use your answer in **(d)(i)** to suggest and explain how the orbit of the electron is affected by the presence of the other electron.

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..... [1]

[Total: 11]

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6 Two parallel metal plates X and Y are separated by a distance of 0.041 m, as shown in Fig. 6.1.

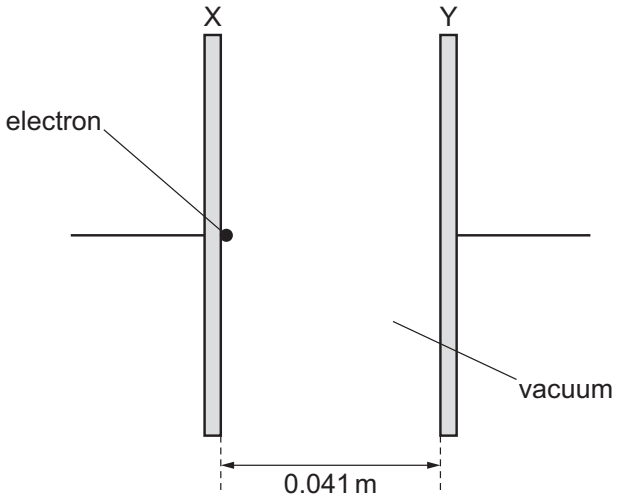


Fig. 6.1

There is a vacuum between the plates. An electron is at rest at the centre of plate X.

A potential difference (p.d.) of 58 kV is applied across the plates. This causes the electron to accelerate towards plate Y.

(a) On Fig. 6.1, use the symbols + and – to indicate which of plates X and Y is the positive plate and which is the negative plate. [1]

(b) **(i)** Calculate the electric field strength *E* between the plates. Give a unit with your answer.

E = ... unit ..... [2]

(ii) Determine the acceleration of the electron.

acceleration = ... $m\,s^{-2}$ [2]

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(c) Many electrons are now accelerated from rest from plate X to plate Y in Fig. 6.1. When the electrons hit plate Y, the absorption of their kinetic energies results in the emission of electromagnetic waves.

(i) Show that the minimum wavelength of these electromagnetic waves is 21 pm.

[3]

(ii) State the region of the electromagnetic spectrum that contains these waves.

..... [1]

(iii) Explain how these electromagnetic waves may be used to form images of internal body structures.

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..... [2]

[Total: 11]

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2 A helium atom may be modelled as a nucleus surrounded by two electrons in diametrically opposite circular orbits, each of radius 170 pm, as shown in Fig. 2.1.

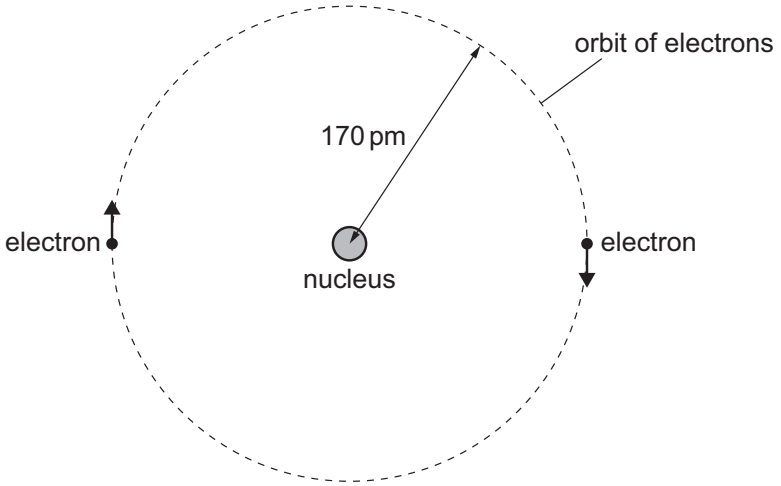


Fig. 2.1

(a) State Coulomb's law.

.....

.....

..... [2]

(b) **(i)** State the charge on the nucleus, in terms of the elementary charge *e*.

charge = ... *e* [1]

(ii) Show that the electric force between the nucleus and one of the electrons is 1.6×10^{-8} N.

[1]

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(c) Assume that the force in **(b)(ii)** is the only force on the electrons.

(i) Calculate the speed of the orbiting electrons.

speed = ... $m\,s^{-1}$ [2]

(ii) Calculate the period of the orbit of the electrons.

period = ... s [2]

(d) In practice, the orbit of each electron is affected by the presence of the other electron.

(i) For the position of one of the electrons, determine the ratio

$$\frac{\text{electric field strength due to the other electron}}{\text{electric field strength due to the nucleus}}.$$

ratio = ... [2]

(ii) Use your answer in **(d)(i)** to suggest and explain how the orbit of the electron is affected by the presence of the other electron.

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..... [1]

[Total: 11]

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5 **(a)** Define electric potential at a point.

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.....

..... [2]

(b) An isolated solid metal sphere of radius r is given a positive charge.

The potential at the surface of the sphere is 9.0×10^4 V. At a distance of $3r$ from the centre of the sphere, the electric field strength is 2.0×10^5 N C^{-1}.

(i) Determine the electric field strength at the surface of the sphere.

electric field strength = ... N C^{-1} [2]

(ii) Show that the radius of the sphere is 5.0 cm.

[2]

(iii) Calculate the charge on the sphere.

charge = ... C [2]

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(iv) Use your answer in **(b)(iii)** to determine the capacitance of the sphere.

capacitance = ... F [2]

[Total: 10]

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5 **(a)** Define electric potential at a point.

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..... [2]

(b) A hydrogen atom may be considered to consist of a proton and an electron separated by a distance of 120 pm, as shown in Fig. 5.1.

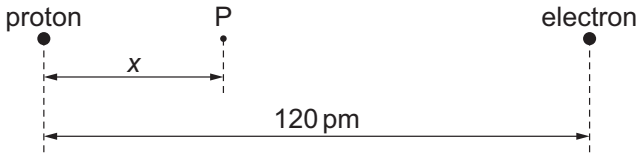


Fig. 5.1

The two particles may be considered as point charges.

Point P lies on the line joining the electron and the proton and is at a variable distance *x* from the proton.

(i) Show that the electric potential *V* at point P when *x* = 10 pm is equal to 130 V.

[2]

(ii) Calculate, to two significant figures, *V* when *x* = 30 pm.

V = ...V [2]

(iii) On Fig. 5.1, draw a cross (×) at one position, other than infinity, where the electric potential is zero. [1]

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(iv) On Fig. 5.2, sketch the variation of V with x between $x = 10\,\text{pm}$ and $x = 110\,\text{pm}$.

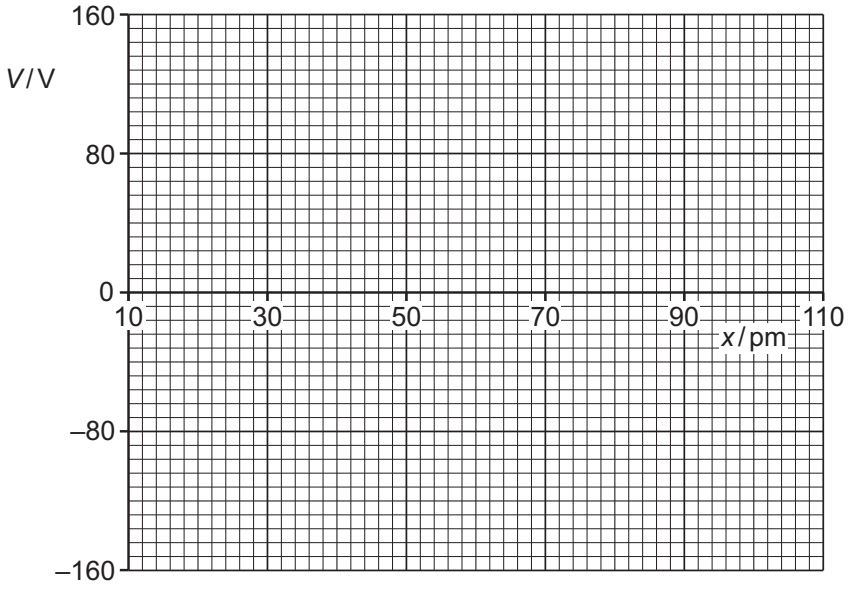


Fig. 5.2

[3]

[Total: 10]

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6 **(a)** Define electric field at a point.

.....

..... [1]

(b) An isolated conducting sphere in a vacuum has a capacitance of 69 pF.
The charge on the sphere is +83 pC.

(i) On Fig. 6.1, draw field lines to represent the electric field outside the sphere due to the charge on the sphere.

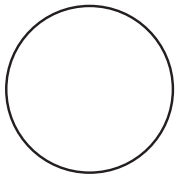


Fig. 6.1

[2]

(ii) Calculate the electric potential at the surface of the sphere.

electric potential = ... V [2]

(iii) Determine the radius of the sphere.

radius = ... m [2]

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(iv) Calculate the electric field strength E at the surface of the sphere. Give a unit with your answer.

E = ... unit ..... [2]

(c) The sphere in **(b)** is discharged by connecting it to earth (0 V) through a resistor of resistance 120 MΩ.

Calculate the time taken for the charge to fall to 26 pC.

time = ... s [2]

[Total: 11]

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5 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) A hydrogen atom may be considered to consist of a proton and an electron separated by a distance of 120 pm, as shown in Fig. 5.1.

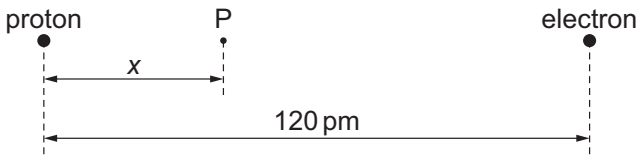


Fig. 5.1

The two particles may be considered as point charges.

Point P lies on the line joining the electron and the proton and is at a variable distance *x* from the proton.

(i) Show that the electric potential *V* at point P when *x* = 10 pm is equal to 130 V.

[2]

(ii) Calculate, to two significant figures, *V* when *x* = 30 pm.

V = ...V [2]

(iii) On Fig. 5.1, draw a cross (×) at one position, other than infinity, where the electric potential is zero. [1]

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(iv) On Fig. 5.2, sketch the variation of V with x between $x = 10\,\text{pm}$ and $x = 110\,\text{pm}$.

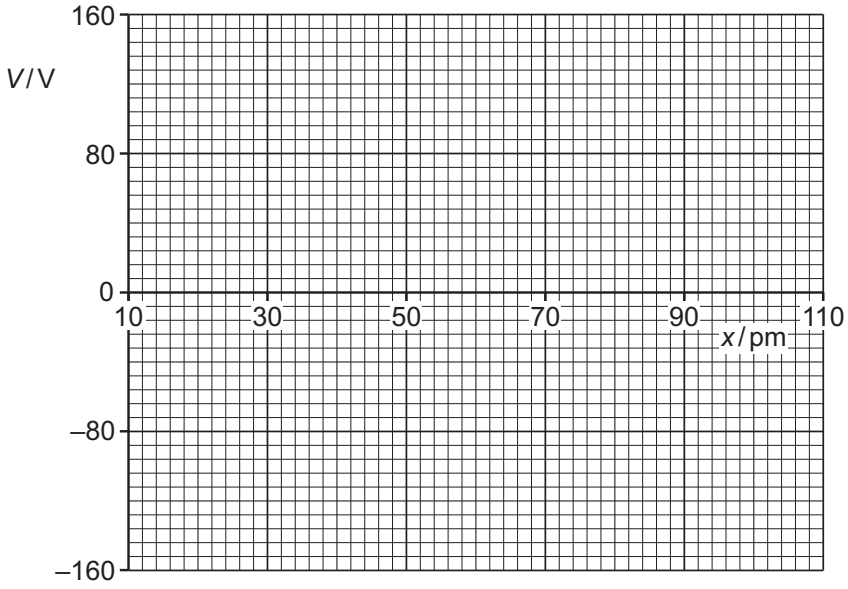


Fig. 5.2

[3]

[Total: 10]

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5 **(a)** Explain why the electric potential near an isolated proton is positive.

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..... [3]

(b) An isolated metal sphere is positively charged and has radius *R*, as shown in Fig. 5.1.

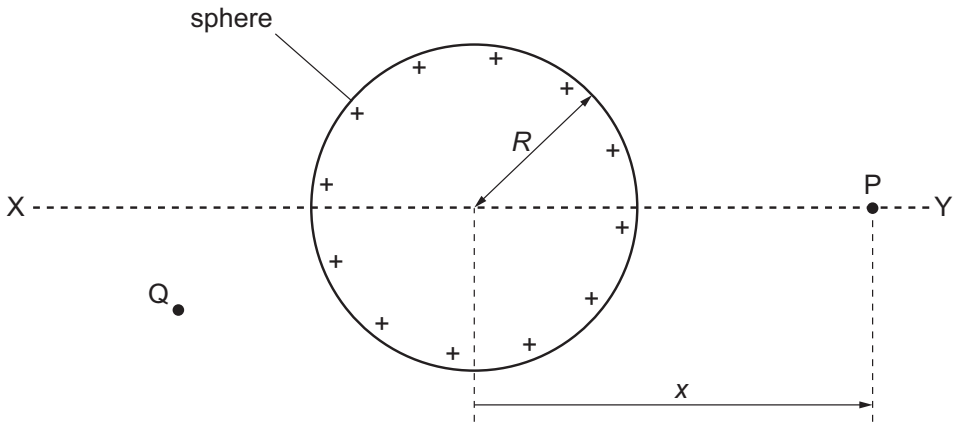


Fig. 5.1

Line XY passes through the centre of the sphere.
Point P lies on line XY at a variable displacement *x* from the centre of the sphere.
Point Q is at a fixed position that is not on line XY.

The electric field strength at the surface of the sphere is E_0.

(i) On Fig. 5.1, draw an arrow at point Q to show the direction of the electric field at that point. [1]

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(ii) On Fig. 5.2, sketch the variation of the electric field E at point P with x for values of x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

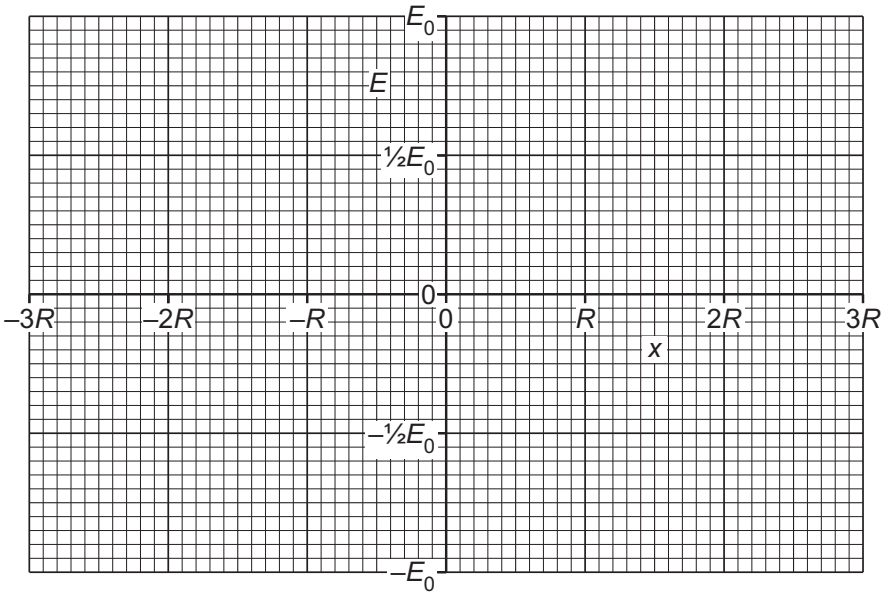


Fig. 5.2

[3]

(c) The proton and the electron in a hydrogen atom are separated by a distance of 5.3×10^{-11} m.

Calculate the electric potential energy of the proton and the electron.

electric potential energy = ... J [2]

[Total: 9]

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4 A hydrogen atom consists of a proton and an electron separated by distance $2x$, as shown in Figure 4.1.

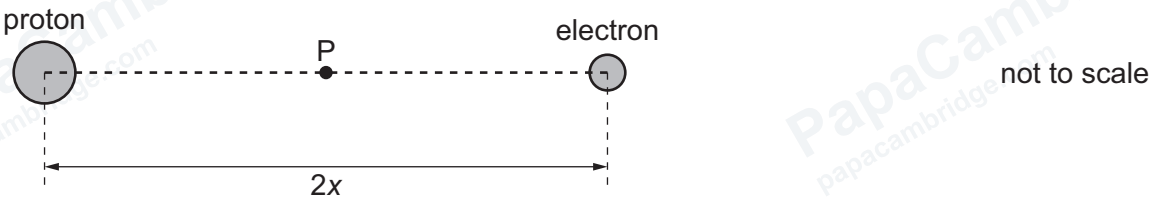


Figure 4.1

Point P is at the midpoint of the line joining the centres of the proton and the electron.

(a) (i) For point P, compare the electric field due to the proton with the electric field due to the electron.

.....

.....

.....

.....

..... [3]

(ii) For point P, compare the electric potential due to the proton with the electric potential due to the electron.

.....

.....

.....

.....

..... [3]

(b) (i) The elementary charge is e and the permittivity of free space is ε_0.

State an expression, in terms of x, e and ε_0, for the magnitude of the electric force F between the proton and the electron.

F = ... [2]

(ii) Determine an expression, in terms of F and x, for the electric potential energy E_P of the proton and the electron.

E_P = ... [1]

[Total: 9]

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6 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) An isolated conducting sphere of radius R is in a vacuum. The sphere has a charge of $+Q$ distributed around its surface, as illustrated in Figure 6.1.

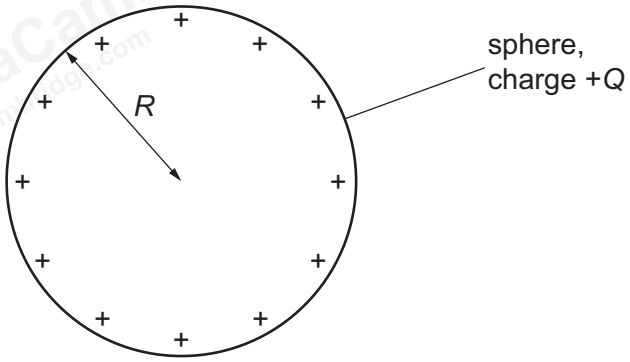


Figure 6.1

(i) State an expression, in terms of R and Q, for the magnitude V_0 of the electric potential at the surface of the sphere. Identify any other symbols that you use.

V_0 = ... [1]

(ii) On Figure 6.2, sketch the variation of the electric potential V at distance x from the centre of the sphere between $x = R$ and $x = 3R$.

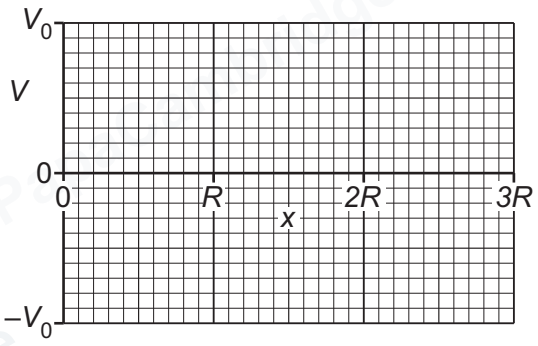


Figure 6.2

[3]

(c) The dome of an electrostatic generator, called a Van de Graaff generator, may be considered to be a conducting sphere of radius 11.9 cm.

The dome is charged so that the electric potential at its surface is -4.80×10^4 V.

(i) Calculate the charge on the surface of the dome.

charge = ... C [2]

(ii) Explain why the dome acts as a capacitor.

.....

..... [1]

(iii) Calculate the capacitance of the dome. Give a unit with your answer.

capacitance = ... unit ..... [2]

(iv) Determine the energy stored by the dome.

energy = ... J [2]

[Total: 13]

4 A hydrogen atom consists of a proton and an electron separated by distance $2x$, as shown in Figure 4.1.

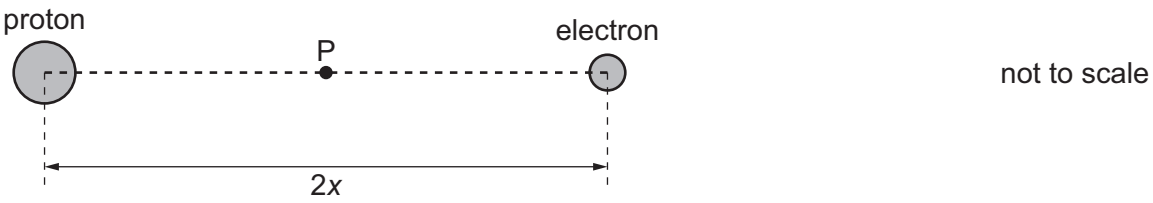


Figure 4.1

Point P is at the midpoint of the line joining the centres of the proton and the electron.

(a) (i) For point P, compare the electric field due to the proton with the electric field due to the electron.

.....

.....

.....

.....

..... [3]

(ii) For point P, compare the electric potential due to the proton with the electric potential due to the electron.

.....

.....

.....

.....

..... [3]

(b) (i) The elementary charge is e and the permittivity of free space is ε_0.

State an expression, in terms of x, e and ε_0, for the magnitude of the electric force F between the proton and the electron.

F = ... [2]

(ii) Determine an expression, in terms of F and x, for the electric potential energy E_P of the proton and the electron.

E_P = ... [1]

[Total: 9]

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1 **(a)** State Coulomb's law.

.....

.....

..... [2]

(b) In a model of an atom, an electron is in a circular orbit around a proton, as shown in Figure 1.1.

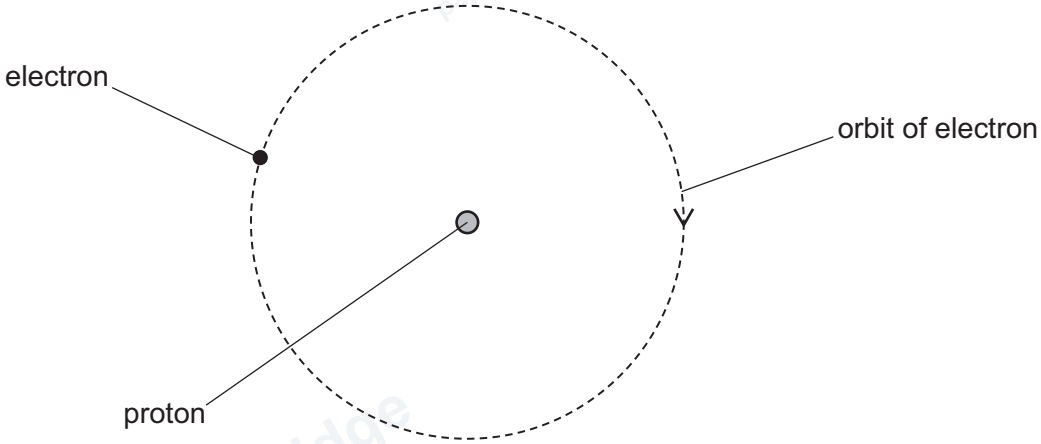


Figure 1.1

The electric potential energy of the electron is -1.09×10^{-18} J.

(i) Show that the radius of the orbit is 2.11×10^{-10} m.

[1]

(ii) Use the information in **1(b)(i)** to determine the magnitude of the electric force between the electron and the proton.

force = ... N [1]

(iii) Calculate the centripetal acceleration of the electron.

acceleration = ... $m\,s^{-2}$ [1]

(iv) Use your answer in **1(b)(iii)** to determine the period of the orbit of the electron.

period = ... s [2]

(c) The electron in **1(b)** emits a photon so that the electric potential energy of the electron decreases. The electron remains in a circular orbit around the proton.

By placing **one** tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for the electron change when the potential energy decreases.

Table 1.1

	decreases	**stays the same**	**increases**
radius of orbit			
centripetal acceleration			
angular speed			

[3]

[Total: 10]

19 Capacitance

10 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 19 Capacitance

考纲要求（9702 · 2025 – 2027 syllabus）

19.1

19.1.1 define capacitance, as applied to both isolated spherical conductors and to parallel plate capacitors

19.1.2 recall and use C = Q / V

19.1.3 derive, using C = Q / V, formulae for the combined capacitance of capacitors in series and in parallel

19.1.4 use the capacitance formulae for capacitors in series and in parallel

19.2

19.2.1 determine the electric potential energy stored in a capacitor from the area under the potential–charge graph

19.2.2 recall and use W = 2 1 QV = 2 1 CV2

19.3

19.3.1 analyse graphs of the variation with time of potential difference, charge and current for a capacitor discharging through a resistor

19.3.2 recall and use τ = RC for the time constant for a capacitor discharging through a resistor

19.3.3 use equations of the form x = x0 e–(t / RC) where x could represent current, charge or potential difference for a capacitor discharging through a resistor

6 Fig. 6.1 shows a capacitor of capacitance C connected in series with a resistor of resistance R.

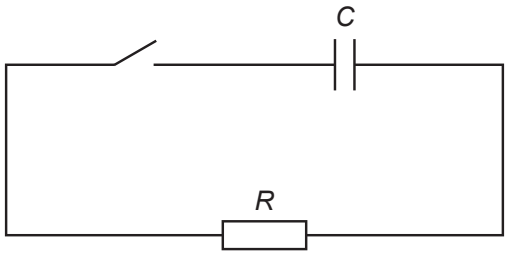


Fig. 6.1

Initially the switch is open and there is a p.d. of 12 V across the capacitor.

At time $t = 0$, the switch is closed so that there is a current I in the resistor.

Fig. 6.2 shows the variation of I with t.

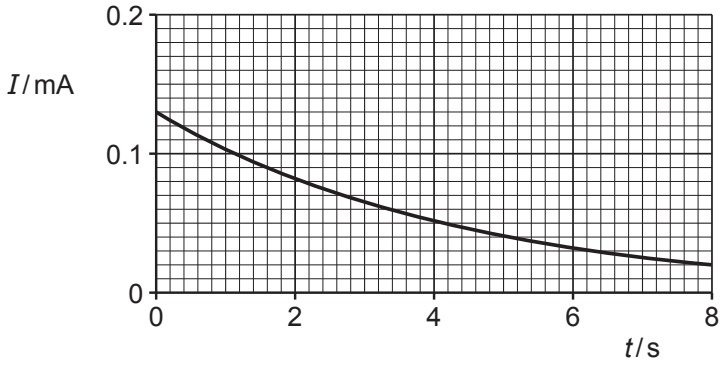


Fig. 6.2

(a) Explain the shape of the line in Fig. 6.2.

.....

.....

.....

.....

..... [3]

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(b) Use Fig. 6.2 to determine:

(i) resistance R

R = ... Ω [2]

(ii) the time constant τ of the circuit in Fig. 6.1.

τ = ... s [3]

(c) Use your answers in **(b)** to determine capacitance C.

C = ... F [2]

[Total: 10]

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6 **(a)** Two capacitors X and Y are connected in series to a power supply of voltage V, as shown in Fig. 6.1.

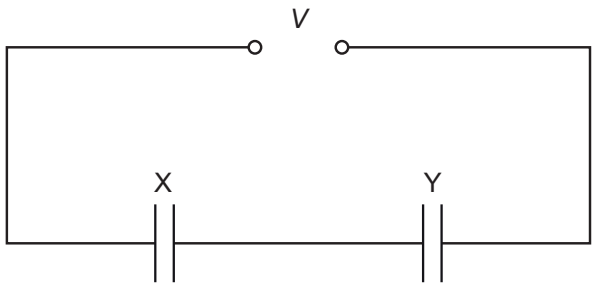


Fig. 6.1

The capacitance of X is C_X and the capacitance of Y is C_Y.

Derive an expression, in terms of C_X and C_Y, for the combined capacitance C_T of the capacitors in this circuit.
Explain your reasoning.

[3]

(b) Two capacitors P and Q are connected in parallel to a power supply of voltage V.
The capacitance of P is 200 μF. The capacitance C_Q of Q can be varied between 0 and 400 μF.
When C_Q = 0, the total energy stored in the capacitors is 2.5 mJ.

(i) Show that the supply voltage V is 5.0 V.

[2]

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(ii) Calculate the total energy, in mJ, stored in the capacitors when C_Q has its maximum value.

total energy = ... mJ [3]

(iii) On Fig. 6.2, sketch the variation of the total energy E stored in the capacitors with C_Q, as C_Q varies from 0 to 400 μF.

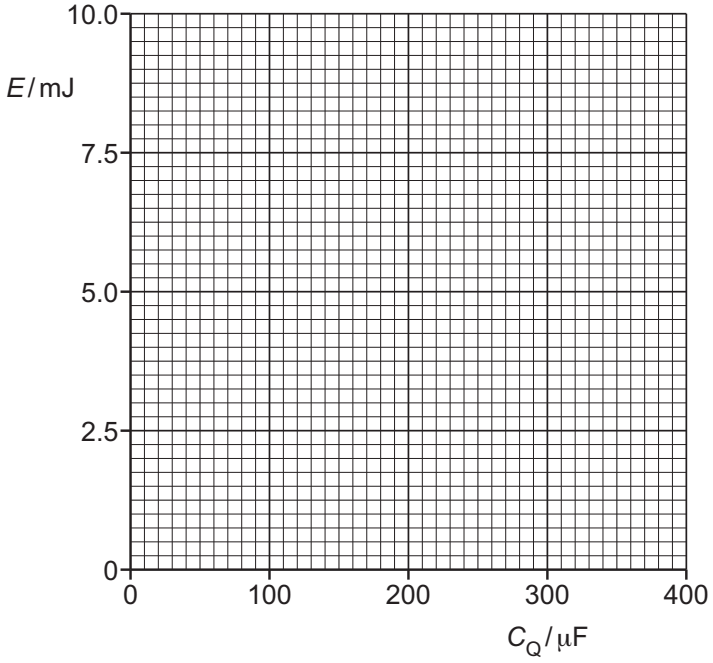


Fig. 6.2

[2]

[Total: 10]

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7 **(a)** Define the capacitance of a parallel-plate capacitor.

.....

.....

..... [2]

(b) An initially uncharged capacitor X, of capacitance *C*, is gradually charged so that the final potential difference (p.d.) between its plates is *V* and the final charge is *Q*.

(i) On Fig. 7.1, sketch the variation of charge with p.d. for capacitor X as the p.d. increases from 0 to *V*.

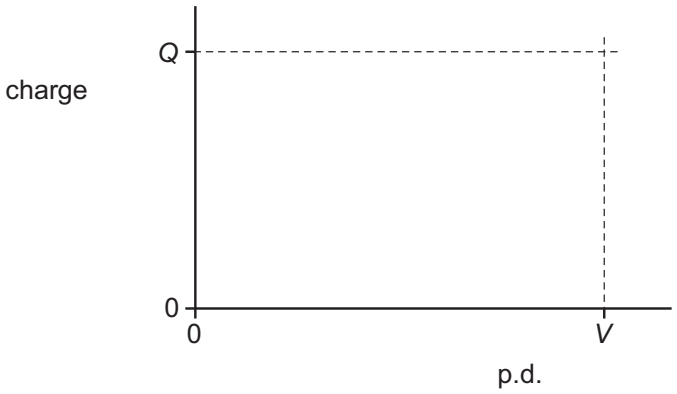


Fig. 7.1

[2]

(ii) Determine an expression, in terms of *Q* and *V*, for the work *W* done on capacitor X during the charging process. Explain your reasoning.

W = ... [2]

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(c) Another capacitor Y is initially uncharged. The fully charged capacitor X in **(b)** is now connected to capacitor Y, as shown in Fig. 7.2.

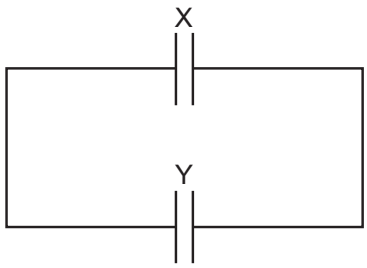


Fig. 7.2

The capacitance of capacitor Y is 3*C*.

(i) Complete Table 7.1 to show expressions, in terms of *Q* and *V*, for the final p.d.s across, and the final charges on, the two capacitors.
Use the space below for any working that you need.

Table 7.1

	X	Y
final p.d.		
final charge		

[3]

(ii) State whether the total energy stored in the two capacitors is less than, the same as, or greater than the energy initially stored in capacitor X.

..... [1]

[Total: 10]

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5 **(a)** A capacitor of capacitance C_1 is connected in series with a second capacitor of capacitance C_2.

Show that the combined capacitance C of the two capacitors is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}.$$

[2]

(b) Three identical capacitors, each of capacitance C, are connected in a network as shown in Fig. 5.1.

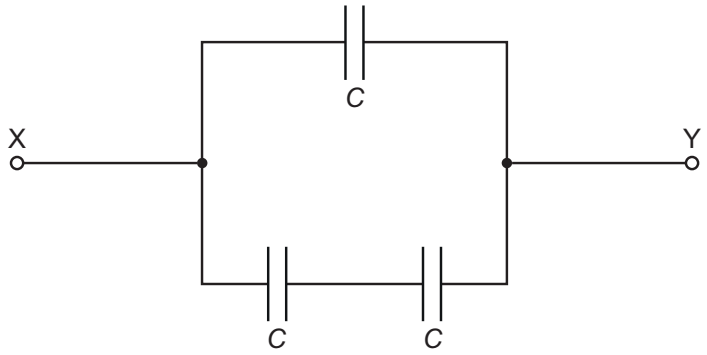


Fig. 5.1

The variation of the charge Q with the potential difference (p.d.) V between the terminals X and Y is shown in Fig. 5.2.

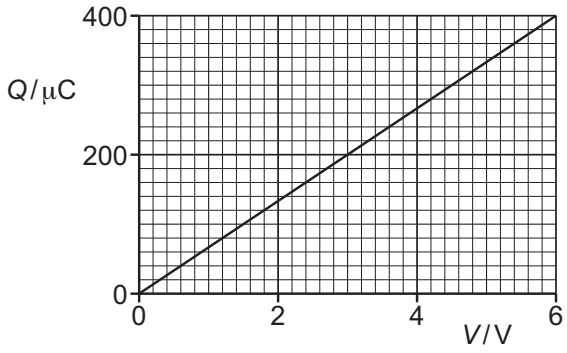


Fig. 5.2

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Show that C is equal to 44 μF.

[3]

(c) The capacitor network in Fig. 5.1 is charged and then connected to a resistor of resistance 54 kΩ. The capacitor network discharges through the resistor.

(i) Determine the time constant τ of the circuit. Give a unit with your answer.

τ = ... unit ..... [2]

(ii) Determine the time taken for the discharge current to reduce to 15% of the initial discharge current.

time = ... s [2]

[Total: 9]

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7 Fig. 7.1 shows a circuit containing a capacitor of capacitance C and a resistor of resistance R.

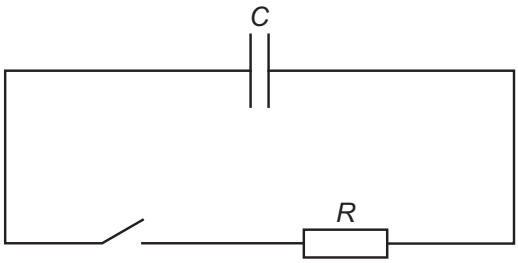


Fig. 7.1

Initially, the switch is open and the potential difference (p.d.) across the capacitor is 12 V.

The switch is closed at time $t = 0$ and the capacitor discharges through the resistor.

Fig. 7.2 shows the variation of the charge Q on the capacitor with the p.d. V_C across the capacitor as the capacitor discharges. Fig. 7.3 shows the variation of the current I in the resistor with the p.d. V_R across the resistor as the capacitor discharges.

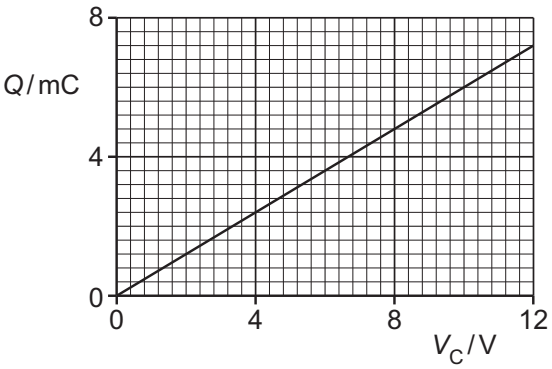


Fig. 7.2

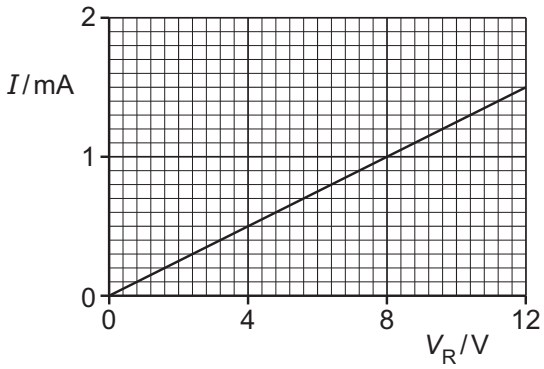


Fig. 7.3

(a) State the relationship between V_C and V_R.

..... [1]

(b) Determine:

(i) the capacitance C, in μF

C = ... μF [2]

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(ii) the resistance R, in kΩ

R = ... kΩ [2]

(iii) the time constant τ of the circuit.

τ = ... s [2]

(c) Use Fig. 7.2, Fig. 7.3 and your answer in **(a)** to explain why the variation of Q with t is exponential in nature.

.....

.....

.....

.....

..... [3]

[Total: 10]

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6 **(a)** Two parallel plate capacitors C_1 and C_2 are connected to a supply that has a potential difference (p.d.) V_S. The capacitors may be connected in series or in parallel.

The supply provides charge Q_S and the plates of the two capacitors acquire charges Q_1 and Q_2 respectively. The p.d.s across the plates of the capacitors are V_1 and V_2 respectively.

Complete Table 6.1 to indicate how Q_S, Q_1 and Q_2 relate to each other, and how V_S, V_1 and V_2 relate to each other, for series and parallel connections of the capacitors to the supply.

Table 6.1

	relationship between charges	relationship between p.d.s
series		
parallel		

[4]

(b) An isolated capacitor of capacitance 470 μF stores 19 mJ of energy.

(i) Calculate the p.d. across the capacitor.

p.d. = ...V [2]

(ii) Calculate the charge on the capacitor.

charge = ... C [2]

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(iii) The capacitor is now connected in parallel with a capacitor of capacitance 180 μF that is initially uncharged.

Determine the total energy, in mJ, now stored in the two capacitors.

energy = ... mJ [3]

[Total: 11]

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6 **(a)** Two parallel plate capacitors C_1 and C_2 are connected to a supply that has a potential difference (p.d.) V_S. The capacitors may be connected in series or in parallel.

The supply provides charge Q_S and the plates of the two capacitors acquire charges Q_1 and Q_2 respectively. The p.d.s across the plates of the capacitors are V_1 and V_2 respectively.

Complete Table 6.1 to indicate how Q_S, Q_1 and Q_2 relate to each other, and how V_S, V_1 and V_2 relate to each other, for series and parallel connections of the capacitors to the supply.

Table 6.1

	relationship between charges	relationship between p.d.s
series		
parallel		

[4]

(b) An isolated capacitor of capacitance 470 μF stores 19 mJ of energy.

(i) Calculate the p.d. across the capacitor.

p.d. = ...V [2]

(ii) Calculate the charge on the capacitor.

charge = ... C [2]

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(iii) The capacitor is now connected in parallel with a capacitor of capacitance 180 μF that is initially uncharged.

Determine the total energy, in mJ, now stored in the two capacitors.

energy = ... mJ [3]

[Total: 11]

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5 **(a)** Define the capacitance of a parallel plate capacitor.

.....

.....

..... [2]

(b) Two capacitors X and Y have capacitances C_X and C_Y respectively. The capacitors are connected in parallel to a supply that has electromotive force (e.m.f.) E, as shown in Figure 5.1.

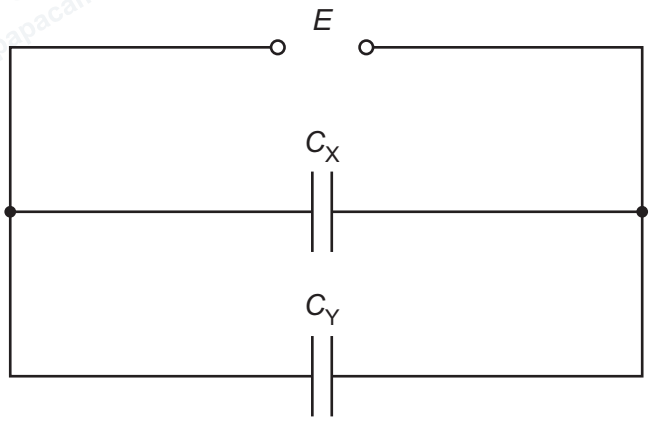


Figure 5.1

Derive the expression for the relationship between C_X, C_Y and the combined capacitance C_T of the circuit.

[3]

(c) Two capacitors are connected in **series** to a supply that has e.m.f. 6.00 V.

One of the capacitors has capacitance C and the other has capacitance $2C$.

The total energy stored by the two capacitors is 1.30 mJ.

Calculate C in μF.

C = ... μF [3]

[Total: 8]

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5 **(a)** Define the capacitance of a parallel plate capacitor.

.....

.....

..... [2]

(b) Two capacitors X and Y have capacitances C_X and C_Y respectively. The capacitors are connected in parallel to a supply that has electromotive force (e.m.f.) E, as shown in Figure 5.1.

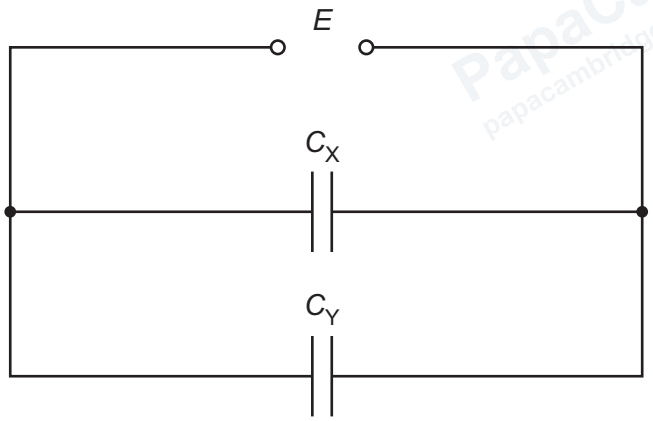


Figure 5.1

Derive the expression for the relationship between C_X, C_Y and the combined capacitance C_T of the circuit.

[3]

(c) Two capacitors are connected in **series** to a supply that has e.m.f. 6.00 V.

One of the capacitors has capacitance C and the other has capacitance $2C$.

The total energy stored by the two capacitors is 1.30 mJ.

Calculate C in μF.

C = ... μF [3]

[Total: 8]

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6 A capacitor of capacitance 47 μF, a cell of electromotive force (e.m.f.) 1.50 V and a resistor R are connected into the circuit of Figure 6.1.

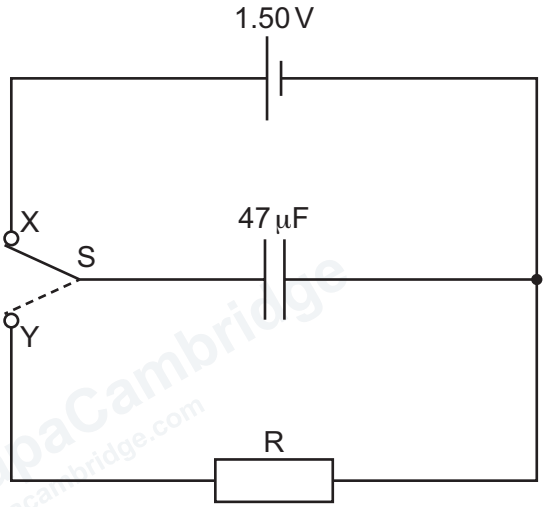


Figure 6.1

The two-way switch S alternates between terminals X and Y with a frequency of 0.20 Hz, as shown in the variation of switch position with time *t* in Figure 6.2.

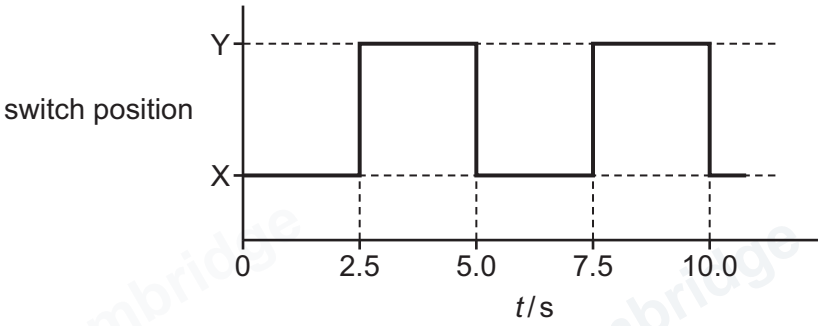


Figure 6.2

The time constant of the discharge circuit is 2.4 s.

(a) (i) Calculate the resistance, in kΩ, of R.

resistance = ... kΩ [2]

(ii) Show that the minimum potential difference (p.d.) across the capacitor is 0.53 V.

[2]

(b) On Figure 6.3, sketch the variation of the p.d. V_C across the capacitor with t between $t = 0$ and $t = 10.0$ s.

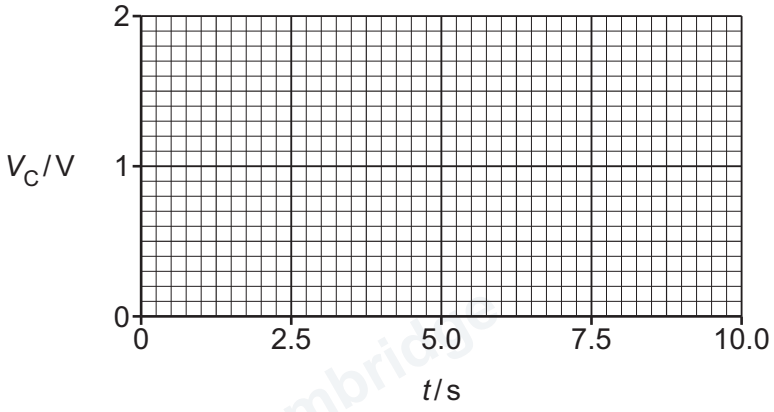


Figure 6.3

[3]

(c) The frequency at which the switch alternates is now doubled.

Explain, without calculation, whether the minimum p.d. across the capacitor is now less than, equal to or greater than 0.53 V.

.....

.....

..... [2]

[Total: 9]

20 Magnetic fields

18 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 20 Magnetic fields

考纲要求（9702 · 2025 – 2027 syllabus）

20.1

20.1.1 understand that a magnetic field is an example of a field of force produced either by moving charges or by permanent magnets

20.1.2 represent a magnetic field by field lines

20.2

20.2.1 understand that a force might act on a current-carrying conductor placed in a magnetic field

20.2.2 recall and use the equation F = BIL sin θ, with directions as interpreted by Fleming’s left-hand rule

20.2.3 define magnetic flux density as the force acting per unit current per unit length on a wire placed at right- angles to the magnetic field

20.3

20.3.1 determine the direction of the force on a charge moving in a magnetic field

20.3.2 recall and use F = BQv sin θ

20.3.3 understand the origin of the Hall voltage and derive and use the expression VH = BI / (ntq), where t = thickness

20.3.4 understand the use of a Hall probe to measure magnetic flux density

20.3.5 describe the motion of a charged particle moving in a uniform magnetic field perpendicular to the direction of motion of the particle

20.3.6 explain how electric and magnetic fields can be used in velocity selection

20.4

20.4.1 sketch magnetic field patterns due to the currents in a long straight wire, a flat circular coil and a long solenoid

20.4.2 understand that the magnetic field due to the current in a solenoid is increased by a ferrous core

20.4.3 explain the origin of the forces between current-carrying conductors and determine the direction of the forces

20.5

20.5.1 define magnetic flux as the product of the magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density

20.5.2 recall and use Φ = BA

20.5.3 understand and use the concept of magnetic flux linkage

20.5.4 understand and explain experiments that demonstrate: that a changing magnetic flux can induce an e.m.f. in a circuit that the induced e.m.f. is in such a direction as to oppose the change producing it the factors affecting the magnitude of the induced e.m.f.

20.5.5 recall and use Faraday’s and Lenz’s laws of electromagnetic induction

5 **(a)** An object travels in a circle at constant speed.

State the names of **two** quantities that vary during the motion of the object.

1 .....

2 .....
[2]

(b) A charged particle of mass m and with charge q enters a region of uniform magnetic field, perpendicular to the field lines. The magnetic flux density is B.

The particle travels in a circle with period T and radius r.

(i) By considering the magnetic force acting on the particle, show that

$$B = \frac{2\pi m}{qT}.$$

[3]

(ii) The particle is an alpha particle. The period of the circular motion is 2.5 μs.

Calculate B.

B = ... T [2]

(iii) A second alpha particle is in the same uniform field. It travels in a circle of radius $2r$.

State and explain how the periods of the motion of the two particles compare.

.....

.....

..... [1]

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(iv) The speed of the alpha particle in **(b)(ii)** is $1.1 \times 10^6\,\text{m s}^{-1}$. An electric field is applied so that this particle now moves with constant velocity.

Use your answer in **(b)(ii)** to calculate the electric field strength E. Give the unit with your answer.

E = ... unit ..... [2]

[Total: 10]

6 **(a)** A small coil C has 64 turns and cross-sectional area 0.71 cm^2. The coil is placed inside a solenoid as shown in Fig. 6.1.

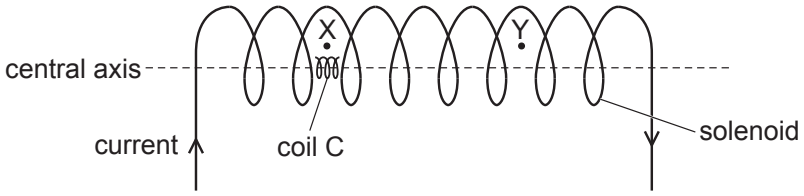


Fig. 6.1

The centre of coil C is on the central axis of the solenoid.

(i) There is a constant current in the solenoid.
Coil C is moved through the solenoid from position X to position Y.

On Fig. 6.2, sketch a line to show the variation of the magnetic flux linkage in coil C with position as it moves from X to Y.

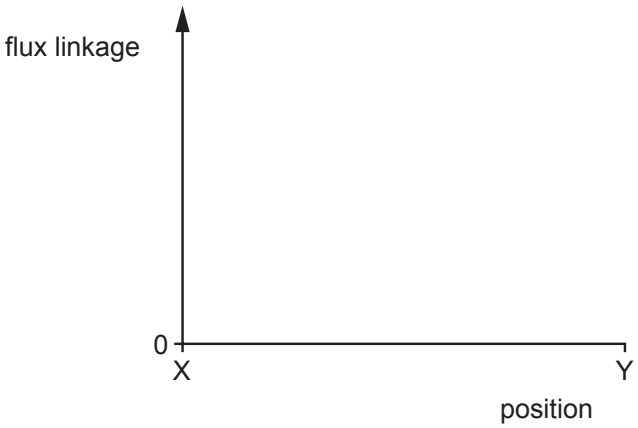


Fig. 6.2

[1]

(ii) Explain the shape of your line in **(a)(i)**.

.....

.....

.....

..... [2]

(iii) Coil C is now held stationary at X. The current in the solenoid varies so that the magnetic flux density *B* at X varies from time 0 to time 4*t* as shown in Fig. 6.3.

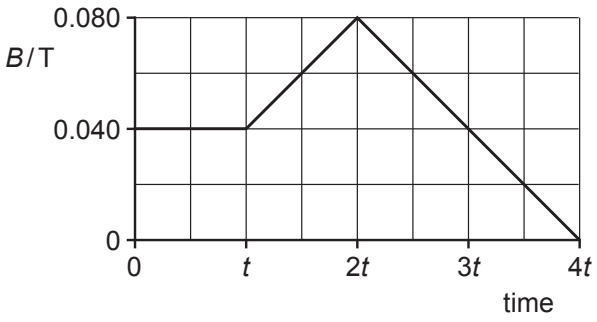


Fig. 6.3

Calculate the maximum magnetic flux linkage in coil C.

flux linkage = ... Wb [2]

(iv) On Fig. 6.4, sketch a line to show the induced electromotive force (e.m.f.) *E* in coil C from time 0 to time 4*t*.

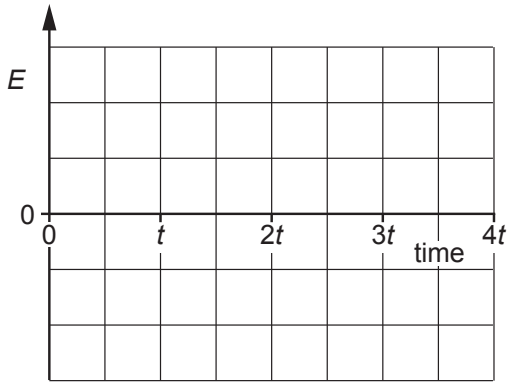


Fig. 6.4

[3]

(b) A metal spring rests on a smooth table. The turns of the spring are equally spaced. The ends of the spring are connected to a d.c. power supply, as shown in Fig. 6.5.

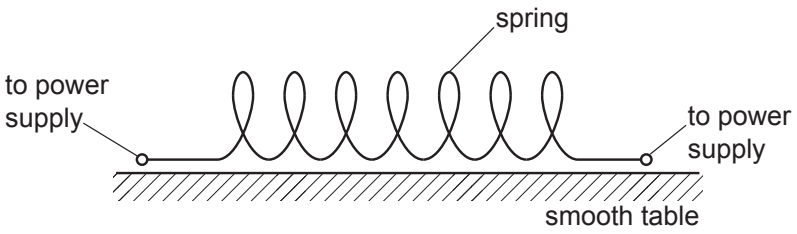


Fig. 6.5

The spring is connected to the d.c. power supply using flexible leads. The spring is not under tension.

With reference to magnetic fields, describe and explain the change in the distance between the turns of the spring when the power supply is first switched on.

.....

.....

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..... [3]

[Total: 11]

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7 **(a)** State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

(b) Fig. 7.1 shows a coil at rest in a uniform magnetic field that is parallel to the axis of the coil.

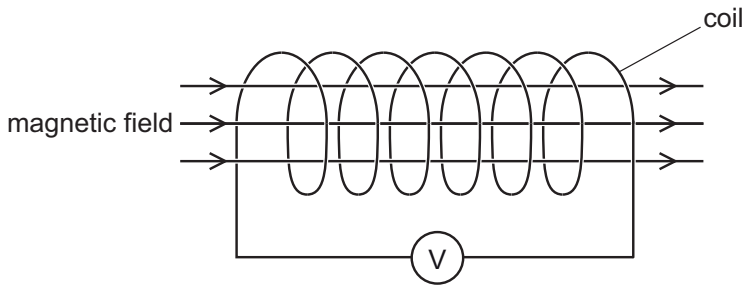


Fig. 7.1

The coil is connected to a centre-zero voltmeter.

The flux density B of the uniform magnetic field varies with time t as shown in Fig. 7.2.

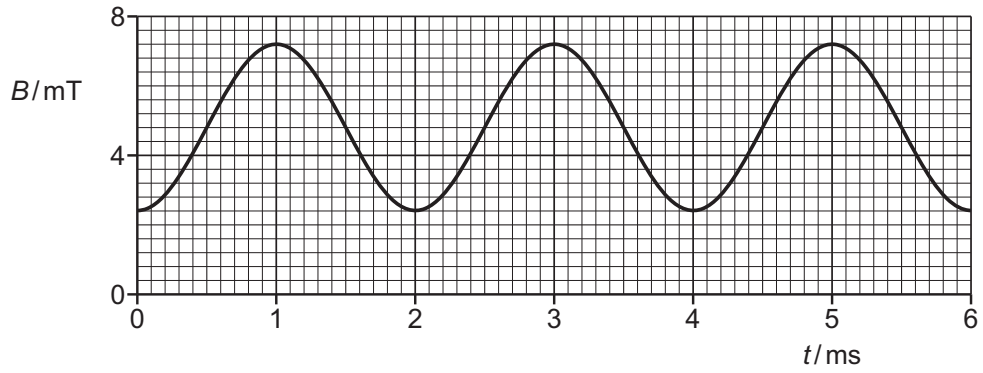


Fig. 7.2

The coil consists of 340 turns, each of cross-sectional area $3.2 \times 10^{-4}\,m^2$.

(i) Calculate the maximum magnetic flux through **one** turn of the coil.

maximum magnetic flux = ... Wb [2]

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(ii) Determine the maximum rate of change of magnetic flux linkage in the coil.

maximum rate of change of flux linkage = ... $Wb\,s^{-1}$ [3]

(iii) State the maximum electromotive force (e.m.f.) V_0 induced across the coil.

V_0 = ... V [1]

(iv) On Fig. 7.3, sketch the variation of the e.m.f. V induced across the coil with t from $t = 0$ to $t = 6.0$ ms.

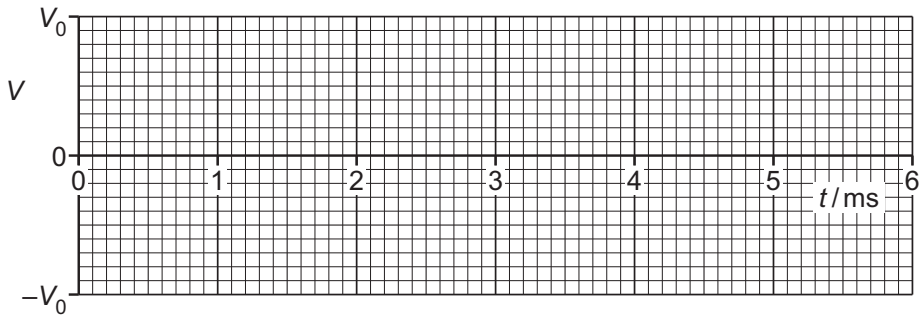


Fig. 7.3

[3]

(v) The variation of V with t can be described by

$$V = A \sin Bt$$

where A and B are constants.

Determine the values of A and B. Give units with your answers.

A = ... unit .....

B = ... unit .....

[3]

[Total: 14]

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7 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) A long, straight wire carries a current into the page, as shown in Fig. 7.1.



Fig. 7.1

On Fig. 7.1, draw four field lines to represent the magnetic field around the wire due to the current in it. [3]

(c) Two identical wires X and Y are placed parallel to each other. The wires both carry current into the page, as shown in Fig. 7.2.



Fig. 7.2

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(i) Explain why the two wires exert a magnetic force on each other.

.....

.....

.....

..... [2]

(ii) On Fig. 7.2, draw an arrow to show the direction of the magnetic force exerted on wire X. Label your arrow F. [1]

(iii) The current in X is double the current in Y.

State how the magnetic force exerted on wire Y compares with the magnetic force exerted on wire X.

.....

.....

..... [2]

(iv) The direction of the current in both wires is now reversed.

State, with a reason, the effect of this change on the direction of the force on wire X.

.....

..... [1]

[Total: 11]

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1 A metal wheel consists of an axle A, eight spokes and a rim, as shown in Fig. 1.1.

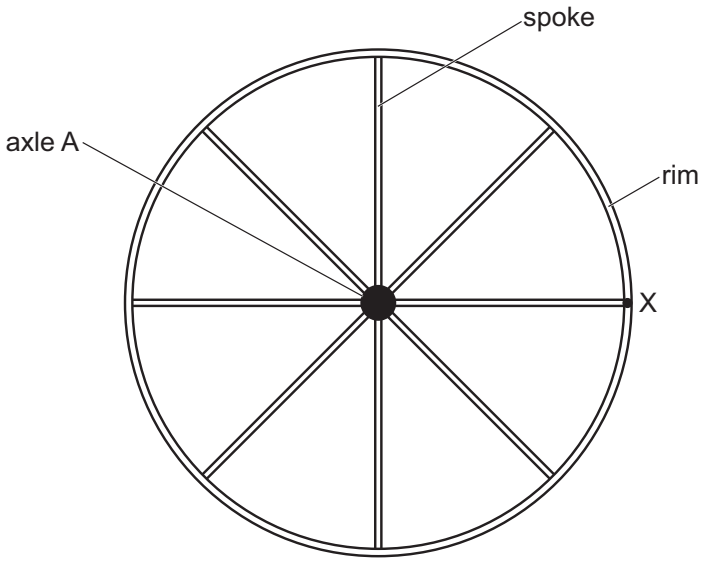


Fig. 1.1

Point X is on the rim at the end of one of the spokes.

The rim has a radius of 0.85 m.
The wheel is rotating clockwise with an angular speed of 140 rad s^{-1}.

(a) For point X, determine:

(i) the speed

speed = ... $m\,s^{-1}$ [2]

(ii) the centripetal acceleration.

acceleration = ... $m\,s^{-2}$ [2]

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(b) There is a uniform magnetic field of flux density 0.18 T into the plane of the page.

(i) State Lenz's law of electromagnetic induction.

.....

.....

..... [2]

(ii) Show that the time taken for point X to complete one revolution is 45 ms.

[1]

(iii) Calculate the magnetic flux cut by spoke AX during one revolution of the wheel.
Give a unit with your answer.

magnetic flux = ... unit ..... [3]

(iv) Determine the magnitude of the electromotive force (e.m.f.) induced across spoke AX.

induced e.m.f. = ... V [2]

(v) Use Lenz's law to explain whether the potential is higher at end A or end X of the spoke.

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.....

..... [1]

[Total: 13]

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6 An electric field and a magnetic field are used to form a velocity selector. Charged particles, called ions, pass into a region of uniform electric and magnetic fields that is between parallel plates, as shown in Fig. 6.1.

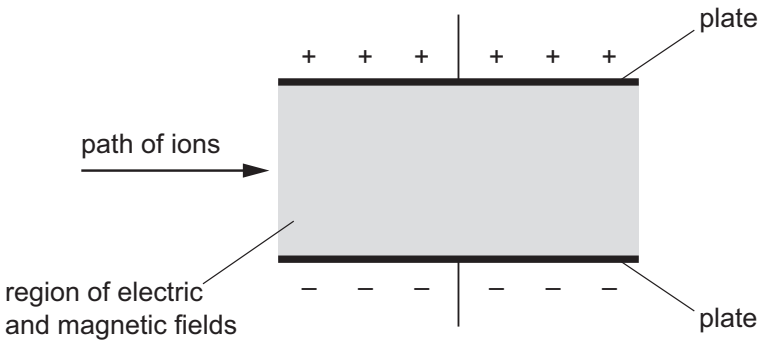


Fig. 6.1

(a) The potential difference (p.d.) between the plates of the velocity selector is V. The separation of the plates is d and the magnetic flux density is B.

Show that the speed u of ions that pass undeviated through the velocity selector is given by

$$u = \frac{V}{Bd}.$$

[2]

(b) Positive ions with kinetic energy 4.1×10^{-17} J and mass 3.2×10^{-27} kg pass undeviated through the velocity selector when V is equal to 980 V and d is equal to 3.6×10^{-2} m.

Determine B.

B = ... T [3]

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(c) A proton passes undeviated through the velocity selector.

An alpha particle enters the velocity selector at the same speed as the proton.

State how the expression in **(a)** predicts that the alpha particle also passes undeviated through the velocity selector.

.....

..... [1]

(d) By reference to Fig. 6.1 and to the forces acting on a positive ion, determine the direction of the magnetic field. Explain your reasoning.

.....

.....

.....

.....

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..... [3]

(e) The positive ions in **(b)** enter the velocity selector with greater kinetic energy.

On Fig. 6.1, sketch the path of these ions. [2]

[Total: 11]

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7 **(a)** State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

(b) A metal rod is accelerated uniformly from rest in a uniform magnetic field as shown in Fig. 7.1.

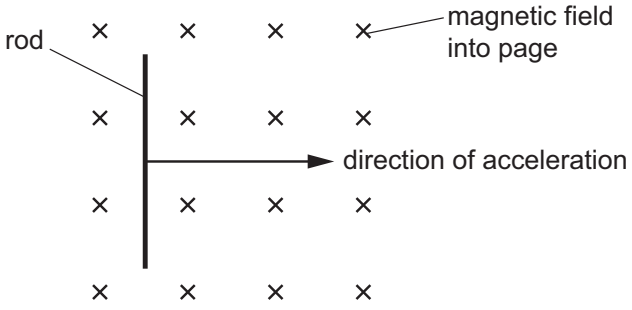


Fig. 7.1

The rod has length l and the flux density of the magnetic field is B.

An electromotive force (e.m.f.) is induced in the rod. The variation with time t of the induced e.m.f. E is shown in Fig. 7.2.

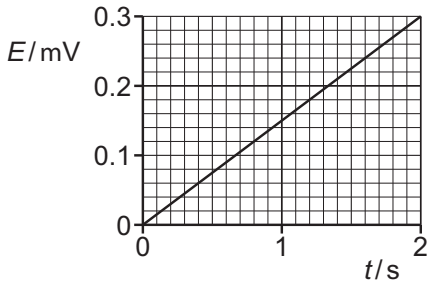


Fig. 7.2

(i) Explain how Fig. 7.2 shows that E is proportional to the velocity v of the rod.

.....

.....

..... [2]

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(ii) Use Faraday's law to show that the variation of E with time t is given by

$$E = Blat$$

where a is the acceleration of the rod.

[3]

(iii) The length of the rod is 0.45 m. The acceleration a of the rod is $7.8\,m\,s^{-2}$.

Determine the value of B.

B = ... T [2]

[Total: 9]

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(c) The capacitor C has a capacitance of 570 μF.

Use your answer in **(b)(iv)** to determine the resistance of resistor R.

resistance = ... Ω [2]

[Total: 12]

7 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) A particle of mass *m* and charge +*Q* moves at speed *v* into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

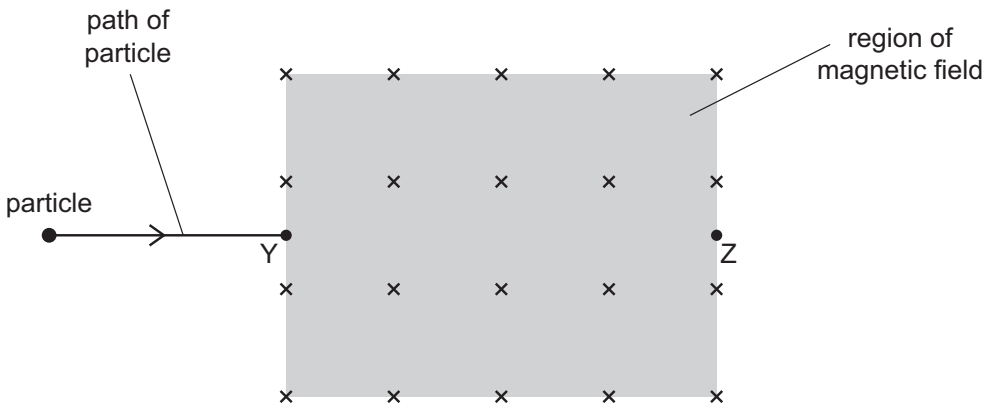


Fig. 7.1

The uniform magnetic field is into the page and has flux density *B*. The particle enters the region of the field at point Y.

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(i) State an expression, in terms of some or all of m, Q, B and v, for the magnetic force F that acts on the particle when it is at point Y.

$F =$ ..... [1]

(ii) On Fig. 7.1, draw an arrow at point Y to indicate the direction of the force in **(b)(i)**. [1]

(iii) On Fig. 7.1, draw a line to show a possible path for the particle through the region of the magnetic field. [1]

(c) (i) Explain how an electric field can be used with the magnetic field to ensure that the particle in **(b)** now passes through point Z.

.....

.....

.....

.....

..... [3]

(ii) Derive an expression for v in terms of B and the electric field strength E.

$v =$ ..... [2]

[Total: 10]

2 **(a)** **(i)** State what is represented by a gravitational field line.

.....

.....

..... [2]

(ii) The Earth may be considered as a uniform sphere, as shown in Fig. 2.1.

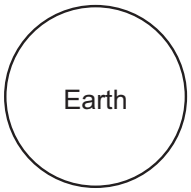


Fig. 2.1

On Fig. 2.1, draw field lines to represent the Earth's gravitational field outside the Earth. [2]

(b) The Earth's magnetic field may be considered as being due to the Earth acting as a long solenoid, as shown in Fig. 2.2.

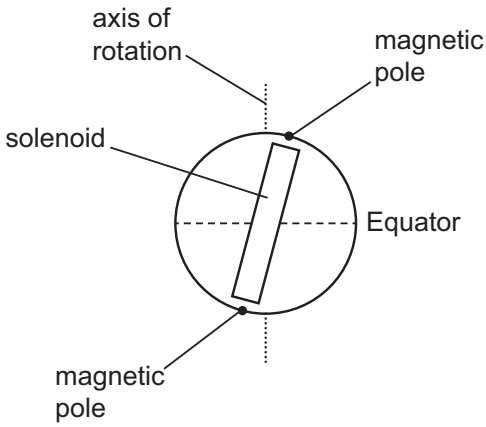


Fig. 2.2

The magnetic poles do not align with the geographic poles, which are on the axis of rotation.

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Fig. 2.3 is a copy of Fig. 2.2 without the labels but with two magnetic field lines shown.

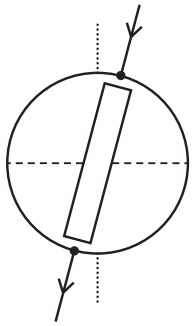


Fig. 2.3

(i) On Fig. 2.3, label the magnetic poles with the letters N and S to indicate which one is the magnetic N pole and which one is the magnetic S pole. [1]

(ii) On Fig. 2.3, draw field lines to represent the Earth's magnetic field outside the Earth. [2]

(c) An observer moves around the surface of the Earth.

(i) Use your answer in **(a)(ii)** to explain why the observed gravitational field of the Earth does not vary around the surface.

.....

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..... [2]

(ii) With reference to your answer in **(b)(ii)**, describe how the observed magnetic field of the Earth varies around the surface.

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..... [3]

[Total: 12]

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(c) The capacitor C has a capacitance of 570 μF.

Use your answer in **(b)(iv)** to determine the resistance of resistor R.

resistance = ... Ω [2]

[Total: 12]

7 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) A particle of mass *m* and charge +*Q* moves at speed *v* into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

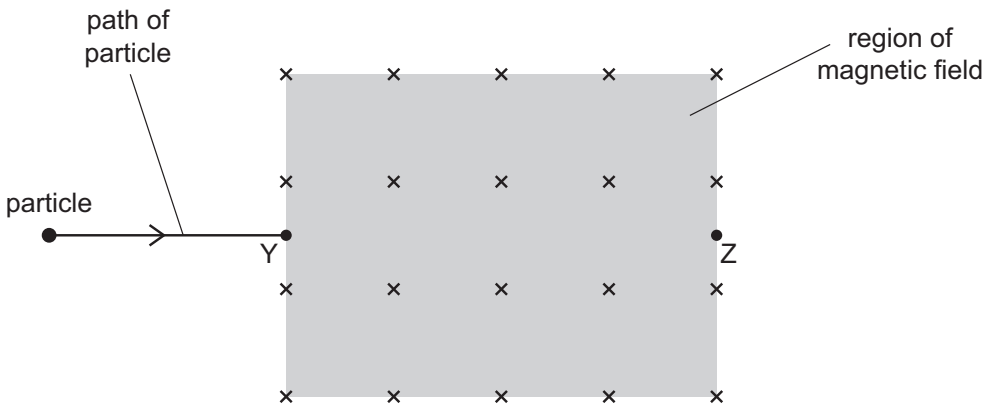


Fig. 7.1

The uniform magnetic field is into the page and has flux density *B*. The particle enters the region of the field at point Y.

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(i) State an expression, in terms of some or all of m, Q, B and v, for the magnetic force F that acts on the particle when it is at point Y.

$F =$ ..... [1]

(ii) On Fig. 7.1, draw an arrow at point Y to indicate the direction of the force in **(b)(i)**. [1]

(iii) On Fig. 7.1, draw a line to show a possible path for the particle through the region of the magnetic field. [1]

(c) (i) Explain how an electric field can be used with the magnetic field to ensure that the particle in **(b)** now passes through point Z.

.....

.....

.....

.....

..... [3]

(ii) Derive an expression for v in terms of B and the electric field strength E.

$v =$ ..... [2]

[Total: 10]

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6 A rectangular coil PQRS of wire is free to rotate about its axis XY, as shown in Fig. 6.1.

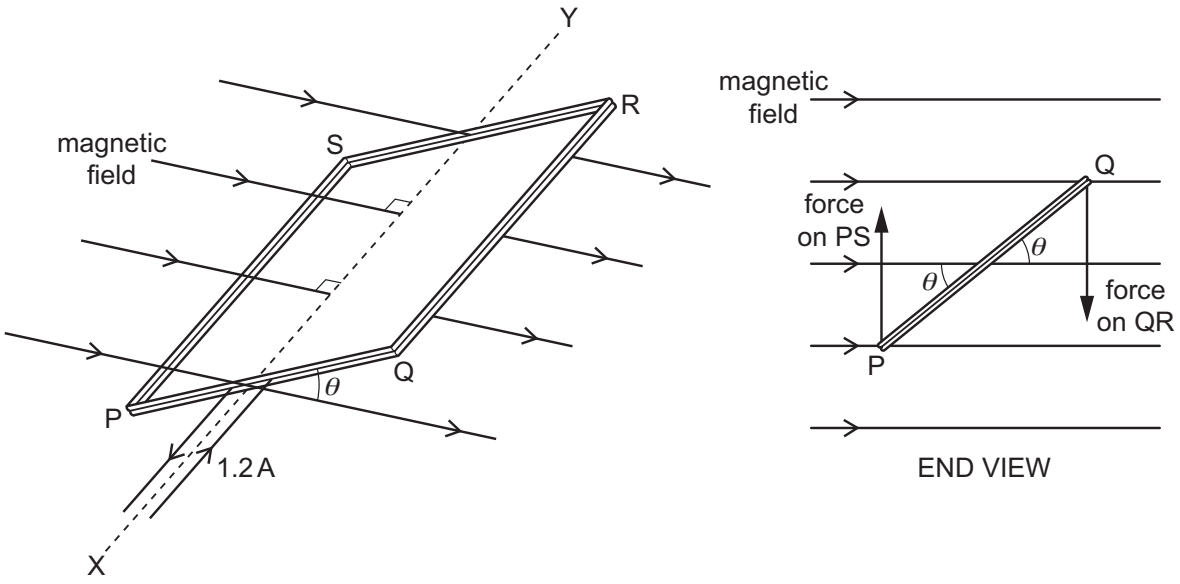


Fig. 6.1 (not to scale)

The coil has length QR of 5.4 cm, width PQ of 2.5 cm and has 190 turns of wire.
The plane of the coil is at an angle θ to a uniform magnetic field of flux density 5.2×10^{-3} T.
The axis XY of the coil is normal to the field.
The current in the coil is 1.2 A.

(a) (i) Calculate the magnitude of the force on side QR of the coil.

force = ... N [3]

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(ii) Use your answer in **(a)(i)** to show that the torque τ on the coil is given by

$$\tau = 1.6 \times 10^{-3} \cos\theta \text{ N m}.$$

[2]

(iii) Using the expression in **(a)(ii)** sketch, on the axes of Fig. 6.2, a graph to show the variation of the torque τ with angle θ for values of θ between 0 and 360°. Label the τ axis with an appropriate scale.

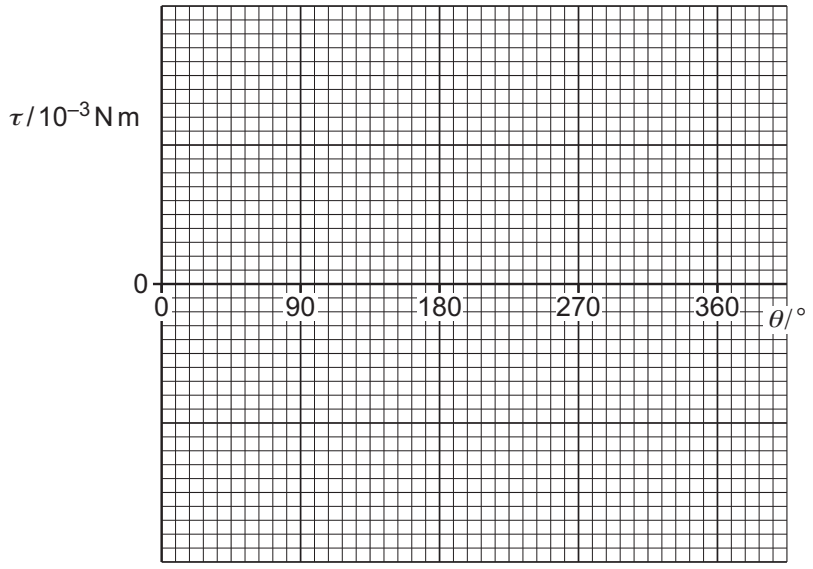


Fig. 6.2

[3]

(b) The coil is now replaced by an identical coil wound on a ferrous core.

Suggest, with a reason, how the torque on this coil compares with the torque on the original coil.

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..... [2]

[Total: 10]

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7 **(a)** State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

(b) An aircraft is flying horizontally at constant speed v through the Earth's magnetic field, as shown in Fig. 7.1.

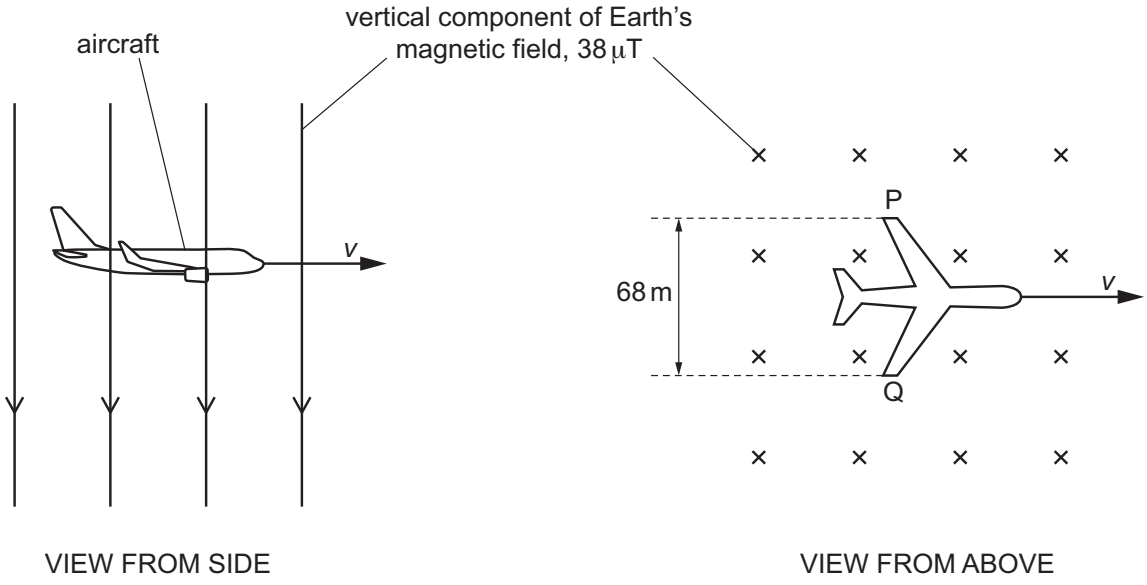


Fig. 7.1

At the location of the aircraft, the vertical component of the Earth's magnetic field is 38 μT towards the ground.

The distance between the wingtips P and Q of the aircraft is 68 m.

As the aircraft moves through the magnetic field, an electromotive force (e.m.f.) of 0.54 V is induced between the wingtips P and Q.

(i) Calculate the magnetic flux cut by the wings of the aircraft in a time of 15 s. Give a unit with your answer.

magnetic flux = ... unit ..... [2]

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(ii) Determine the area of flux cut by the wings in a time of 15 s.

area = ...m^2 [2]

(iii) Use your answer in **(b)(ii)** to determine the speed v of the aircraft.

v = ...$m\,s^{-1}$ [2]

(iv) Use Lenz's law of electromagnetic induction to explain which of the wingtips P and Q is at the higher induced potential.

.....

.....

.....

.....

..... [3]

[Total: 11]

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7 **(a)** State Faraday's law of electromagnetic induction.

.....

.....

..... [2]

(b) An aircraft is flying horizontally at constant speed v through the Earth's magnetic field, as shown in Fig. 7.1.

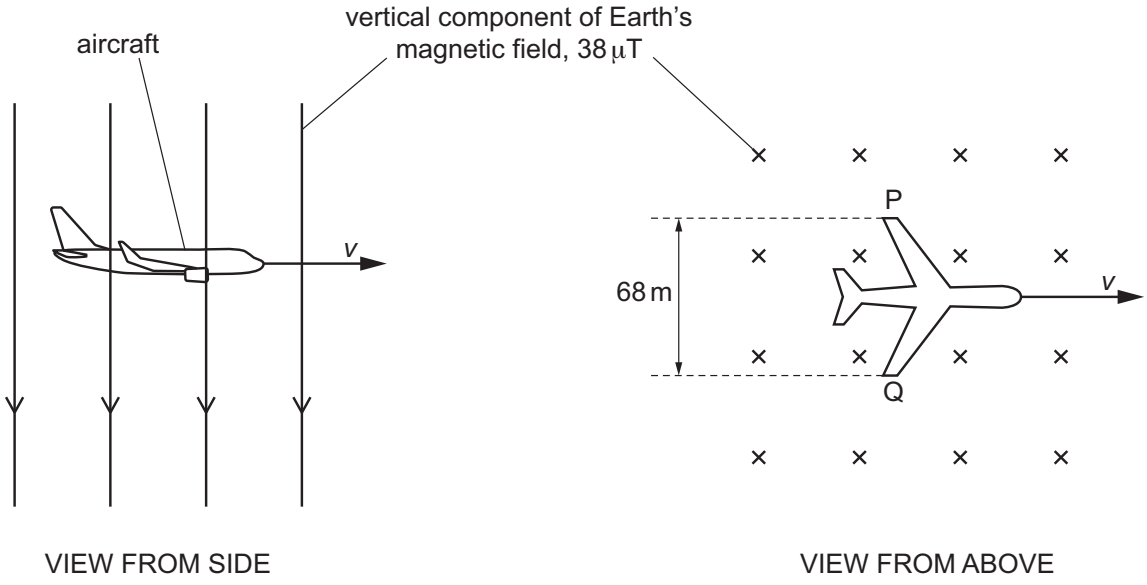


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magnetic flux = ... unit ..... [2]

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(ii) Determine the area of flux cut by the wings in a time of 15 s.

area = ...m^2 [2]

(iii) Use your answer in **(b)(ii)** to determine the speed v of the aircraft.

v = ...$m\,s^{-1}$ [2]

(iv) Use Lenz's law of electromagnetic induction to explain which of the wingtips P and Q is at the higher induced potential.

.....

.....

.....

.....

..... [3]

[Total: 11]

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7 **(a)** State Lenz's law of electromagnetic induction.

.....

.....

..... [2]

(b) A helicopter hovering in stationary equilibrium has four rotors, each of length 12 m, as shown in the view from above in Fig. 7.1.

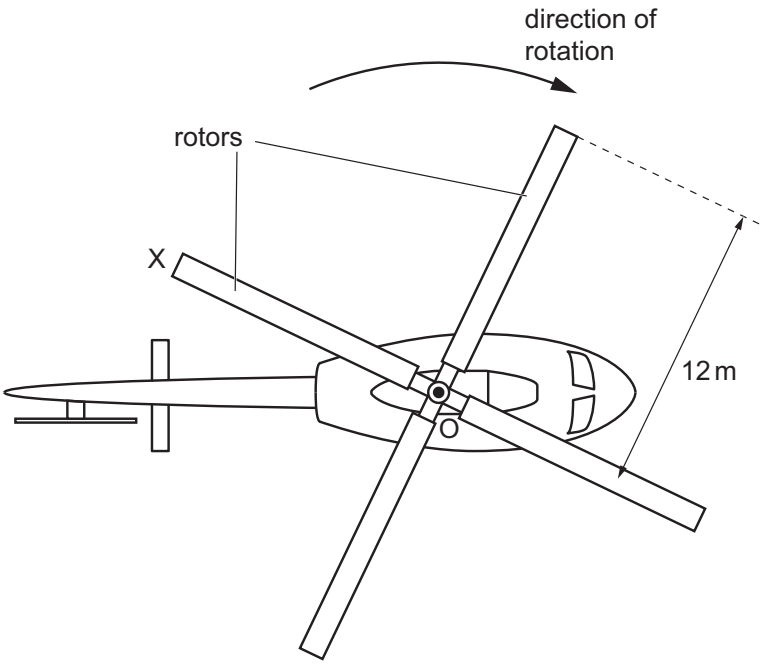


Fig. 7.1

The vertical component of the Earth's magnetic field at the helicopter is downwards with a flux density of 0.047 mT.

The rotors each rotate in a horizontal plane in the direction shown with a frequency of 85 Hz.

(i) Calculate the magnetic flux Φ cut by rotor OX during one complete rotation. Give a unit with your answer.

Φ = ... unit ..... [3]

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(ii) Determine the magnitude of the electromotive force (e.m.f.) induced across the length of rotor OX.

e.m.f. = ... V [2]

(iii) Use Lenz's law to explain whether end O or end X of the rotor is at the higher potential.

.....

.....

.....

..... [2]

[Total: 9]

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6 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) Figure 6.1 shows a beam of charged particles in a device known as a mass spectrometer.

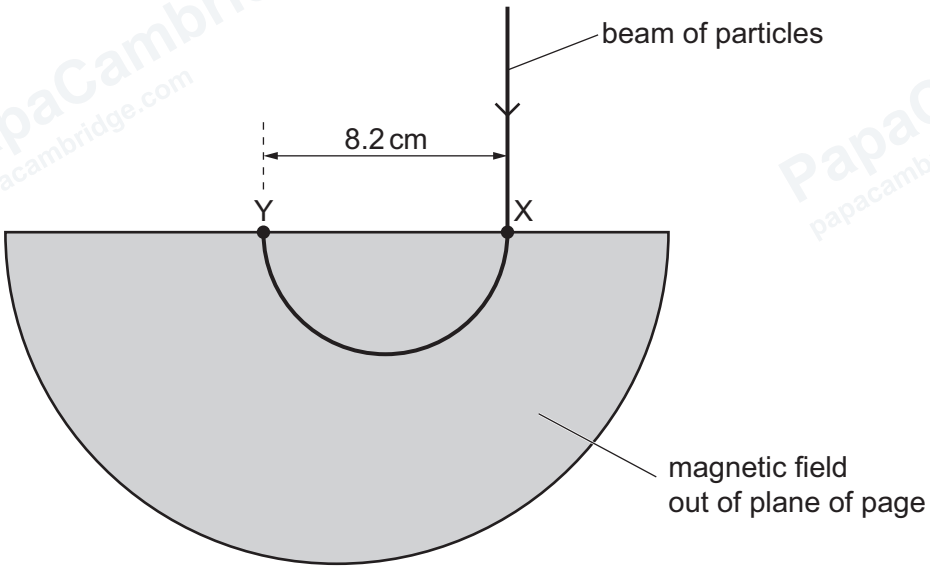


Figure 6.1

The shaded region contains a uniform magnetic field that is out of the plane of the page.

The particles enter the region of the magnetic field at point X and strike a detector at point Y.

(i) Explain whether the sign of the charge on the particles is positive or negative.

.....

.....

.....

..... [2]

(ii) Explain why the path followed by the beam of particles inside the magnetic field is circular.

.....

.....

..... [2]

(c) Each of the particles in **6(b)** has a mass of 2.0×10^{-26} kg and a charge of magnitude 3.2×10^{-19} C. All of the particles have a speed of $7.3 \times 10^{4}\,m\,s^{-1}$.

The distance XY is 8.2 cm.

(i) Show that the centripetal acceleration of one of the particles is $1.3 \times 10^{11}\,m\,s^{-2}$.

[2]

(ii) Use the information in **6(c)(i)** to determine the flux density B of the uniform magnetic field. Give a unit with your answer.

B = ... unit ..... [2]

(d) The particles in **6(b)** are replaced with particles that have half of the charge and a quarter of the mass of the particles in **6(b)**. The sign of the charge and the speed of the particles are both unchanged.

On Figure 6.1, draw the path of this beam of particles. [2]

[Total: 12]

7 A Hall probe consists of a slice of conducting material, as shown in Figure 7.1.

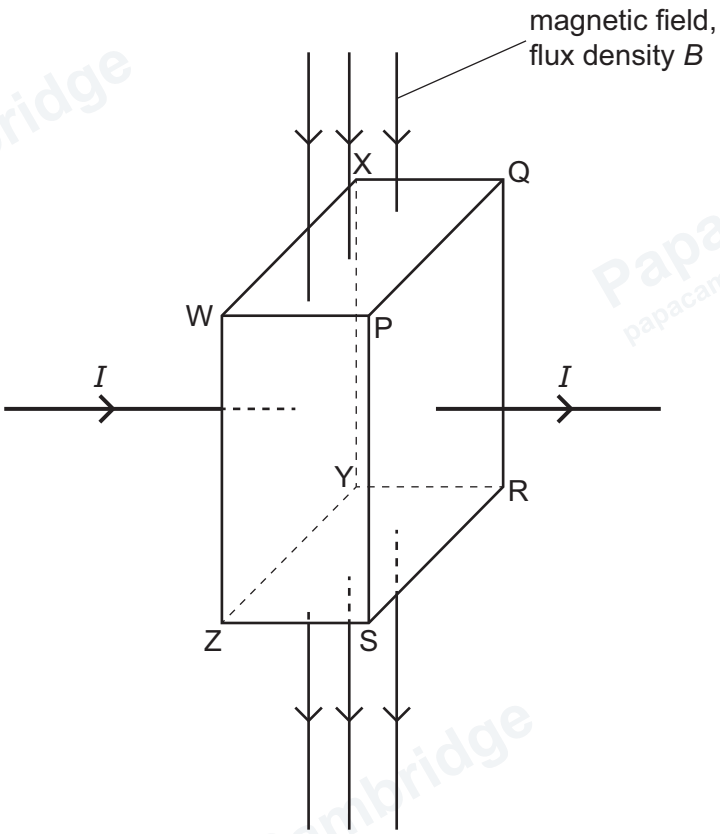


Figure 7.1

Current I enters the slice at face WXYZ and leaves it at face PQRS.

The slice is in a uniform magnetic field of flux density B. This magnetic field is perpendicular to faces WXQP and ZYRS.

A Hall voltage V_H is induced between two of the faces of the slice.

(a) (i) Using the letters P, Q, R, S, W, X, Y and Z, identify the faces of the slice between which the Hall voltage is induced.

..... and ..... [1]

(ii) The magnitude of the Hall voltage in Figure 7.1 is given by the equation

$$V_H = \frac{BI}{nqt}.$$

State the meanings of the symbols n, q and t in this equation.

n .....

q .....

t .....

[3]

(b) (i) Explain the origin of the Hall voltage.

.....

.....

.....

.....

.....

..... [3]

(ii) Identify which of the faces in **7(a)(i)** is at the higher potential. Explain your reasoning.

.....

.....

..... [1]

[Total: 8]

6 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) Figure 6.1 shows a beam of charged particles in a device known as a mass spectrometer.

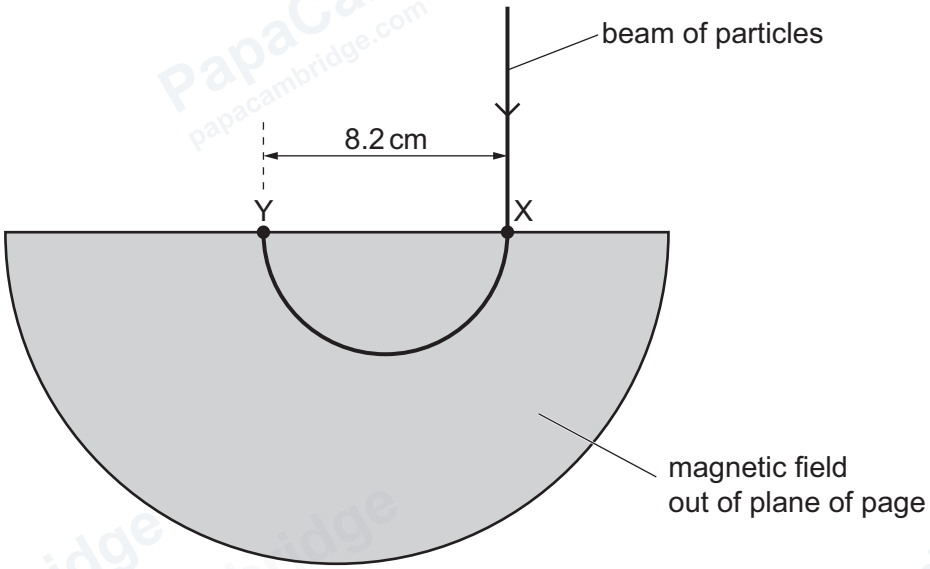


Figure 6.1

The shaded region contains a uniform magnetic field that is out of the plane of the page.

The particles enter the region of the magnetic field at point X and strike a detector at point Y.

(i) Explain whether the sign of the charge on the particles is positive or negative.

.....

.....

.....

..... [2]

(ii) Explain why the path followed by the beam of particles inside the magnetic field is circular.

.....

.....

..... [2]

(c) Each of the particles in **6(b)** has a mass of 2.0×10^{-26} kg and a charge of magnitude 3.2×10^{-19} C. All of the particles have a speed of 7.3×10^{4} m s^{-1}.

The distance XY is 8.2 cm.

(i) Show that the centripetal acceleration of one of the particles is 1.3×10^{11} m s^{-2}.

[2]

(ii) Use the information in **6(c)(i)** to determine the flux density B of the uniform magnetic field. Give a unit with your answer.

B = ... unit ..... [2]

(d) The particles in **6(b)** are replaced with particles that have half of the charge and a quarter of the mass of the particles in **6(b)**. The sign of the charge and the speed of the particles are both unchanged.

On Figure 6.1, draw the path of this beam of particles. [2]

[Total: 12]

7 **(a)** State what is meant by an alternating current.

.....

..... [1]

(b) A coil of wire is wound onto a hollow insulating tube to form a solenoid, as shown in Figure 7.1.

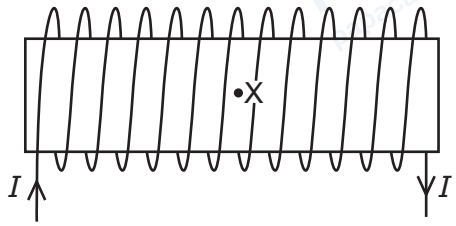


Figure 7.1

There is a constant current I in the solenoid in the direction indicated. There is a magnetic field inside and outside the solenoid due to this current.

(i) On Figure 7.1, draw five field lines to represent the magnetic field inside and outside the solenoid. [3]

(ii) On Figure 7.1, indicate, with the letters N and S, the magnetic poles of the solenoid. [1]

(c) The current I in the solenoid in **7(b)** now varies sinusoidally so that the flux density B of the magnetic field at point X varies with time t according to

$$B = 2.0 \cos 200\pi t$$

where B is in mT and t is in s.

(i) Show that the period of the alternating magnetic field is 0.010 s.

[1]

(ii) On Figure 7.2, sketch the variation of B with t between $t = 0$ and $t = 0.020\,\text{s}$.

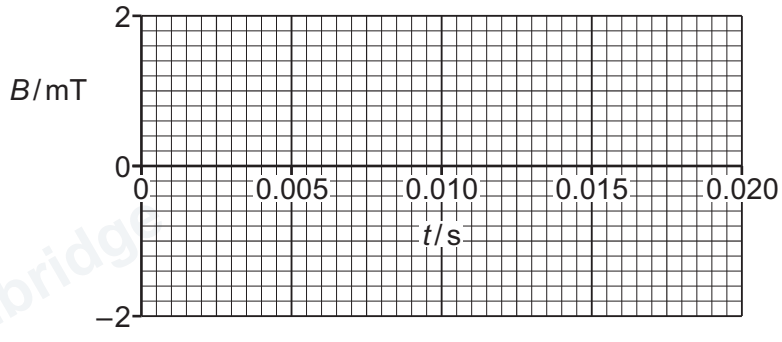


Figure 7.2

[2]

(d) A small coil is placed at point X inside the solenoid in **7(c)**, so that its axis is parallel to the axis of the solenoid.

The small coil has 45 turns, each with a radius of $2.8 \times 10^{-3}\,\text{m}$.

(i) Calculate the maximum magnetic flux Φ linked by the small coil. Give a unit with your answer.

Φ = ... unit ..... [3]

(ii) Describe and explain how the variation of B with t in Figure 7.2 may be used to determine the maximum electromotive force (e.m.f.) induced in the small coil. No calculations are required.

.....

.....

.....

.....

..... [3]

[Total: 14]

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21 Alternating currents

13 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 21 Alternating currents

考纲要求（9702 · 2025 – 2027 syllabus）

21.1

21.1.1 understand and use the terms period, frequency and peak value as applied to an alternating current or voltage

21.1.2 use equations of the form x = x0 sin ωt representing a sinusoidally alternating current or voltage

21.1.3 recall and use the fact that the mean power in a resistive load is half the maximum power for a sinusoidal alternating current

21.1.4 distinguish between root-mean-square (r.m.s.) and peak values and recall and use I r.m.s. = I0 / 2 and Vr.m.s. = V0 / 2 for a sinusoidal alternating current

21.2

21.2.1 distinguish graphically between half-wave and full-wave rectification

21.2.2 explain the use of a single diode for the half-wave rectification of an alternating current

21.2.3 explain the use of four diodes (bridge rectifier) for the full-wave rectification of an alternating current

21.2.4 analyse the effect of a single capacitor in smoothing, including the effect of the values of capacitance and the load resistance

4 **(a)** Three capacitors are connected as shown in Fig. 4.1.

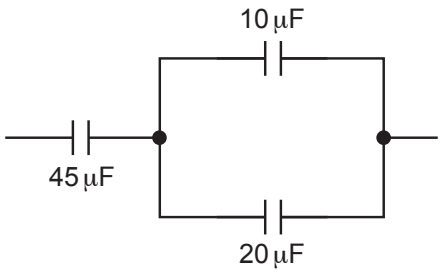


Fig. 4.1

Determine the total capacitance, in μF, of the network of three capacitors.

capacitance = ... μF [2]

(b) A capacitor of capacitance 45 μF is connected to a variable power supply initially set at 8.0 V. The output of the power supply increases so that the potential difference (p.d.) across the capacitor increases to 9.6 V.

Calculate the increase in energy ΔE stored in the capacitor.

ΔE = ... J [2]

(c) A sinusoidal a.c. power supply is connected to the input of a bridge rectifier.
The output of the rectifier is connected to a load resistor.

(i) Complete the circuit in Fig. 4.2 by adding a capacitor to smooth the p.d. across the load resistor.

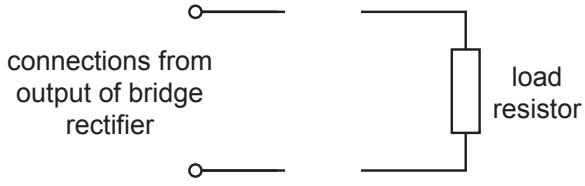


Fig. 4.2

[1]

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(ii) The variation with time t of the p.d. V of the smoothed output is shown in Fig. 4.3.

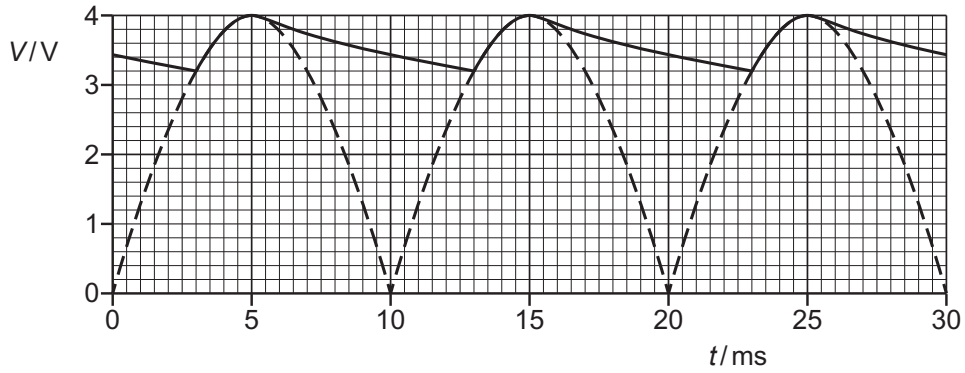


Fig. 4.3

Determine the time constant, in ms, of the smoothing circuit.

time constant = ... ms [3]

(d) A sinusoidal a.c. power supply has a maximum power of 16 W.

State the value of the mean power when the output of the power supply is:

(i) full-wave rectified

mean power = ... W [1]

(ii) half-wave rectified.

mean power = ... W [1]

[Total: 10]

7 A circuit contains a power supply that provides a sinusoidal alternating input voltage V_{IN}. There is an output voltage V_{OUT} across a load resistor R, as shown in Fig. 7.1.

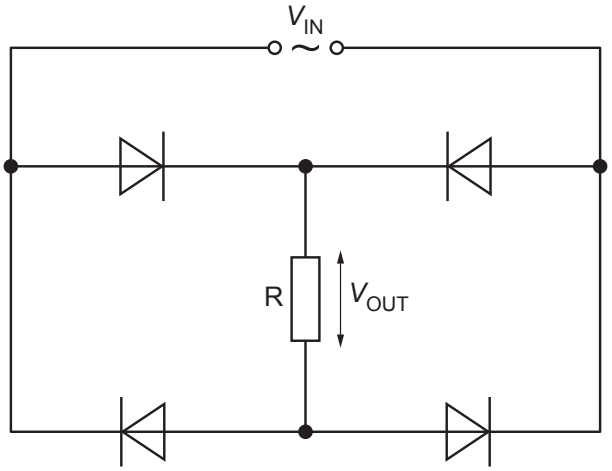


Fig. 7.1

(a) State the purpose of the circuit in Fig. 7.1.

.....

.....

..... [2]

(b) Fig. 7.2 shows the variation of V_{OUT} with time t.

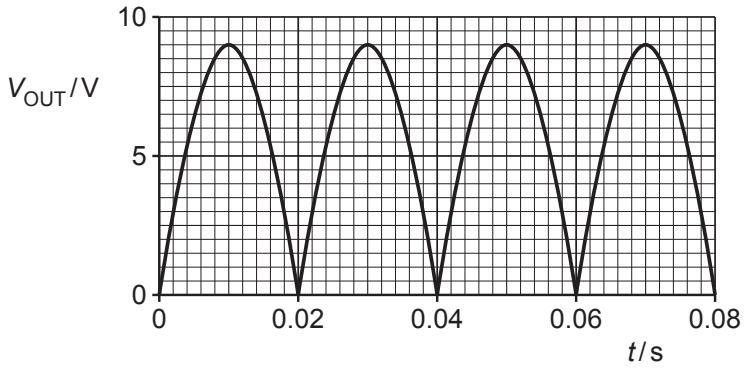


Fig. 7.2

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(i) The load resistor R has a resistance of 370 Ω.

Show that the maximum power dissipated in R is 0.22 W.

[2]

(ii) On Fig. 7.3, sketch the variation with *t* of the power *P* dissipated in R.

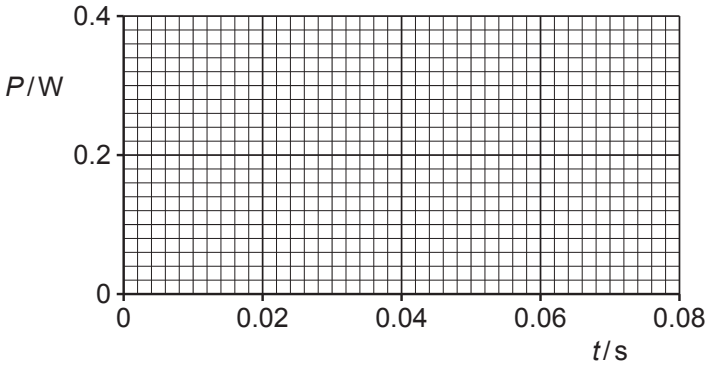


Fig. 7.3

[3]

(iii) Calculate the mean power dissipated in R.

mean power = ... W [1]

(c) The circuit of Fig. 7.1 is disconnected, and R is connected directly across the power supply.

Explain, without calculation, how the mean power now dissipated in R compares with the answer in **(b)(iii)**.

.....

.....

..... [2]

[Total: 10]

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6 **(a)** **(i)** State what is meant by rectification of an alternating voltage.

.....

..... [1]

(ii) State the difference between half-wave rectification and full-wave rectification.

.....

.....

..... [2]

(b) **(i)** Complete Fig. 6.1 to show a circuit that produces half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT} across the resistor R.

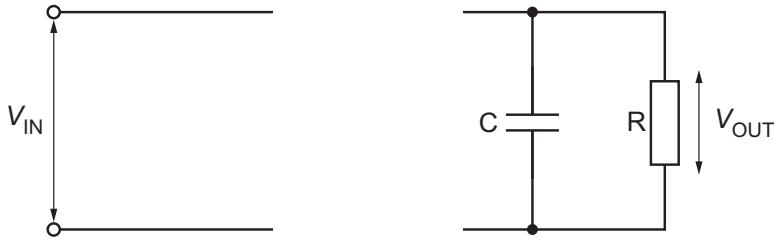


Fig. 6.1

[2]

(ii) State the purpose of the capacitor C in the circuit of Fig. 6.1.

.....

..... [1]

(c) The input voltage V_{IN} in Fig. 6.1 is a square wave. Fig. 6.2 shows the variation of V_{IN} with time t.

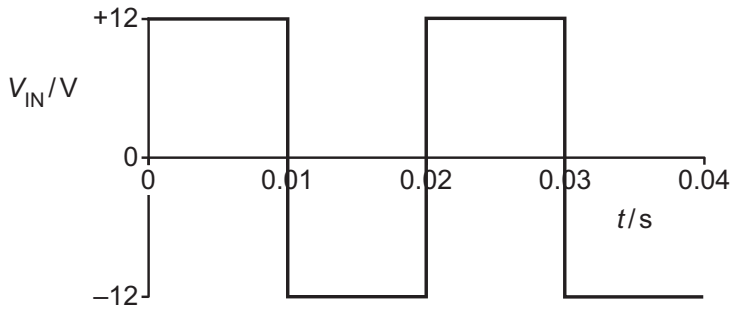


Fig. 6.2

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Fig. 6.3 shows the variation of V_{OUT} with t.

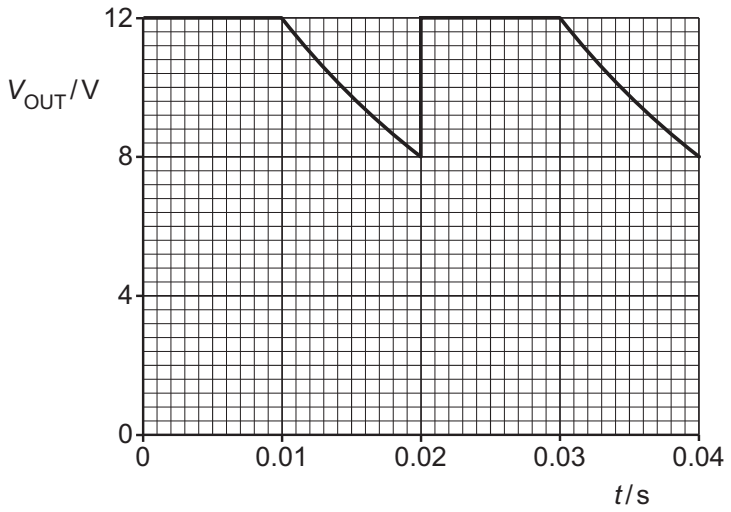


Fig. 6.3

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is 570 μF.

[2]

(ii) Determine the resistance of R.

resistance = ... Ω [3]

[Total: 11]

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8 **(a)** State what is meant by the frequency of an alternating current.

.....

..... [1]

(b) An alternating current I in a resistor of resistance 680 Ω varies with time t according to

$$I = 3.5\sin(40\pi t)$$

where I is in A and t is in s.

(i) Show that the period of the alternating current is 50 ms.

[1]

(ii) On Fig. 8.1, sketch the variation of I with t between $t = 0$ and $t = 100$ ms.

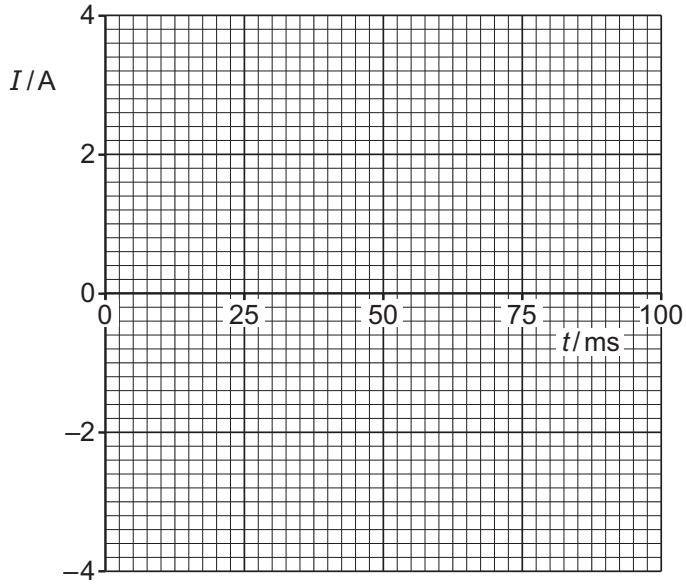


Fig. 8.1

[3]

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(iii) Determine the root-mean-square (r.m.s.) current in the resistor.

r.m.s. current = ...A [1]

(c) Use data from **(b)**, including your answer in **(b)(iii)**, to show by calculation that the mean power in the 680 Ω resistor is half of the peak power.

[3]

[Total: 9]

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6 Fig. 6.1 shows a circuit that rectifies an alternating input voltage V_{IN} and produces an output voltage V_{OUT} across a resistor R.

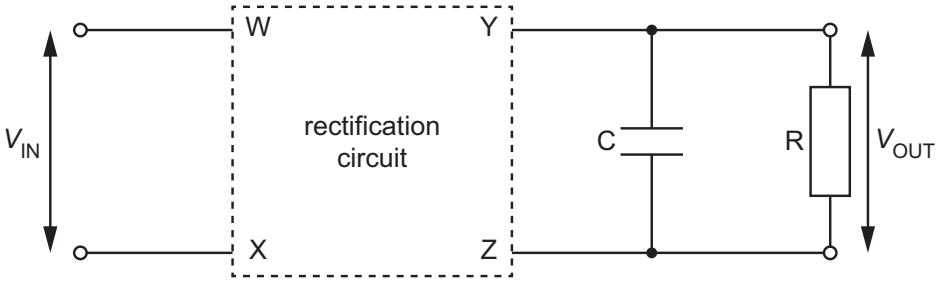


Fig. 6.1

The four terminals of the rectification circuit are labelled W, X, Y and Z.
A capacitor C is connected in parallel with resistor R.

(a) (i) State what is meant by rectification.

.....

..... [1]

(ii) State the purpose of capacitor C.

.....

..... [1]

(b) Fig. 6.2 shows the variations with time t of the potential differences (p.d.s) V_{IN} and V_{OUT}.

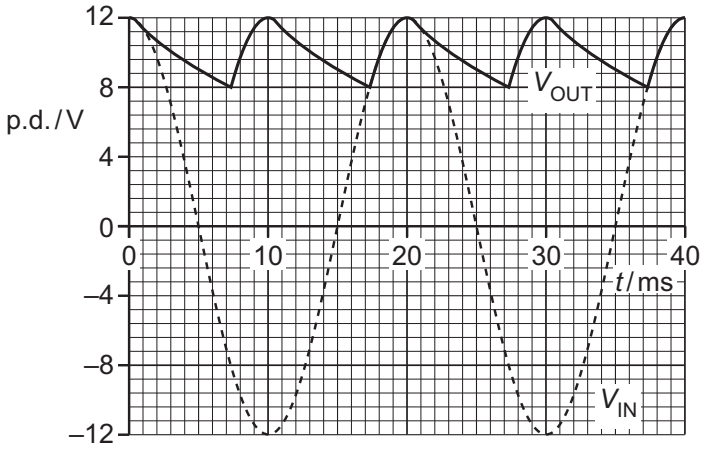


Fig. 6.2

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(i) The variation of V_{IN} with t can be represented by

$$V_{IN} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B. Give a unit with your answer for A.

A = ... unit .....

B = ... $rad\,s^{-1}$ [2]

(ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

(iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

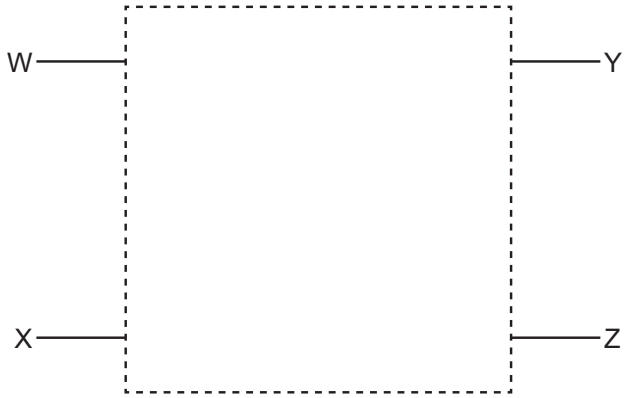


Fig. 6.3

[2]

(iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = ... s [3]

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(c) The capacitor C has a capacitance of 570 μF.

Use your answer in **(b)(iv)** to determine the resistance of resistor R.

resistance = ... Ω [2]

[Total: 12]

7 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) A particle of mass *m* and charge +*Q* moves at speed *v* into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

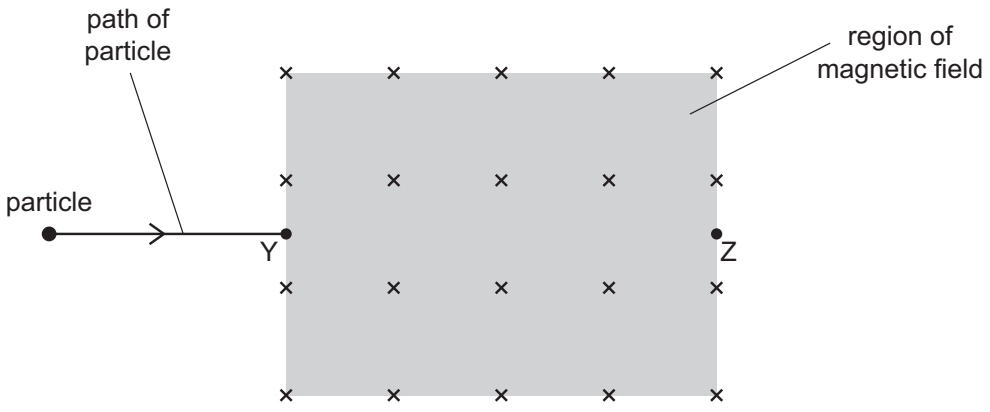


Fig. 7.1

The uniform magnetic field is into the page and has flux density *B*. The particle enters the region of the field at point Y.

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8 Fig. 8.1 shows a circuit that produces rectification of an alternating input voltage.

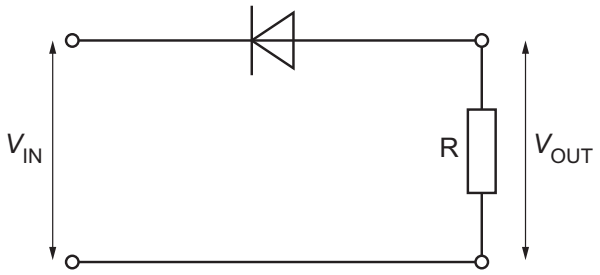


Fig. 8.1

The input voltage V_{IN} is sinusoidal. The rectified output voltage V_{OUT} is applied across resistor R.

The variation of V_{IN} with time t has amplitude V_0 and period T, as shown in Fig. 8.2.

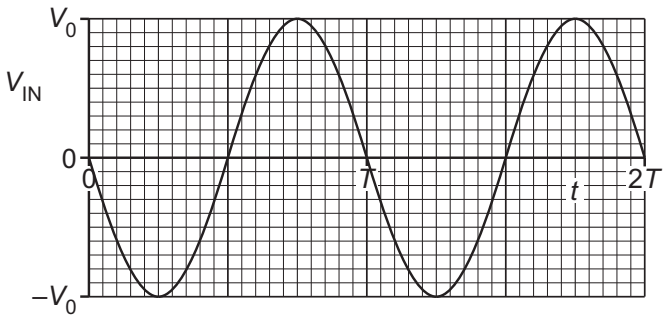


Fig. 8.2

The root-mean-square (r.m.s.) value of V_{IN} is 6.0 V.

(a) (i) State the type of rectification produced by the circuit of Fig. 8.1.

..... [1]

(ii) Calculate V_0.

V_0 = ... V [1]

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(b) Resistor R has resistance 45 Ω.

Assume that there is no p.d. across the diode when it is conducting.

(i) Determine the peak power P_0 in the resistor.

P_0 = ... W [2]

(ii) On Fig. 8.3, sketch the variation of the power P in the resistor with t between $t = 0$ and $t = 2T$.

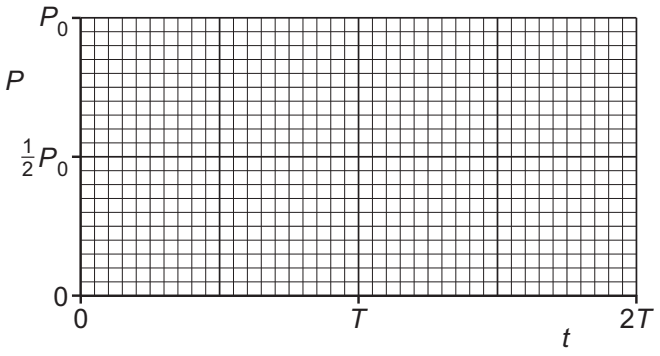


Fig. 8.3

[3]

(iii) Use the answer in **(b)(ii)** to explain why the mean power in the resistor is $\frac{1}{4}P_0$.

.....

.....

..... [2]

(iv) Use the information in **(b)(iii)** to determine the r.m.s. value of V_{OUT}.

r.m.s. voltage = ... V [1]

[Total: 10]

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6 Fig. 6.1 shows a circuit that rectifies an alternating input voltage V_{IN} and produces an output voltage V_{OUT} across a resistor R.

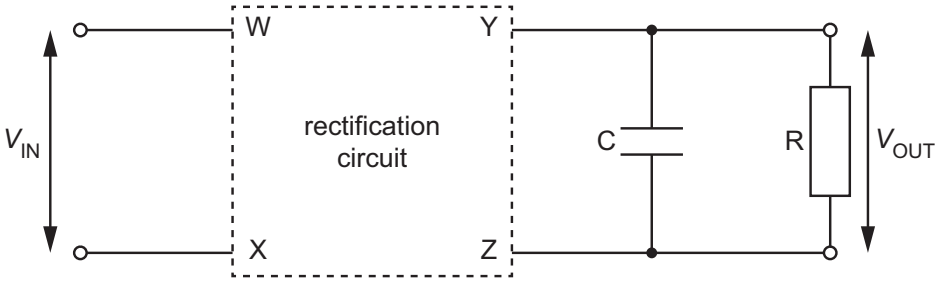


Fig. 6.1

The four terminals of the rectification circuit are labelled W, X, Y and Z.
A capacitor C is connected in parallel with resistor R.

(a) (i) State what is meant by rectification.

.....

..... [1]

(ii) State the purpose of capacitor C.

.....

..... [1]

(b) Fig. 6.2 shows the variations with time t of the potential differences (p.d.s) V_{IN} and V_{OUT}.

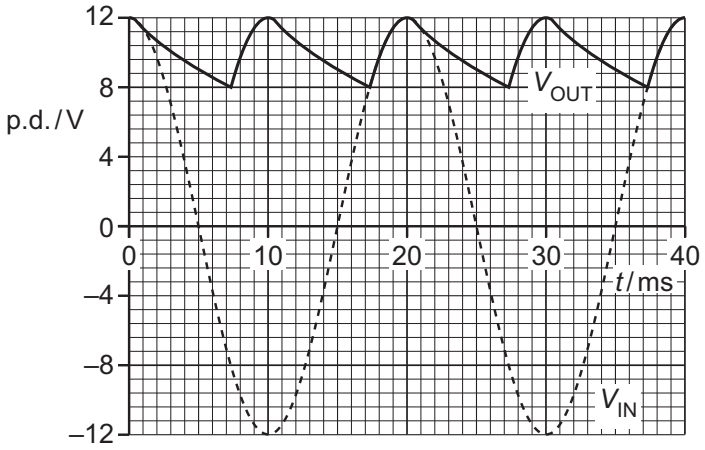


Fig. 6.2

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(i) The variation of V_{IN} with t can be represented by

$$V_{IN} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B. Give a unit with your answer for A.

A = ... unit .....

B = ... rad s^{-1} [2]

(ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

(iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

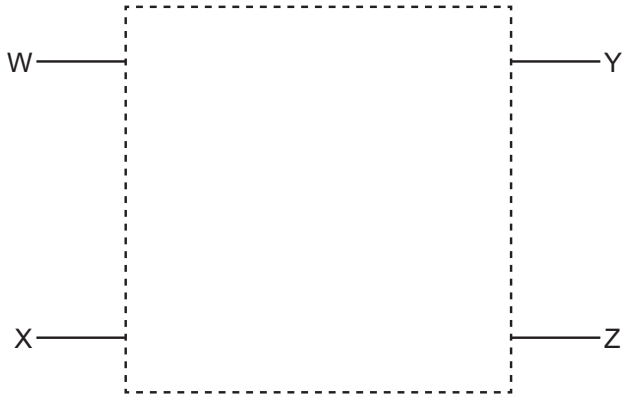


Fig. 6.3

[2]

(iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = ... s [3]

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(c) The capacitor C has a capacitance of 570 μF.

Use your answer in **(b)(iv)** to determine the resistance of resistor R.

resistance = ... Ω [2]

[Total: 12]

7 **(a)** Define magnetic flux density.

.....

.....

..... [2]

(b) A particle of mass *m* and charge +*Q* moves at speed *v* into a region where there is a uniform magnetic field, as shown in Fig. 7.1.

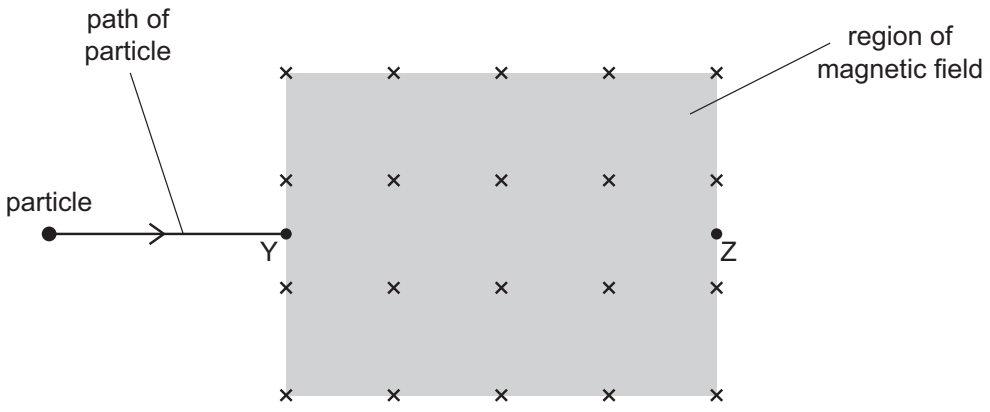


Fig. 7.1

The uniform magnetic field is into the page and has flux density *B*. The particle enters the region of the field at point Y.

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8 An incomplete circuit diagram of a bridge rectifier is shown in Fig. 8.1.

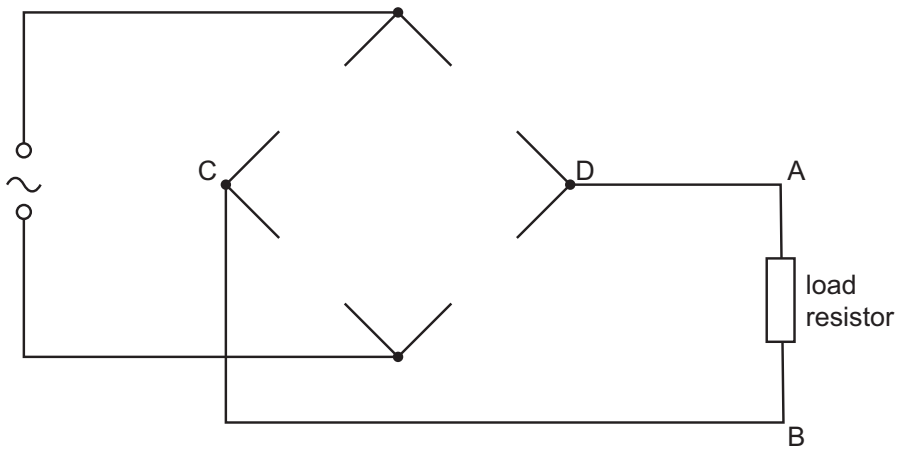


Fig. 8.1

(a) Complete Fig. 8.1 for the bridge rectifier such that the point A is at a positive potential with respect to point B. [2]

(b) The variation with time t of the potential difference (p.d.) V across the load resistor is shown in Fig. 8.2.

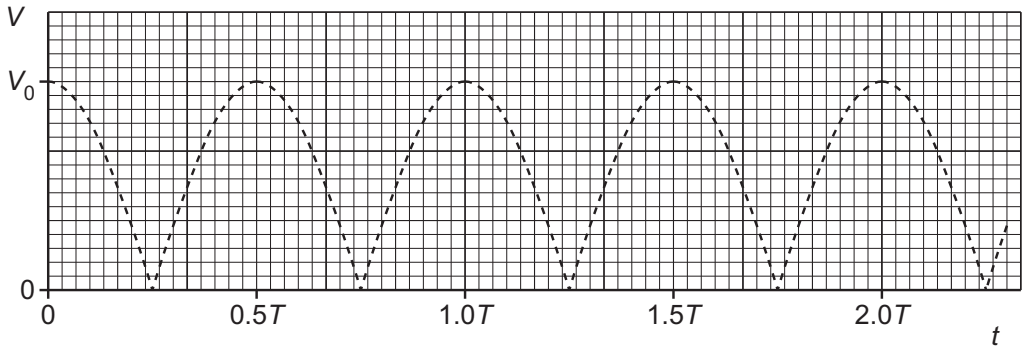


Fig. 8.2

A capacitor is now connected between points C and D of the bridge rectifier. This results in smoothing of the p.d. across the load resistor. The difference between the maximum and minimum values of the smoothed p.d. is 33% of the peak p.d. V_0.

(i) On Fig. 8.2, draw a line to show the variation of the potential difference V across the load resistor with time t. Your line should extend from $t = 0.5T$ to $t = 2.0T$. [3]

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(ii) Use your line in **(b)(i)** to determine, in terms of T, the time constant of the smoothing circuit.

time constant = ... T [3]

(iii) The resistance of the load resistor is now increased. The capacitance of the capacitor is unchanged.

State and explain the effect of this change on the smoothed output p.d.

.....

.....

..... [2]

[Total: 10]

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7 An alternating voltage V varies with time t according to

$$V = 18\cos 40\pi t$$

where V is in V and t is in s.

(a) For the alternating voltage:

(i) show that the period is 0.050 s

[1]

(ii) determine the root-mean-square (r.m.s.) voltage.

r.m.s. voltage = ... V [1]

(b) On Fig. 7.1, sketch the variation of V with t for values of t from $t = 0$ to $t = 100$ ms.

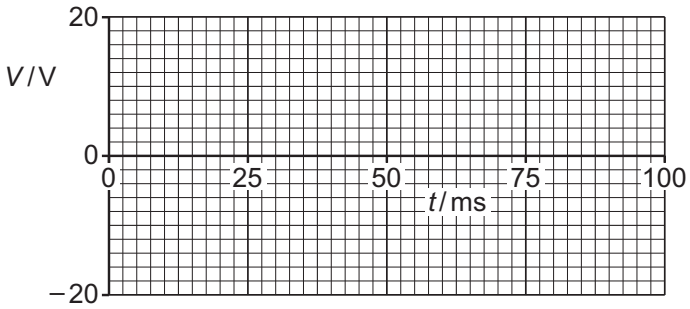


Fig. 7.1

[3]

(c) The alternating voltage is rectified to produce an output voltage across a load resistor R, as shown in Fig. 7.2.

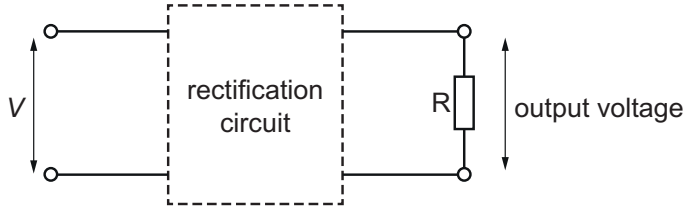


Fig. 7.2

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Fig. 7.3 shows the variation with t of the power P in the load resistor.

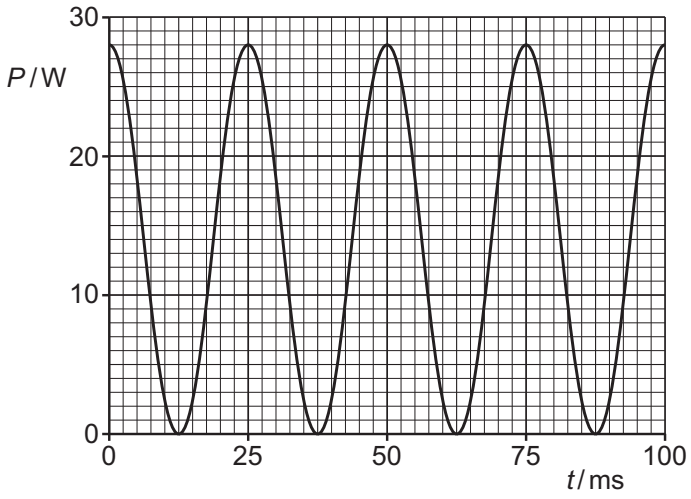


Fig. 7.3

State **three** conclusions that can be drawn from Fig. 7.3. The conclusions may be qualitative or quantitative. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 8]

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6 Fig. 6.1 shows part of a bridge rectifier circuit that can be used for rectification of an alternating input voltage V_{IN}.

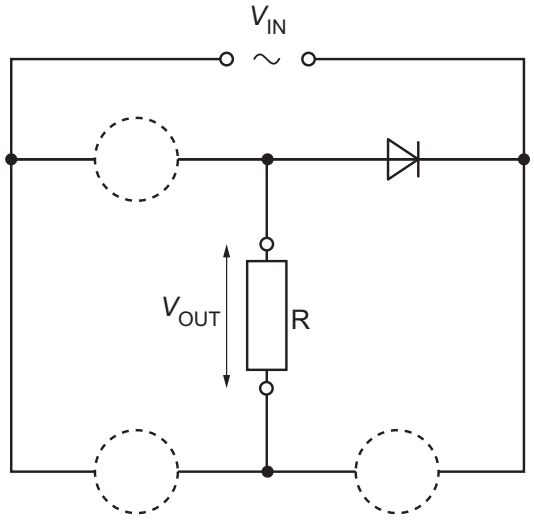


Fig. 6.1

The circuit contains four diodes, one of which is shown.
The rectified output voltage V_{OUT} is applied across load resistor R.

(a) (i) State what is meant by rectification.

.....

..... [1]

(ii) State the name of the type of rectification produced by a bridge rectifier circuit.

..... [1]

(iii) Complete the circuit in Fig. 6.1 by drawing the three missing diodes inside the dashed circles. [2]

(b) The input voltage varies with time t according to the equation

$$V_{IN} = 34 \sin 18t$$

where V_{IN} is in V and t is in s.

(i) Show that the period of the input voltage is 0.35 s.

[2]

(ii) Calculate the root-mean-square (r.m.s.) input voltage.

r.m.s. voltage = ... V [1]

(iii) On Fig. 6.2, sketch the variation of V_{OUT} with t from $t = 0$ to $t = 0.35$ s.

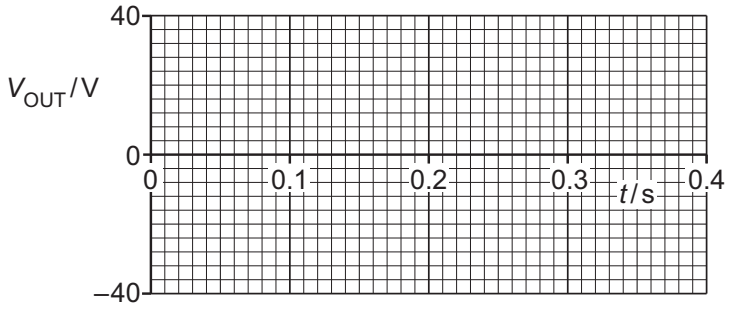


Fig. 6.2

[3]

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(c) Resistor R has a resistance of 56 kΩ. A capacitor of capacitance 12 μF is connected into the circuit of Fig. 6.1 in order to smooth the output voltage.

(i) On Fig. 6.1, draw the capacitor correctly connected into the circuit. [1]

(ii) Calculate the time constant of the smoothing circuit.

time constant = ... s [2]

(iii) During each discharge cycle, the time for which the capacitor is discharging is 0.14 s.

Determine the minimum value of the smoothed output voltage.

minimum voltage = ... V [2]

[Total: 15]

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7 **(a)** **(i)** State what is meant by half-wave rectification of an alternating current.

.....

.....

..... [2]

(ii) Figure 7.1 shows an incomplete diagram of the circuit used to achieve half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT}.

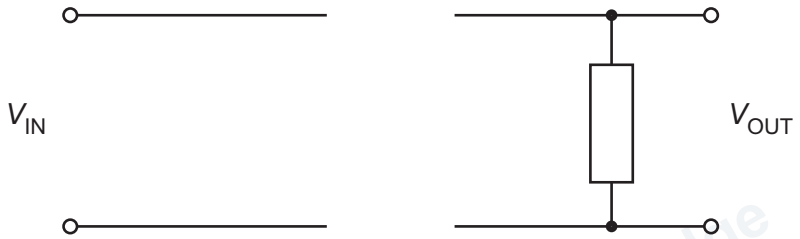


Figure 7.1

Complete the circuit diagram in Figure 7.1. [2]

(b) Two alternating voltages V_1 and V_2 have the same peak voltage V_0 and the same period T. One is sinusoidal and the other is a square wave, as shown by the variations with time t in Figure 7.2 and Figure 7.3.

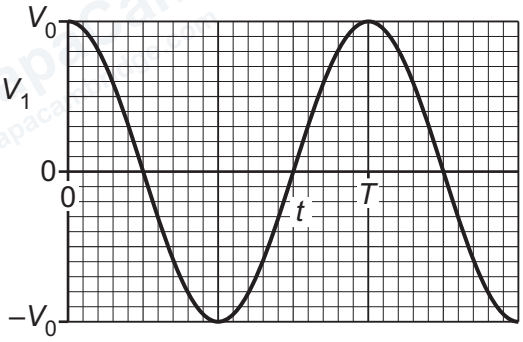


Figure 7.2

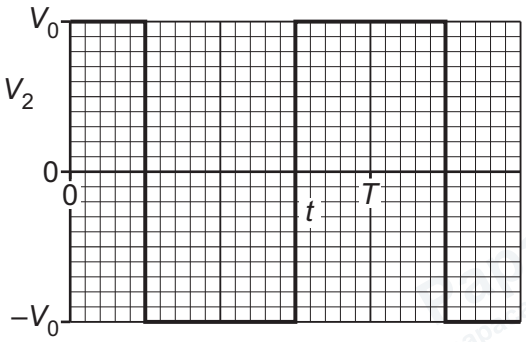


Figure 7.3

The alternating voltages are applied in turn to the input terminals in the completed circuit of Figure 7.1.

Complete Table 7.1 to show, in terms of V_0, expressions for the root-mean-square (r.m.s.) values of the two unrectified and the two half-wave rectified voltages.

Table 7.1

	r.m.s. value	
	V_1	V_2
unrectified		
half-wave rectified		

[4]

[Total: 8]

8 A sinusoidal alternating input voltage V_{IN} is supplied to a circuit that produces a rectified and smoothed output voltage V_{OUT}.

The variations with time t of V_{IN} and V_{OUT} are shown in Figure 8.1 and Figure 8.2 respectively.

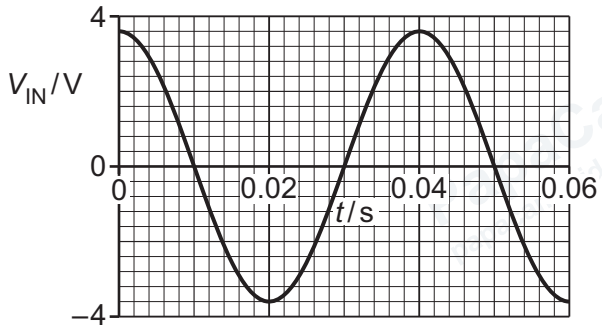


Figure 8.1

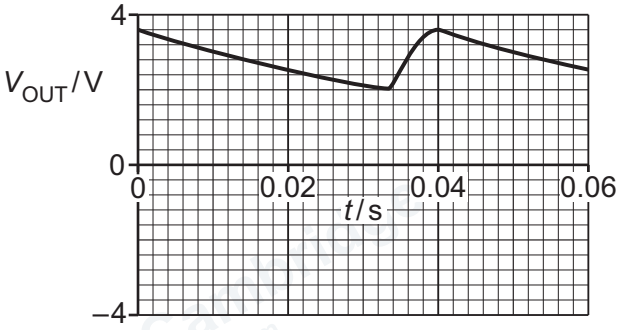


Figure 8.2

(a) (i) State the name of the component in the circuit that produces the rectification.

..... [1]

(ii) State the name of the component in the circuit that produces the smoothing.

..... [1]

(b) (i) Determine the peak value of V_{IN}.

peak value of V_{IN} = ... V [1]

(ii) Determine the period of V_{IN}.

period of V_{IN} = ... s [1]

(c) Apart from the peak value and period of V_{IN}, determine **three** other conclusions that can be drawn from Figure 8.1 and Figure 8.2 about the input and output voltages and about the circuit. The conclusions may be qualitative or quantitative. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 7]

7 **(a)** **(i)** State what is meant by half-wave rectification of an alternating current.

.....

.....

..... [2]

(ii) Figure 7.1 shows an incomplete diagram of the circuit used to achieve half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT}.



Figure 7.1

Complete the circuit diagram in Figure 7.1. [2]

(b) Two alternating voltages V_1 and V_2 have the same peak voltage V_0 and the same period T. One is sinusoidal and the other is a square wave, as shown by the variations with time t in Figure 7.2 and Figure 7.3.

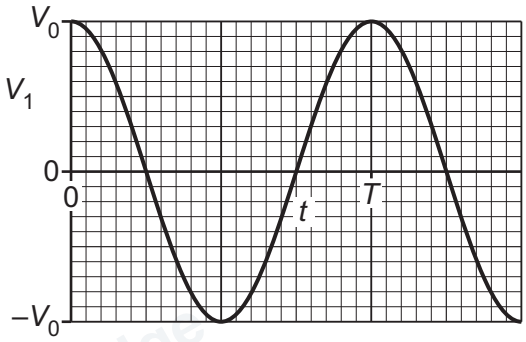


Figure 7.2

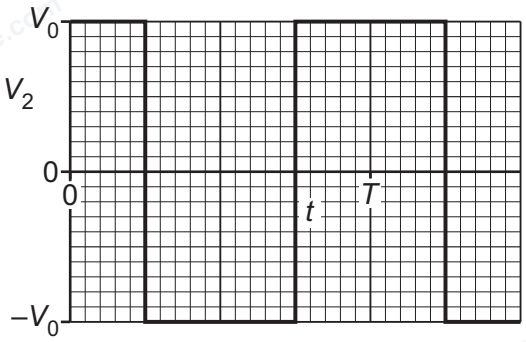


Figure 7.3

The alternating voltages are applied in turn to the input terminals in the completed circuit of Figure 7.1.

Complete Table 7.1 to show, in terms of V_0, expressions for the root-mean-square (r.m.s.) values of the two unrectified and the two half-wave rectified voltages.

Table 7.1

	r.m.s. value	
	V_1	V_2
unrectified		
half-wave rectified		

[4]

[Total: 8]

22 Quantum physics

14 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 22 Quantum physics

考纲要求（9702 · 2025 – 2027 syllabus）

22.1

22.1.1 understand that electromagnetic radiation has a particulate nature

22.1.2 understand that a photon is a quantum of electromagnetic energy

22.1.3 recall and use E = hf

22.1.4 use the electronvolt (eV) as a unit of energy

22.1.5 understand that a photon has momentum and that the momentum is given by p = E / c

22.2

22.2.1 understand that photoelectrons may be emitted from a metal surface when it is illuminated by electromagnetic radiation

22.2.2 understand and use the terms threshold frequency and work function

22.2.3 explain photoelectric emission in terms of photon energy and work function energy

22.2.4 recall and use $hf = \Phi + \frac{1}{2}mv^2_{max}$

22.2.5 explain why the maximum kinetic energy of photoelectrons is independent of intensity, whereas the photoelectric current is proportional to intensity

22.3

22.3.1 understand that the photoelectric effect provides evidence for a particulate nature of electromagnetic radiation while phenomena such as interference and diffraction provide evidence for a wave nature

22.3.2 describe and interpret qualitatively the evidence provided by electron diffraction for the wave nature of particles

22.3.3 understand the de Broglie wavelength as the wavelength associated with a moving particle

22.3.4 recall and use $\lambda = h / p$

22.4

22.4.1 understand that there are discrete electron energy levels in isolated atoms (e.g. atomic hydrogen)

22.4.2 understand the appearance and formation of emission and absorption line spectra

22.4.3 recall and use hf = E1 – E2

7 **(a)** A photon has an energy of 3.11×10^{-19} J.

Calculate the momentum of the photon.

momentum = ... N s [2]

(b) A laser beam has a power of 350 mW. The light from the laser has a wavelength of 640 nm.

(i) Determine the number of photons emitted by the laser in a time of 1.0 s.

number = ... [2]

(ii) The laser beam is incident normally on a surface that absorbs all of the photons.

Show that the force F exerted on the surface by the laser beam is given by

$$F = \frac{P}{c}$$

where P is the power of the laser beam and c is the speed of light.

[2]

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(c) Light of a single wavelength is incident on the surface of different metals. The work function energy of the metals is given in Table 7.1.

Table 7.1

metal	work function energy / eV
tungsten	4.49
magnesium	3.68
potassium	2.26

(i) Explain the term threshold wavelength.

.....

.....

..... [1]

(ii) For the metals in Table 7.1, calculate the value of the largest threshold wavelength.

threshold wavelength = ... m [2]

[Total: 9]

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8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) Fig. 8.1 shows a tube in which X-rays are produced at a metal target.

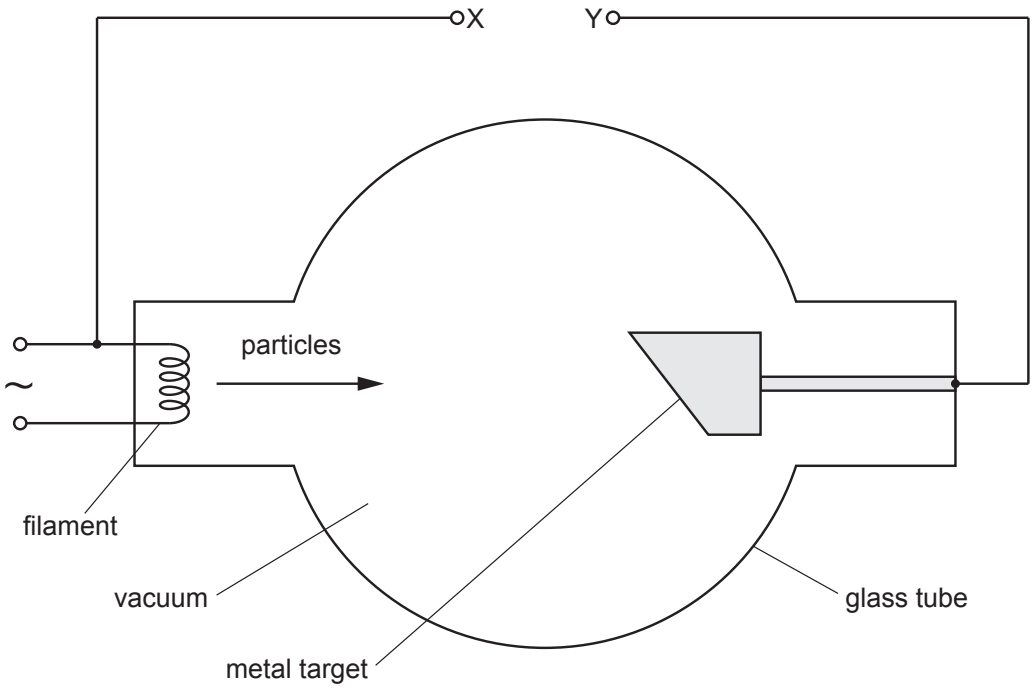


Fig. 8.1

Particles are accelerated from the filament to the target by a constant high voltage applied across the terminals X and Y.

(i) State the name of the particles.

..... [1]

(ii) On Fig. 8.1, use + and – signs to label terminals X and Y to indicate the polarity of the high voltage. [1]

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(c) For an accelerating voltage of 32 kV in Fig. 8.1, determine:

(i) the maximum energy, in MeV, of an X-ray photon produced at the target

maximum photon energy = ... MeV [1]

(ii) the maximum momentum of an X-ray photon produced at the target

maximum photon momentum = ... N s [2]

(iii) the minimum wavelength of X-rays produced at the target.

minimum wavelength = ... m [3]

(d) Explain why X-rays can be used to produce images of internal body structures that have good contrast.

.....

.....

.....

.....

..... [3]

[Total: 13]

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8 A polished sheet of magnesium in a vacuum emits electrons when it is illuminated by ultraviolet radiation.

(a) State the name of this phenomenon.

..... [1]

(b) For emission of electrons to occur, the frequency of the ultraviolet radiation must be at least 8.8×10^{14} Hz.

(i) Calculate the work function energy of magnesium.

work function energy = ... J [2]

(ii) For ultraviolet radiation with a frequency of 11×10^{14} Hz, calculate the maximum speed of the emitted electrons.

maximum speed = ... $m\,s^{-1}$ [3]

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(c) The frequency f of the ultraviolet radiation incident on the magnesium sheet is varied between 8.0×10^{14} Hz and 11×10^{14} Hz.

On Fig. 8.1, sketch the variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons. Use the space below for any working that you need.

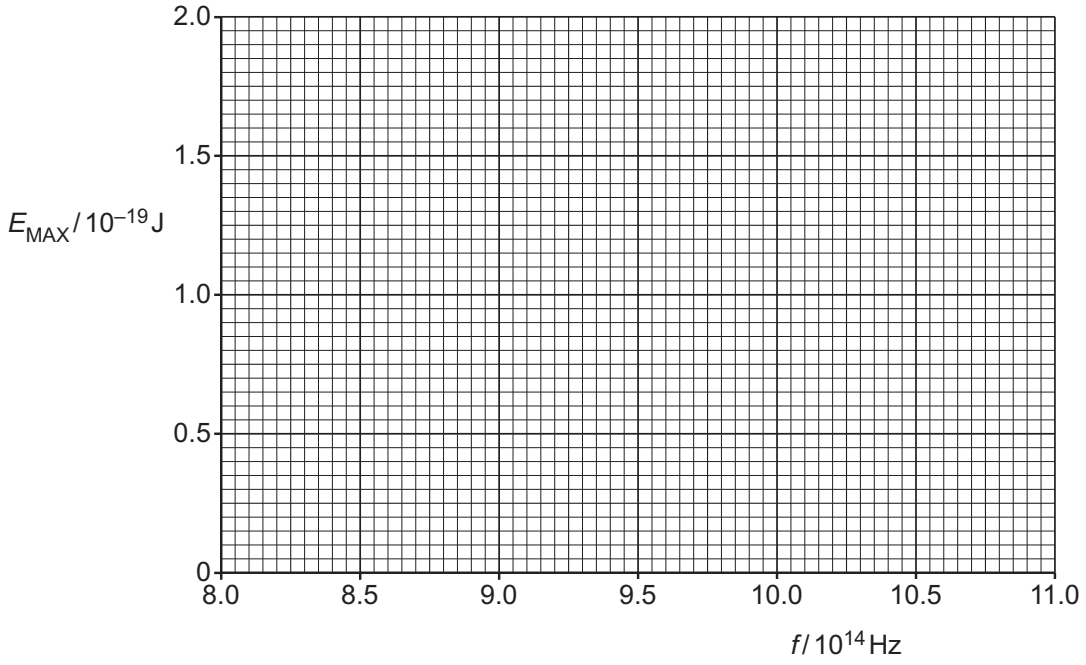


Fig. 8.1

[3]

[Total: 9]

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9 Electrons in a vacuum are accelerated from rest through a potential difference (p.d.) V to form a beam. The electrons each have mass m and charge q.

The beam is incident on a graphite crystal that acts as a diffraction grating. After passing through the crystal, the beam reaches a fluorescent screen. An interference pattern is observed on this screen.

(a) Explain what this observation shows about the nature of electrons.

.....

.....

..... [1]

(b) Determine an expression, in terms of m, q and V, for the momentum p of an electron in the beam.

p = ... [3]

(c) The p.d. through which the electrons are accelerated is now increased to a greater value.

Describe and explain the effect of this change on the interference pattern observed.

.....

.....

..... [2]

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(d) The electrons are now accelerated through different values of V, resulting in pairs of corresponding values for p and the de Broglie wavelength λ.

(i) On Fig. 9.1, sketch the variation of p with $\frac{1}{\lambda}$.

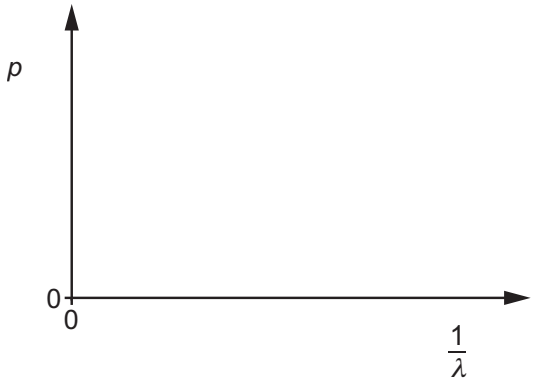


Fig. 9.1

[2]

(ii) State the name of the quantity represented by the gradient of the line in Fig. 9.1.

..... [1]

[Total: 9]

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8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) A laser emits red light of a single wavelength. The light is produced when electrons move from a higher energy level to a lower energy level. The difference in energy between the two levels is 1.96 eV.

(i) Calculate the wavelength of the light.

wavelength = ... m [3]

(ii) The power of the beam emitted by the laser is 1.0×10^{-2} W.

Calculate the number of photons emitted per unit time by the laser.

number per unit time = ... s^{-1} [1]

(iii) The photons are incident normally on a surface. Half of the number of photons are absorbed by the surface, and half are reflected.

Determine the average force exerted by the beam of photons on the surface.

average force = ... N [4]

[Total: 10]

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[Turn over

8 **(a)** State what is meant by the de Broglie wavelength.

.....

..... [1]

(b) Calculate the de Broglie wavelength of an electron moving at a speed of $4.9 \times 10^7\,\text{m s}^{-1}$.

wavelength = ... m [2]

(c) State **one** similarity and **one** difference between an electron and a positron.

similarity: .....

.....

difference: .....

.....
[2]

(d) An electron moving at a speed of $4.9 \times 10^7\,\text{m s}^{-1}$ collides with a positron that is travelling at the same speed in the opposite direction. As a result of the collision, two gamma-ray photons are produced.

(i) State the name of this type of reaction.

..... [1]

(ii) State what happens to the electron and to the positron.

.....

.....

..... [2]

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(iii) Explain why two gamma-ray photons are produced, rather than just one.

.....

..... [1]

(iv) Show that the kinetic energy of the electron before the collision is 1.1×10^{-15} J.

[1]

(v) Use the information in **(d)(iv)** to determine, to three significant figures, the wavelength associated with the gamma radiation emitted in the collision.

wavelength = ... m [3]

[Total: 13]

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9 **(a)** State what is meant by the photoelectric effect.

.....

.....

..... [2]

(b) The photoelectric effect is investigated in two stages using the circuit shown in Fig. 9.1.

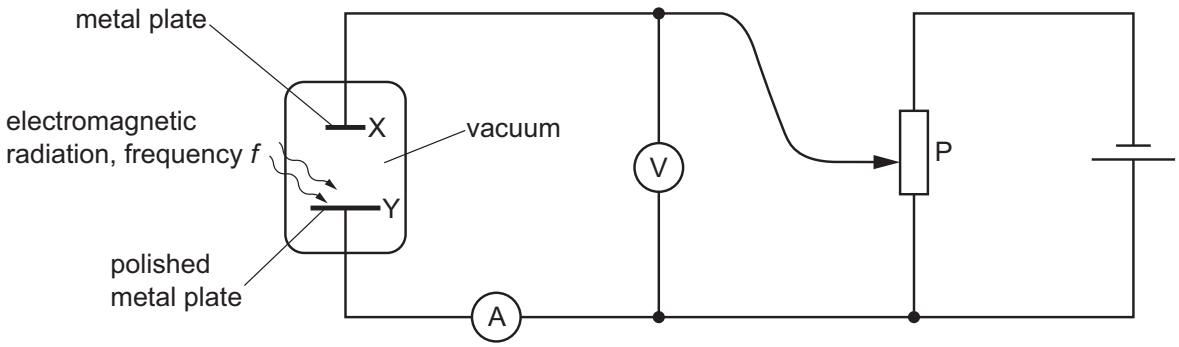


Fig. 9.1

The polished metal plate Y is illuminated with electromagnetic radiation of frequency f and constant power.

In stage 1 of the investigation, frequency f is set to a constant value of 2.5×10^{15} Hz. The current I in the ammeter is varied by adjusting the potentiometer P. Fig. 9.2 shows the variation of I with the voltmeter reading V. There is a value V_S of V at which the current just falls to zero.

In stage 2 of the investigation, stage 1 is repeated for different values of frequency. As frequency f is varied, the voltmeter reading V_S at which the current just falls to zero is measured. Fig. 9.3 shows the variation of V_S with f.

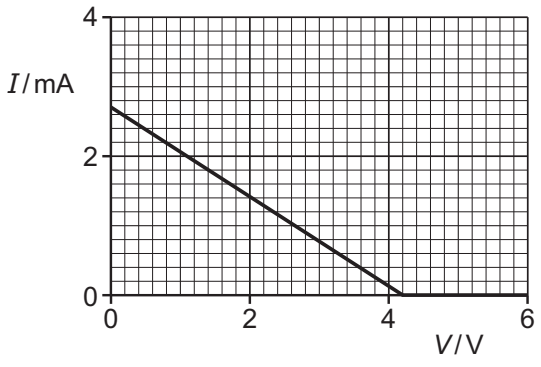


Fig. 9.2

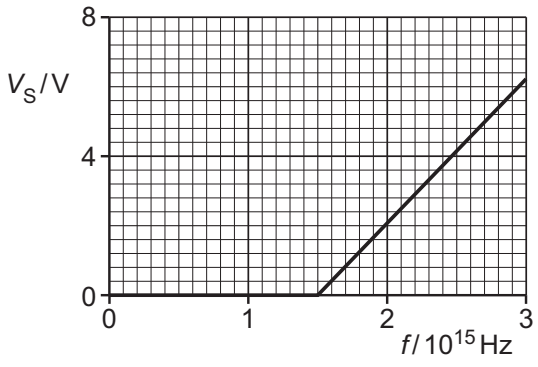


Fig. 9.3

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(i) Explain, with reference to photons, why V_S depends on the frequency of the incident electromagnetic radiation.

.....

.....

.....

.....

.....

..... [3]

(ii) State **three** quantitative conclusions that can be drawn from the results in Fig. 9.2 and Fig. 9.3. Use the space for any working.

1 .....

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2 .....

.....

3 .....

.....

[3]

[Total: 8]

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8 **(a)** State what is meant by the de Broglie wavelength.

.....

..... [1]

(b) Calculate the de Broglie wavelength of an electron moving at a speed of $4.9 \times 10^7\,\text{m s}^{-1}$.

wavelength = ... m [2]

(c) State **one** similarity and **one** difference between an electron and a positron.

similarity: .....

.....

difference: .....

.....
[2]

(d) An electron moving at a speed of $4.9 \times 10^7\,\text{m s}^{-1}$ collides with a positron that is travelling at the same speed in the opposite direction. As a result of the collision, two gamma-ray photons are produced.

(i) State the name of this type of reaction.

..... [1]

(ii) State what happens to the electron and to the positron.

.....

.....

..... [2]

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(iii) Explain why two gamma-ray photons are produced, rather than just one.

.....

..... [1]

(iv) Show that the kinetic energy of the electron before the collision is 1.1×10^{-15} J.

[1]

(v) Use the information in **(d)(iv)** to determine, to three significant figures, the wavelength associated with the gamma radiation emitted in the collision.

wavelength = ... m [3]

[Total: 13]

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9 **(a)** **(i)** Describe what is meant by wave–particle duality.

.....

.....

..... [2]

(ii) State the relationship between the de Broglie wavelength λ of a particle and its momentum p. State the meaning of any other symbols that you use.

.....

..... [2]

(b) A narrow beam of electrons, all with the same speed, is incident normally on a carbon film. The electrons then move on to a fluorescent screen, as illustrated in Fig. 9.1.

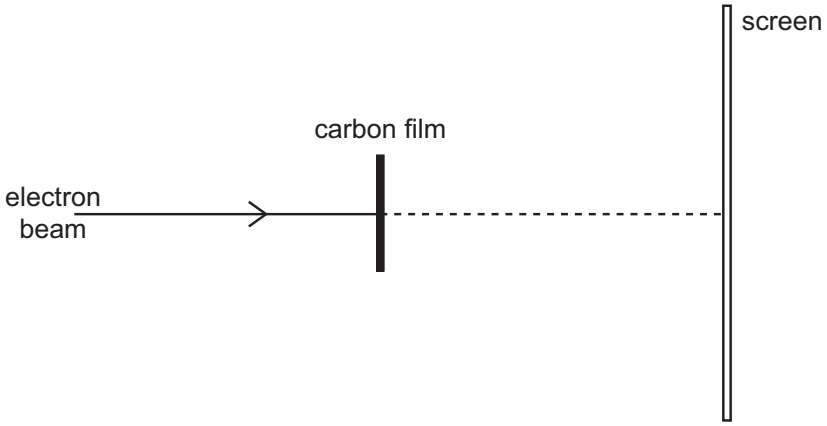


Fig. 9.1

The apparatus is in a vacuum.
The pattern produced on the screen is shown in Fig. 9.2.

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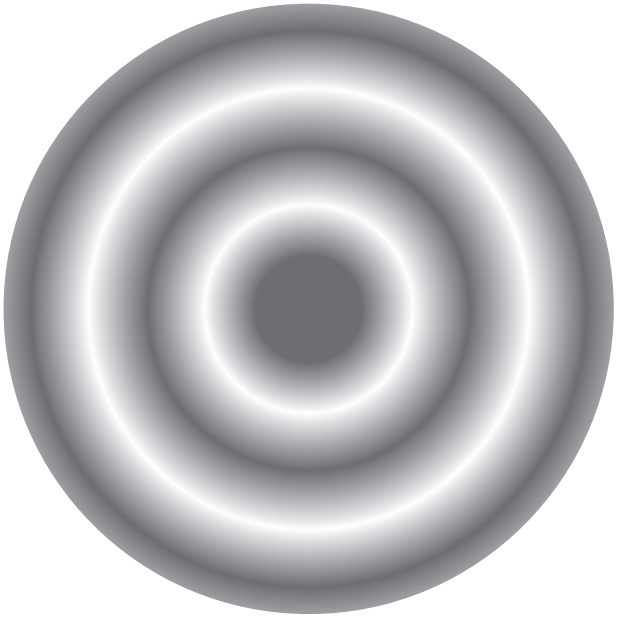


Fig. 9.2 (not to scale)

(i) Explain why the pattern in Fig. 9.2 provides experimental evidence to indicate a wave nature for the electrons.

.....

.....

..... [2]

(ii) The speed of the electrons is increased.

Suggest, with a reason, how this change affects the pattern observed on the screen.

.....

.....

..... [2]

[Total: 8]

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8 Fig. 8.1 shows the three lowest-frequency lines in the part of the emission spectrum for hydrogen that relates to electron transitions to the ground state (level $n = 1$).

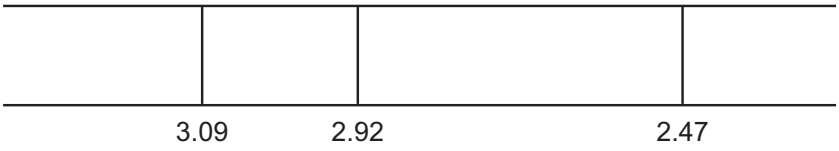


Fig. 8.1 (not to scale)

The numbers represent the frequencies, in 10^{15} Hz, associated with the spectral lines.

(a) Use the photon model of electromagnetic radiation to explain how the existence of spectral lines in the emission spectrum provides evidence for discrete electron energy levels in the hydrogen atom.

.....

.....

.....

.....

..... [3]

(b) The energy of the ground state (level $n = 1$) in a hydrogen atom is –13.6 eV.

(i) Calculate the energy, in J, of the ground state.

energy = ... J [1]

(ii) Show that the energy difference between levels $n = 1$ and $n = 2$ is 10.2 eV.

[2]

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(iii) Complete Table 8.1 to show the energy differences from the ground state, and the energies of the levels up to $n = 4$, in the hydrogen atom. Use the space for any working.

Table 8.1

level	(energy difference from $n = 1$) / eV	energy / eV
$n = 4$		
$n = 3$		
$n = 2$	10.2	
$n = 1$	0.0	–13.6

[4]

[Total: 10]

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9 **(a)** State what is meant by the photoelectric effect.

.....

.....

..... [2]

(b) The photoelectric effect is investigated using two clean metal plates. One plate is made from metal X and the other is made from metal Y.

Metal X has work function energy Φ. Metal Y has work function energy 2Φ.
Metal X has threshold frequency F.

State expressions, in terms of either or both of Φ and F, for

(i) the threshold frequency of metal Y

threshold frequency = ... [1]

(ii) the Planck constant.

Planck constant = ... [1]

(c) The maximum kinetic energy E_K of photoelectrons is determined for each of the plates in **(b)** for different frequencies f of incident radiation.

On Fig. 9.1, sketch the variation of E_K with f for each plate. Label your lines X and Y to identify which line relates to which plate.

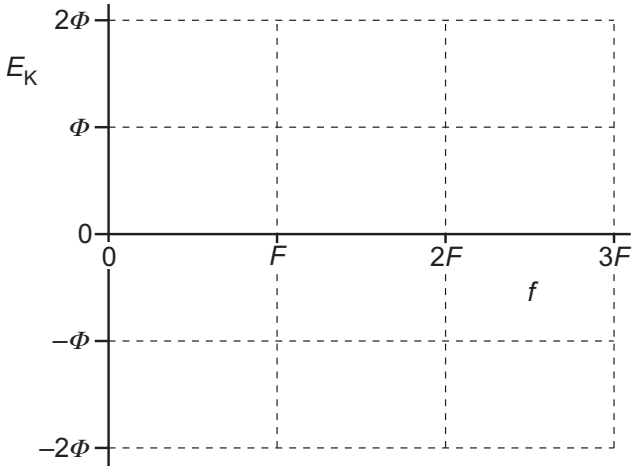


Fig. 9.1

[4]

[Total: 8]

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9 **(a)** State what is meant by the de Broglie wavelength.

.....

.....

..... [1]

(b) Complete Table 9.1 to identify experimental evidence for the wave nature and the particulate nature of matter and of electromagnetic radiation.

Table 9.1

	evidence for	
	wave nature	**particle nature**
matter	....	....
electromagnetic radiation	....	....

[4]

(c) An electron in a vacuum is accelerated from rest by a potential difference (p.d.) of 3.5 kV.

For the accelerated electron, calculate

(i) the speed

speed = ... $m\,s^{-1}$ [3]

(ii) the de Broglie wavelength.

wavelength = ... m [2]

[Total: 10]

8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) X-rays for use in medical diagnosis are produced by accelerating charged particles towards a metal target.

(i) State the name of the charged particles.

..... [1]

(ii) Describe how X-rays are formed at the target.

.....

.....

..... [2]

(iii) For any given accelerating voltage, there is a range of wavelengths of X-rays produced that has a minimum value.

Explain, with reference to photons, why there is a range of wavelengths and why this range has a minimum value.

.....

.....

.....

.....

..... [3]

(iv) Calculate the minimum wavelength of X-rays produced when the accelerating voltage is 52.0 kV.

minimum wavelength = ... m [3]

(c) X-rays are used to create an image of the internal structure of an object that is made from two types of material of similar thickness.

Describe and explain how a property of the materials determines whether or not the image produced has good contrast.

.....

.....

.....

..... [2]

[Total: 13]

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8 **(a)** When a clean metal surface is illuminated by electromagnetic radiation, electrons may be emitted from the surface.

(i) State the name of this phenomenon.

..... [1]

(ii) State what is meant by the threshold frequency of the metal.

.....

..... [1]

(iii) Explain why the existence of the threshold frequency provides evidence for the photon model of electromagnetic radiation.

.....

.....

.....

.....

..... [3]

(b) When a clean metal sheet is illuminated by radiation of wavelength 254 nm, electrons are emitted with a maximum kinetic energy of 1.91×10^{-19} J.

When the same metal sheet is illuminated by radiation of wavelength 178 nm, the maximum kinetic energy of the emitted electrons is 5.27×10^{-19} J.

(i) Use these data to calculate a value, to three significant figures, for the Planck constant.

Planck constant = ... J s [3]

(ii) Determine a value, in eV, for the work function energy of the metal. Give your answer to two significant figures.

work function energy = ... eV [3]

[Total: 11]

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23 Nuclear physics

16 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 23 Nuclear physics

考纲要求（9702 · 2025 – 2027 syllabus）

23.1

23.1.1 understand the equivalence between energy and mass as represented by E = mc2 and recall and use this equation

23.1.2 represent simple nuclear reactions by nuclear equations of the form N He O H

23.1.3 understand the concept of mass defect, and that it is the difference between the total mass of a nucleus and the total mass of its constituent nucleons

23.1.4 understand the origin of nuclear binding energy and recall and use $\Delta E = \Delta mc^2$

23.1.5 understand that binding energy per nucleon is a measure of the stability of a nucleus

23.1.6 understand and interpret the graph of binding energy per nucleon against nucleon number

23.1.7 use the binding energy per nucleon curve to explain the release of energy in nuclear fission and nuclear fusion

23.2

23.2.1 understand that fluctuations in count rate provide evidence for the random nature of radioactive decay

23.2.2 understand that radioactive decay is both spontaneous and random

23.2.3 define activity and decay constant, and recall and use $A = \lambda N$

23.2.4 define half-life

23.2.5 use $\lambda = 0.693 / t$

23.2.6 understand the exponential nature of radioactive decay, and sketch and use the relationship x = x0e–λt, where x could represent activity, number of undecayed nuclei or received count rate

8 **(a)** State what is meant by the binding energy of a nucleus.

.....

.....

..... [2]

(b) A nucleus of uranium-235 absorbs a neutron and becomes unstable. It then undergoes a fission reaction. One possible reaction is

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{142}_{54}\text{Xe} + ^{90}_{38}\text{Sr} + \text{neutrons}.$$

(i) Determine the number of neutrons produced in this fission reaction.

number = ... [1]

(ii) Data for the binding energies per nucleon for this fission reaction are given in Table 8.1.

Table 8.1

isotope	binding energy per nucleon / MeV
uranium-235	7.59
xenon-142	8.37
strontium-90	8.72

Calculate the energy released, in MeV, from the fission of one nucleus of uranium-235.

energy = ... MeV [2]

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(iii) The isotope xenon-142 is unstable. The isotope xenon-132 is stable.

Suggest a reason why xenon-142 is unstable.

.....

..... [1]

(iv) Xenon-142 decays into the isotope caesium-142.

A sample initially contains only nuclei of xenon-142. After a time equal to 6.0 s, the ratio

$$\frac{\text{number of decayed nuclei of xenon-142}}{\text{number of undecayed nuclei of xenon-142}}$$

is equal to 31.

Calculate the half-life of xenon-142. Show your working.

half-life = ... s [3]

[Total: 9]

9 **(a)** Define half-life of a radioactive isotope.

.....

.....

..... [1]

(b) Radioactive isotope X decays to isotope Y.

A sample contains only nuclei of X at time t = 0. Fig. 9.1 shows the variation with t of the numbers of nuclei of X and of Y as the sample decays.

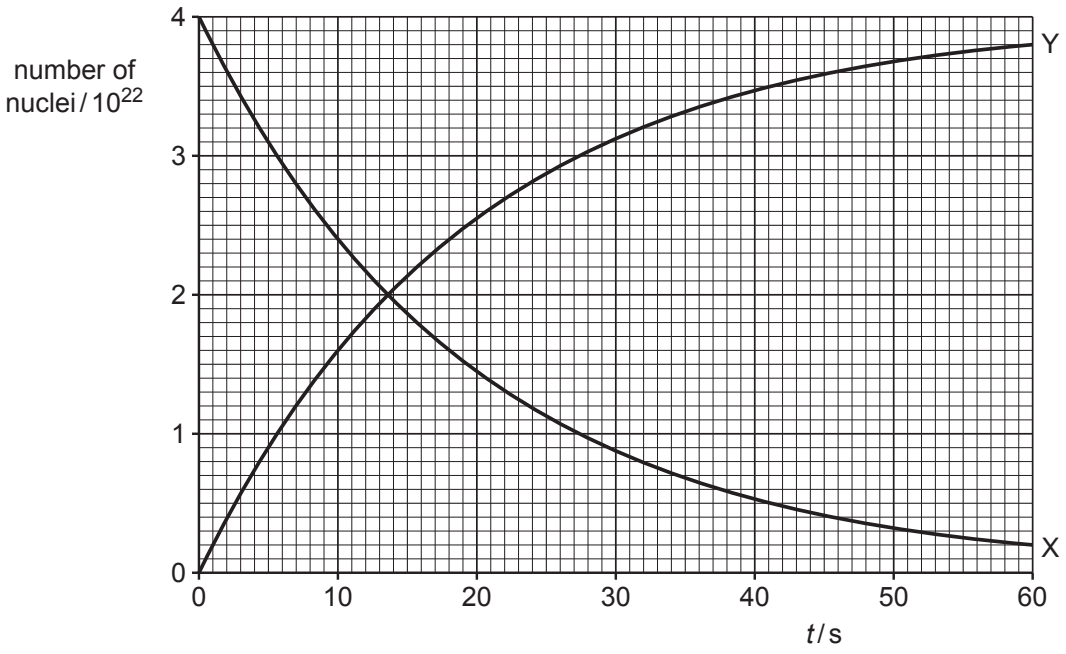


Fig. 9.1

(i) State the name of the quantity represented by the magnitude of the gradient of line X in Fig. 9.1.

..... [1]

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(ii) State **three** conclusions about X or Y that may be drawn from Fig. 9.1. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) The mass of radioactive isotope X in the sample in **(b)** is 7.3×10^{-4} kg at time $t = 0$.

Determine the nucleon number of isotope X.

nucleon number = ... [3]

[Total: 8]

* 0019655342320 *

9 **(a)** State what is meant by the binding energy of a nucleus.

.....

.....

..... [2]

(b) Table 9.1 shows the masses of two sub-atomic particles and a polonium-212 ($^{212}_{84}$Po) nucleus.

Table 9.1

	mass / u
proton	1.007 276
neutron	1.008 665
polonium-212 nucleus	211.942 749

For the polonium-212 nucleus, determine:

(i) the mass defect Δm, in kg

Δm = ... kg [3]

(ii) the binding energy

binding energy = ... J [2]

(iii) the binding energy per nucleon.

binding energy per nucleon = ... J [1]

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(c) (i) On Fig. 9.1, sketch the variation with nucleon number *A* of binding energy per nucleon for values of *A* from 1 to 250.

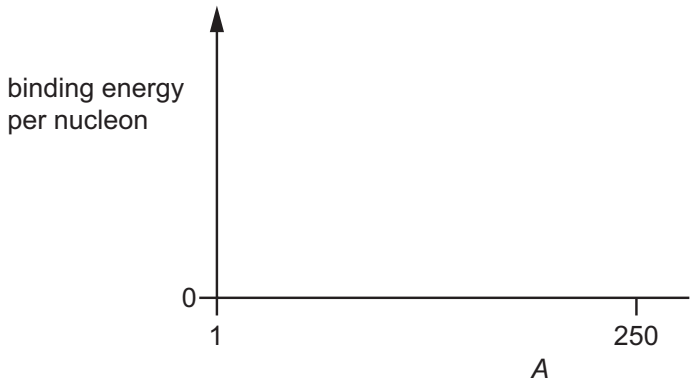


Fig. 9.1

[2]

(ii) On your line in Fig. 9.1, draw an X to show the approximate position of polonium-212.

[1]

(iii) Polonium-212 is radioactive and undergoes alpha-decay.

Suggest and explain, with reference to Fig. 9.1, why the alpha-decay of polonium-212 results in a release of energy.

.....

.....

..... [2]

[Total: 13]

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10 **(a)** Radioactive decay is both random and spontaneous.

(i) State what is meant by random.

.....

..... [1]

(ii) State what is meant by spontaneous.

.....

..... [1]

(iii) State **one** piece of evidence for the random nature of decay.

.....

..... [1]

(b) **(i)** Describe the differences between nuclear fission and nuclear fusion.

.....

.....

.....

.....

..... [3]

(ii) Explain, with reference to the variation of binding energy per nucleon with nucleon number, why the processes of nuclear fission and nuclear fusion both result in a release of energy.

.....

.....

..... [2]

[Total: 8]

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9 Polonium-193 ($^{193}_{84}$Po) is an unstable nuclide. A nucleus of polonium-193 decays to a nucleus of lead-189 ($^{189}_{82}$Pb) by emitting an alpha-particle.

(a) Radioactive decay is both random and spontaneous.

State what is meant by:

(i) random

.....

..... [1]

(ii) spontaneous.

.....

..... [1]

(b) Define half-life.

.....

.....

..... [1]

(c) Data for the binding energy per nucleon of the particles involved in the decay of a nucleus of polonium-193 are given in Table 9.1.

Table 9.1

particle	binding energy per nucleon / eV
$^{193}_{84}$Po	7.774
$^{189}_{82}$Pb	7.826
$^{4}_{2}\alpha$	7.074

Determine the energy, in eV, released when a nucleus of polonium-193 decays into a nucleus of lead-189.

energy = ... eV [2]

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(d) A pure sample of polonium-193 contains N_0 nuclei. After a time t the sample contains N nuclei of polonium-193. The variation of $\ln(N/N_0)$ with t is shown in Fig. 9.1.

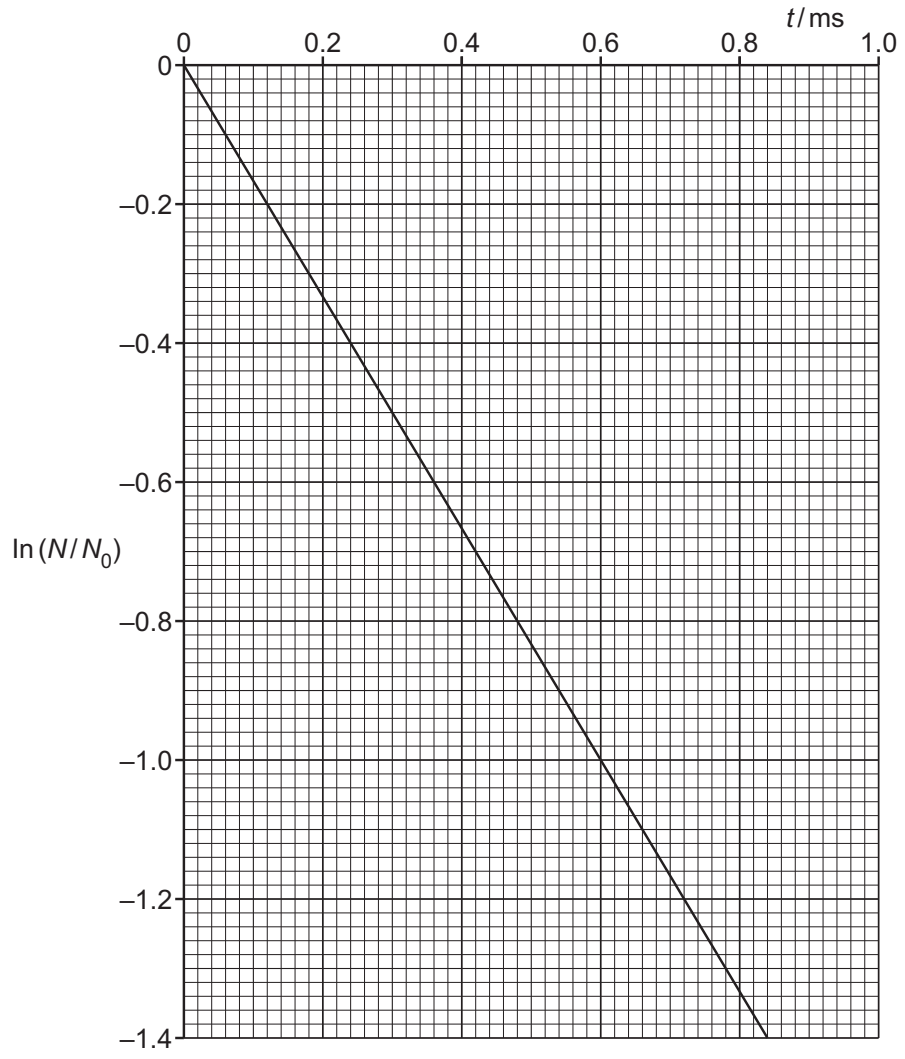


Fig. 9.1

(i) State the name of the quantity that is represented by the magnitude of the gradient of the line in Fig. 9.1.

..... [1]

(ii) Use Fig. 9.1 to determine the half-life, in ms, of polonium-193.

half-life = ... ms [2]

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(e) Positron emission tomography (PET scanning) uses a radioactive tracer.

(i) State what happens to the positrons emitted by the tracer.

.....

..... [1]

(ii) Explain why a tracer with a half-life of approximately 2 hours is a suitable tracer to use.

.....

..... [1]

[Total: 10]

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9 **(a)** Define activity of a radioactive sample.

.....

..... [1]

(b) Explain why the variation with time of the activity of a radioactive sample is exponential in nature.

.....

.....

.....

.....

..... [3]

(c) A sample contains a single radioactive isotope that decays to form a stable isotope.

The sample has an activity of 180 Bq at time $t = 0$.
At a time 8.4 minutes later, the activity is 120 Bq.

(i) Determine the decay constant, in min^{-1}, of the radioactive isotope.

decay constant = ... min^{-1} [2]

(ii) Use your answer in **(c)(i)** to determine the half-life, in min, of the radioactive isotope.

half-life = ... min [1]

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(iii) On Fig. 9.1, sketch the variation of the activity A of the sample with t for values of t between $t = 0$ and $t = 24$ min.

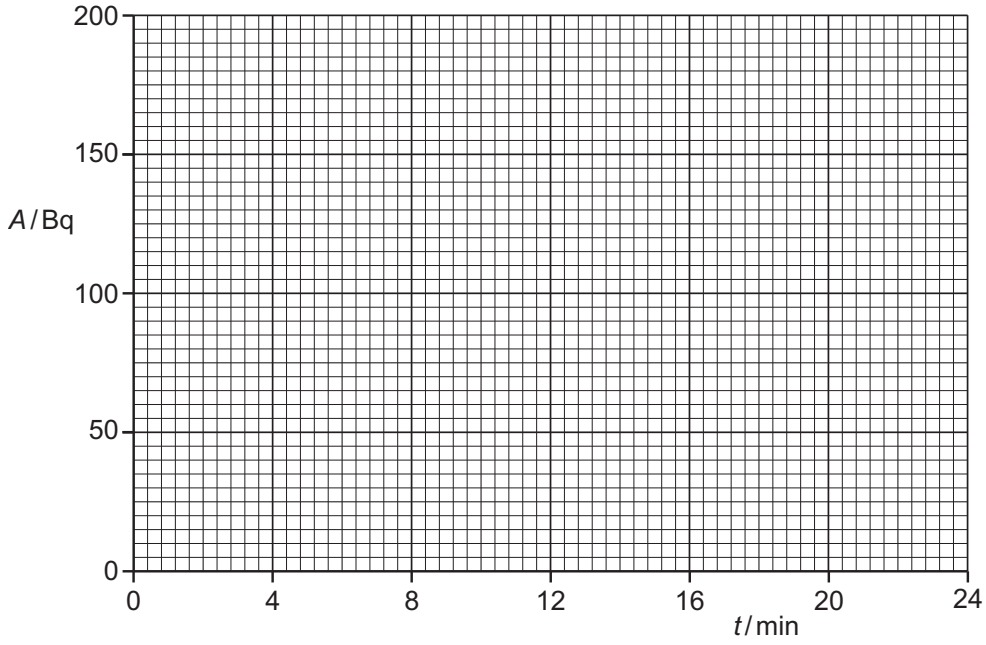


Fig. 9.1

[3]

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* 0000800000022 * DFD

10 **(a)** Radioactive decay is a spontaneous process.

State the meaning, in this context, of the term spontaneous.

.....

..... [1]

(b) Two radioactive isotopes X and Y each decay to form a stable isotope.
A sample initially contains only atoms of isotope X. At this time, its activity is $4A$.
Another sample initially contains only atoms of Y. At this time, its activity is A.

Fig. 10.1 shows the variation of the activity of each sample with time t between $t = 0$ and $t = 6T$.

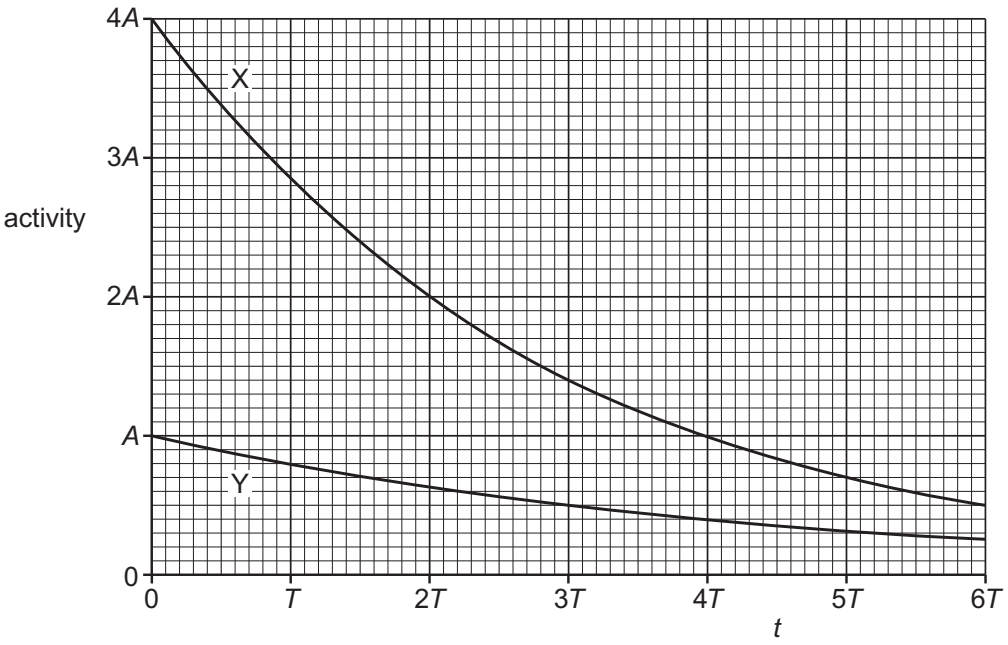


Fig. 10.1

(i) Complete Table 10.1 to give expressions, in terms of either or both of A and T, for the quantities indicated for each of the samples.

Table 10.1

sample	half-life	decay constant	initial activity	initial number of nuclei
X			$4A$	
Y			A	

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(ii) Determine, in terms of T, the time at which the two samples will have equal activities.

time = ... T [3]

(c) A radiation detector is placed near to one of the samples in **(b)**.

Explain why the count rate measured by the detector is less than the activity of the sample.

.....

.....

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[Total: 9]

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9 **(a)** Define activity of a radioactive sample.

.....

..... [1]

(b) Explain why the variation with time of the activity of a radioactive sample is exponential in nature.

.....

.....

.....

.....

..... [3]

(c) A sample contains a single radioactive isotope that decays to form a stable isotope.

The sample has an activity of 180 Bq at time $t = 0$.
At a time 8.4 minutes later, the activity is 120 Bq.

(i) Determine the decay constant, in min^{-1}, of the radioactive isotope.

decay constant = ... min^{-1} [2]

(ii) Use your answer in **(c)(i)** to determine the half-life, in min, of the radioactive isotope.

half-life = ... min [1]

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(iii) On Fig. 9.1, sketch the variation of the activity A of the sample with t for values of t between $t = 0$ and $t = 24$ min.

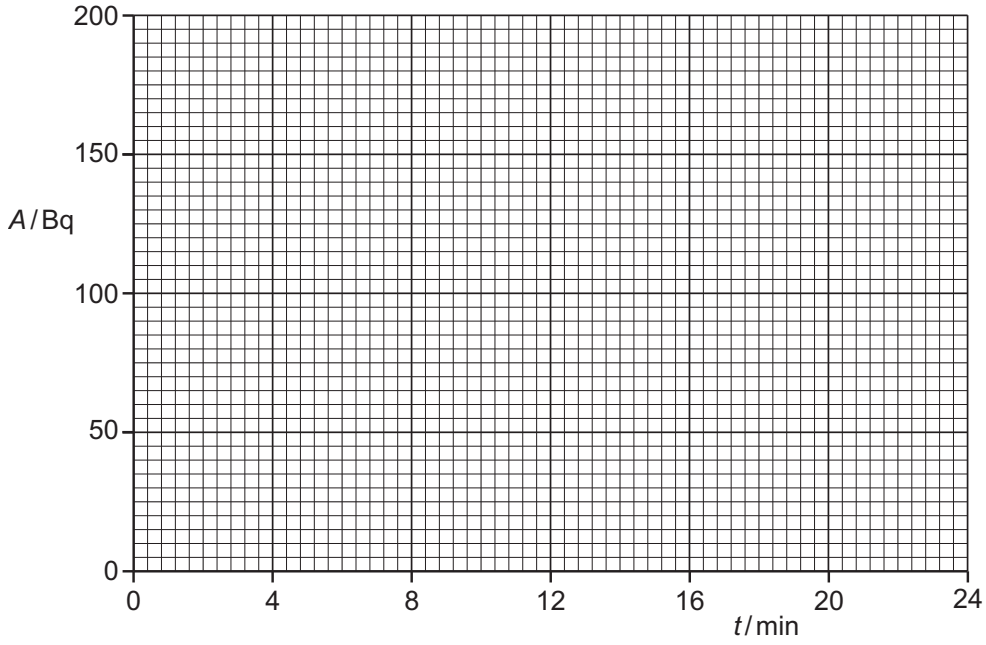


Fig. 9.1

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11 The deuterium nucleus ($^{2}_{1}$H) has a mass defect of 0.002 388 u. The helium-4 nucleus ($^{4}_{2}$He) has a mass defect of 0.030 377 u. Helium-4 is formed from deuterium in a nuclear reaction that can be represented by the equation

$$3\,{}^{2}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{1}\mathrm{p} + {}^{1}_{0}\mathrm{n}.$$

(a) (i) State the name of this type of nuclear reaction.

..... [1]

(ii) Show that the energy released when one nucleus of helium-4 is formed from deuterium is 3.47×10^{-12} J.

[3]

(b) A star has a radius of 6.96×10^{8} m. Helium-4 is produced in this star, from deuterium, at a mass rate of 7.34×10^{11} kg s^{-1}. All the energy released from this process is radiated away from the star. All the energy that is radiated from the star is released by this process.

(i) Calculate the luminosity of the star.

luminosity = ... W [3]

(ii) Use your answer in **(b)(i)** to determine the surface temperature of the star.

temperature = ... K [2]

[Total: 9]

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8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) A stationary nucleus of uranium-238 ($^{238}_{92}$U) undergoes alpha decay to produce a nucleus of thorium-234 ($^{234}_{90}$Th). The kinetic energy of the emitted alpha particle is 4.200 MeV. A gamma-ray photon is also emitted during the decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Table 8.1 shows the masses of the nuclides involved in the decay reaction. The mass of the uranium-238 nuclide is missing.

Table 8.1

nuclide	nuclide mass / u
$^{4}_{2}\alpha$	4.000 407
$^{234}_{90}$Th	233.915 174
$^{238}_{92}$U	

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

(i) Calculate the mass, in u, of the uranium-238 nuclide. Give your answer to five decimal places.

mass = ... u [3]

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(ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

wavelength = ... m [3]

(iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

.....

.....

..... [1]

(c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}$Co) decay by beta emission, and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

.....

.....

..... [2]

[Total: 11]

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9 **(a)** State what is meant by the mass defect of a nucleus.

.....

.....

..... [2]

(b) The nuclear fusion reaction for the formation of helium-4 from deuterium is represented by

$$^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He}.$$

Table 9.1 shows the masses of the nuclides involved in this reaction.

Table 9.1

nuclide	nuclide mass / u
$^{2}_{1}\mathrm{H}$	2.013 553
$^{4}_{2}\mathrm{He}$	4.001 505

Calculate the energy released in the formation of 1.00 mol of helium-4.

energy = ... J [4]

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(c) The star Sirius has a radius of 1.19×10^9 m and loses mass due to nuclear fusion at a rate of 1.09×10^{11} kg s^{-1}. Assume that the power of the radiation emitted by the star is equal to the power released by this process.

(i) Determine a value for the luminosity of Sirius. Give a unit with your answer.

luminosity = ... unit ..... [2]

(ii) Use your answer in **(c)(i)** to determine the surface temperature of Sirius.

surface temperature = ... K [2]

(d) Explain how cosmologists use standard candles to estimate the distance of a galaxy from the Earth.

.....

.....

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.....

..... [3]

[Total: 13]

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8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) A stationary nucleus of uranium-238 ($^{238}_{92}$U) undergoes alpha decay to produce a nucleus of thorium-234 ($^{234}_{90}$Th). The kinetic energy of the emitted alpha particle is 4.200 MeV. A gamma-ray photon is also emitted during the decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Table 8.1 shows the masses of the nuclides involved in the decay reaction. The mass of the uranium-238 nuclide is missing.

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$^{234}_{90}$Th	233.915 174
$^{238}_{92}$U	

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

(i) Calculate the mass, in u, of the uranium-238 nuclide. Give your answer to five decimal places.

mass = ... u [3]

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(ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

wavelength = ... m [3]

(iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

.....

.....

..... [1]

(c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}$Co) decay by beta emission, and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

.....

.....

..... [2]

[Total: 11]

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8 Oxygen-15 ($^{15}_{8}O$) is radioactive and has a half-life of 2.04 minutes.

The decay of oxygen-15 produces positrons. For this reason, oxygen-15 is sometimes used as a tracer in positron emission tomography (PET scanning).

(a) State what is meant by a tracer.

.....

.....

..... [2]

(b) The equation for the decay of oxygen-15 is

$$^{15}_{8}O \rightarrow {}^{P}_{Q}X + {}^{R}_{S}\beta^{+} + Z$$

where X is the nucleus formed during the decay and Z is another particle.

(i) State the values of the integers *P*, *Q*, *R* and *S*.

P = ... *R* = ...

Q = ... *S* = ...

[2]

(ii) State the name of particle Z.

..... [1]

(c) **(i)** Define the activity of a sample.

.....

..... [1]

(ii) Calculate the decay constant of oxygen-15. Give a unit with your answer.

decay constant = ... unit ..... [2]

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(iii) Determine the rate at which positrons are produced in a sample of oxygen-15 that has a mass of 2.85×10^{-6} kg.

rate = ... s^{-1} [4]

(d) The particles that are emitted from the body and detected outside it during PET scanning are not positrons but another type of particle.

(i) State the name of the particles that are detected.

..... [1]

(ii) Explain how these particles are formed inside the body.

.....

.....

.....

..... [2]

[Total: 15]

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9 **(a)** Define the mass defect of a nucleus.

.....

.....

..... [2]

(b) Thorium-228 ($^{228}_{90}$Th) is radioactive.

Table 9.1 shows the mass defects of thorium-228 and some other nuclides.

Table 9.1

nuclide	mass defect / u
$^{4}_{2}$He	0.030 377
$^{224}_{88}$Ra	1.846 840
$^{228}_{89}$Ac	1.869 820
$^{228}_{90}$Th	1.871 290
$^{228}_{91}$Pa	1.866 129

(i) Use the data in Table 9.1 to explain why thorium-228 is an α-emitter and **not** a β^- emitter or a β^+ emitter.

.....

.....

..... [4]

(ii) Calculate the energy, in MeV, released by the α-decay of one thorium-228 nucleus. Give your answer to three significant figures.

energy = ... MeV [3]

[Total: 9]

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10 **(a)** Define the binding energy of a nucleus.

.....

.....

..... [2]

(b) A star emits radiation that is produced in a series of nuclear reactions that form helium from hydrogen. The reactions may be summarised by the overall equation

$$4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{0}_{1}\mathrm{X}$$

where X is a particle that has no binding energy.

All of the radiation emitted by the star is produced by this series of nuclear reactions.

The mass defect of the ${}^{4}_{2}\mathrm{He}$ nuclide is 0.030 377 u, where u is the unified atomic mass unit.

(i) State the name of particle X.

..... [1]

(ii) Show that the binding energy of the ${}^{4}_{2}\mathrm{He}$ nuclide is 4.54×10^{-12} J.

[2]

(c) The star in **10(b)** has a luminosity of 7.98×10^{27} W. An observer on the Earth detects radiation from this star at a distance away of 8.15×10^{16} m.

(i) Calculate the radiant flux intensity F detected by the observer.

F = ... $\mathrm{W\,m^{-2}}$ [2]

(ii) The mass of hydrogen in the star decreases continuously as a result of the series of nuclear reactions.

Determine the rate of decrease of mass of hydrogen in the star.

rate of decrease of mass = ... $kg\,s^{-1}$ [4]

[Total: 11]

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9 **(a)** Define the mass defect of a nucleus.

.....

.....

..... [2]

(b) Thorium-228 ($^{228}_{90}$Th) is radioactive.

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(i) Use the data in Table 9.1 to explain why thorium-228 is an α-emitter and **not** a β^- emitter or a β^+ emitter.

.....

.....

..... [4]

(ii) Calculate the energy, in MeV, released by the α-decay of one thorium-228 nucleus. Give your answer to three significant figures.

energy = ... MeV [3]

[Total: 9]

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24 Medical physics

9 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 24 Medical physics

考纲要求（9702 · 2025 – 2027 syllabus）

24.1

24.1.1 understand that a piezo-electric crystal changes shape when a p.d. is applied across it and that the crystal generates an e.m.f. when its shape changes

24.1.2 understand how ultrasound waves are generated and detected by a piezoelectric transducer

24.1.3 understand how the reflection of pulses of ultrasound at boundaries between tissues can be used to obtain diagnostic information about internal structures

24.1.4 define the specific acoustic impedance of a medium as Z = ρc, where c is the speed of sound in the medium

24.1.5 use IR / I0 = (Z1 – Z2)2 / (Z1 + Z2)2 for the intensity reflection coefficient of a boundary between two media

24.1.6 recall and use I = I0e–μx for the attenuation of ultrasound in matter

24.2

24.2.1 explain that X-rays are produced by electron bombardment of a metal target and calculate the minimum wavelength of X-rays produced from the accelerating p.d.

24.2.2 understand the use of X-rays in imaging internal body structures, including an understanding of the term contrast in X-ray imaging

24.2.3 recall and use I = I0e–μx for the attenuation of X-rays in matter

24.2.4 understand that computed tomography (CT) scanning produces a 3D image of an internal structure by first combining multiple X-ray images taken in the same section from different angles to obtain a 2D image of the section, then repeating this process along an axis and combining 2D images of multiple sections

24.3

24.3.1 understand that a tracer is a substance containing radioactive nuclei that can be introduced into the body and is then absorbed by the tissue being studied

24.3.2 recall that a tracer that decays by β+ decay is used in positron emission tomography (PET scanning)

24.3.3 understand that annihilation occurs when a particle interacts with its antiparticle and that mass–energy and momentum are conserved in the process

24.3.4 explain that, in PET scanning, positrons emitted by the decay of the tracer annihilate when they interact with electrons in the tissue, producing a pair of gamma-ray photons travelling in opposite directions

24.3.5 calculate the energy of the gamma-ray photons emitted during the annihilation of an electron-positron pair

24.3.6 understand that the gamma-ray photons from an annihilation event travel outside the body and can be detected, and an image of the tracer concentration in the tissue can be created by processing the arrival times of the gamma-ray photons

9 **(a)** Electrons in a vacuum are accelerated through a potential difference of 84 kV. The electrons then strike a metal target and X-rays are produced.

(i) Calculate the minimum wavelength of the X-rays that are produced.

wavelength = ... m [2]

(ii) The melting points of two metals are given in Table 9.1.

Table 9.1

metal	melting point / °C
copper	1090
tungsten	3420

Suggest why the metal target is made from tungsten rather than copper.

.....

.....

..... [2]

(b) An X-ray beam is incident normally on a sample of soft tissue and bone as shown in Fig. 9.1.

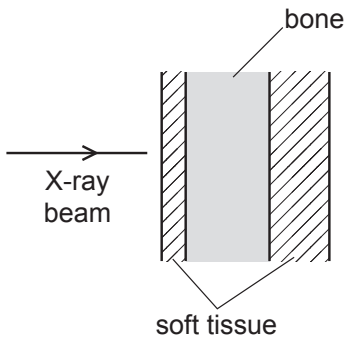


Fig. 9.1

Data for the two materials are given in Table 9.2.

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Table 9.2

medium	linear attenuation coefficient μ / cm^{-1}	specific acoustic impedance Z / 10^6 kg m^{-2} s^{-1}
soft tissue	0.22	1.7
bone	3.0	7.8

The total thickness of soft tissue is *x*. The total thickness of bone is also *x*.

The incident intensity of the X-ray beam is I_0. The transmitted intensity of the X-ray beam is 13% of the incident intensity.

Determine *x*, in cm.

x = ... cm [3]

(c) (i) Define the specific acoustic impedance of a medium.

.....

.....

..... [2]

(ii) Use data from Table 9.2 to calculate the percentage of the intensity of ultrasound that is transmitted at a boundary between soft tissue and bone.

percentage transmitted = ...% [2]

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(iii) The ultrasound is now incident on the sample of soft tissue and bone shown in Fig. 9.1.

Suggest **two** reasons why the transmitted intensity through the sample is less than the answer in **(c)(ii)**.

1 .....

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2 .....

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[2]

[Total: 13]

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10 **(a)** Describe how reflected ultrasound pulses may be used to obtain diagnostic information about internal structures.

.....

.....

.....

..... [2]

(b) **(i)** Define specific acoustic impedance of a medium.

.....

.....

..... [2]

(ii) Table 10.1 shows some data for water and for glass.

Table 10.1

	density / kg m^{-3}	speed of sound / m s^{-1}
water	1000	1420
glass	2500	4560

Determine the intensity reflection coefficient for ultrasound that is incident on a water–glass boundary.

intensity reflection coefficient = ... [3]

[Total: 7]

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9 Fluorine-18 ($^{18}_{9}F$) decays by beta-plus (β^+) emission with a half-life of 110 minutes.

(a) (i) State the name of the beta-plus particle.

..... [1]

(ii) Show that the decay constant of fluorine-18 is $1.05 \times 10^{-4}\,s^{-1}$.

[1]

(iii) Determine the activity of 2.1×10^{-12} kg of fluorine-18.

activity = ... Bq [3]

(b) A small sample of fluorine-18 injected into the body acts as a tracer for use in medical imaging.

(i) Describe how the interaction of a β^+ particle with an electron in the body enables the formation of an image.

.....

.....

.....

.....

..... [3]

(ii) Suggest why 110 minutes is a suitable half-life for a nuclide used as a tracer in medical diagnosis.

.....

.....

..... [2]

[Total: 10]

10 **(a)** Describe how the piezoelectric crystal in a transducer generates ultrasound waves for use in medical diagnosis.

.....

.....

.....

.....

..... [3]

(b) A parallel ultrasound beam is incident on the boundary between two media, as illustrated in Fig. 10.1.

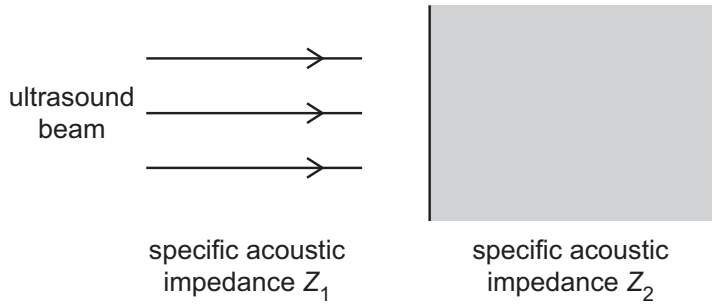


Fig. 10.1

The media have specific acoustic impedances Z_1 and Z_2.

At the boundary, a fraction α of the incident intensity of the ultrasound beam is reflected. The remainder is transmitted.

(i) State what is meant by specific acoustic impedance.

.....

.....

..... [2]

(ii) Describe how α depends on the relative values of Z_1 and Z_2.

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.....

..... [2]

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(c) A parallel ultrasound beam of intensity I_0 enters a region of soft tissue. After passing a distance of 2.1 cm through this tissue, the intensity of the ultrasound is $0.62\,I_0$.

Calculate the linear attenuation coefficient μ of ultrasound in the soft tissue. Give a unit with your answer.

μ = ... unit ..... [2]

[Total: 9]

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10 **(a)** Define specific acoustic impedance.

.....

.....

..... [2]

(b) Explain how ultrasound waves are detected by a piezoelectric crystal.

.....

.....

..... [2]

(c) Table 10.1 shows the specific acoustic impedance Z for body tissue, water and steel.

Table 10.1

material	Z / kg m^{-2} s^{-1}
body tissue	1.38×10^{6}
water	1.48×10^{6}
steel	4.04×10^{7}

(i) Calculate the intensity reflection coefficient for ultrasound incident on a water–steel boundary.

intensity reflection coefficient = ... [2]

(ii) Explain, without calculation, what is likely to happen when ultrasound is incident on a body tissue–water boundary.

.....

.....

..... [2]

[Total: 8]

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10 **(a)** State what is meant by contrast in an X-ray image.

.....

..... [1]

(b) X-rays of intensity I_0 are incident normally on a structure, as shown in Fig. 10.1.

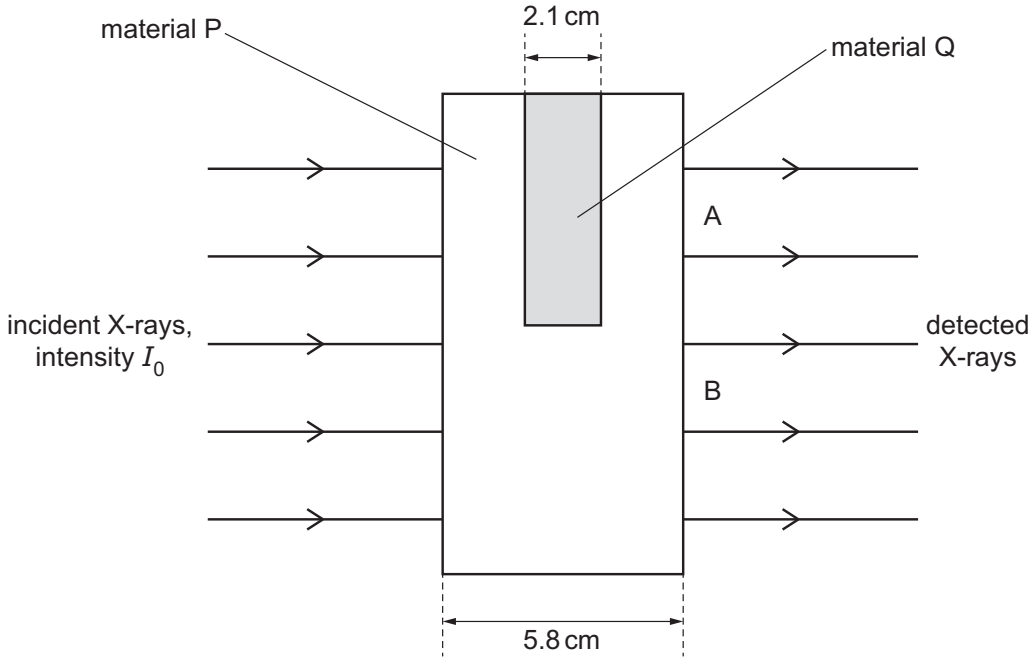


Fig. 10.1

Material P has a linear attenuation coefficient of $0.35\,\text{cm}^{-1}$.
The X-rays emerging from the structure in region A have an intensity of $0.053I_0$.

(i) Show that the intensity of the X-rays emerging in region B is $0.13I_0$.

[1]

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(ii) Determine the linear attenuation coefficient μ of material Q.

μ = ... cm^{-1} [3]

(iii) Use the information in **(b)(i)** to suggest why the X-rays emerging from the structure form an image that has poor contrast.

.....

.....

..... [1]

(c) Explain how X-rays are used in computed tomography (CT) scanning to produce a three-dimensional image of an internal structure.

.....

.....

.....

.....

..... [3]

[Total: 9]

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10 **(a)** Define specific acoustic impedance.

.....

.....

..... [2]

(b) Explain how ultrasound waves are detected by a piezoelectric crystal.

.....

.....

..... [2]

(c) Table 10.1 shows the specific acoustic impedance Z for body tissue, water and steel.

Table 10.1

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water	1.48×10^{6}
steel	4.04×10^{7}

(i) Calculate the intensity reflection coefficient for ultrasound incident on a water–steel boundary.

intensity reflection coefficient = ... [2]

(ii) Explain, without calculation, what is likely to happen when ultrasound is incident on a body tissue–water boundary.

.....

.....

..... [2]

[Total: 8]

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8 **(a)** State what is meant by a photon.

.....

.....

..... [2]

(b) X-rays for use in medical diagnosis are produced by accelerating charged particles towards a metal target.

(i) State the name of the charged particles.

..... [1]

(ii) Describe how X-rays are formed at the target.

.....

.....

..... [2]

(iii) For any given accelerating voltage, there is a range of wavelengths of X-rays produced that has a minimum value.

Explain, with reference to photons, why there is a range of wavelengths and why this range has a minimum value.

.....

.....

.....

.....

..... [3]

(iv) Calculate the minimum wavelength of X-rays produced when the accelerating voltage is 52.0 kV.

minimum wavelength = ... m [3]

(c) X-rays are used to create an image of the internal structure of an object that is made from two types of material of similar thickness.

Describe and explain how a property of the materials determines whether or not the image produced has good contrast.

.....

.....

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..... [2]

[Total: 13]

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9 **(a)** Explain how ultrasound waves are generated by a piezoelectric transducer.

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..... [3]

(b) Ultrasound waves are used to obtain information about a structure that consists of two materials, as shown in Figure 9.1.

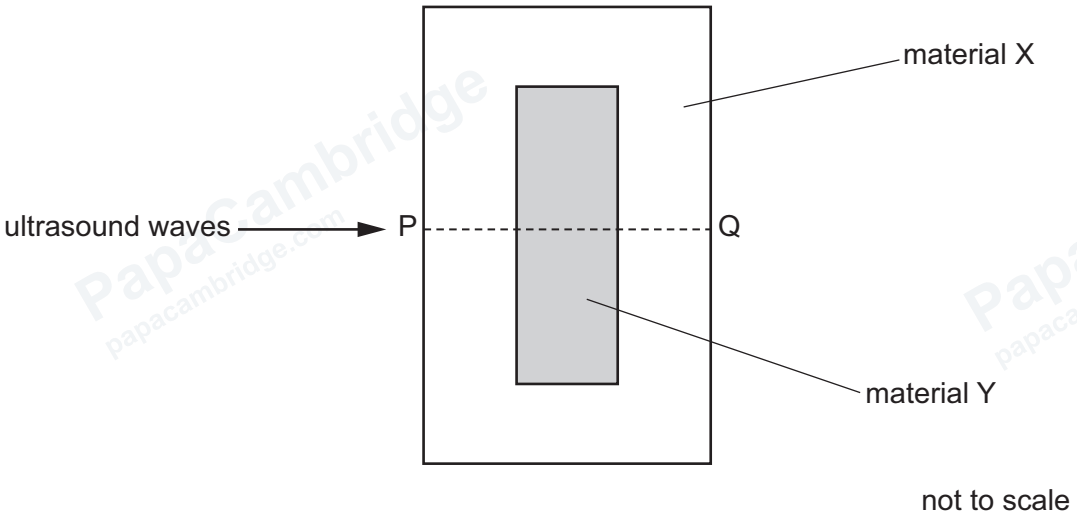


Figure 9.1

The waves enter the structure at point P and travel through the structure to point Q.

Material Y has a density of 8960 kg m^{-3}. The speed of sound in material Y is 4720 m s^{-1}.

(i) Calculate the specific acoustic impedance Z of material Y. Give a unit with your answer.

Z = ... unit ..... [2]

(ii) The intensity of ultrasound entering the structure at point P is I_0.
The transmitted intensity of the ultrasound between P and Q at distance *x* from P is I.

Figure 9.2 shows the variation with *x* of the value of $\ln(I/I_0)$.

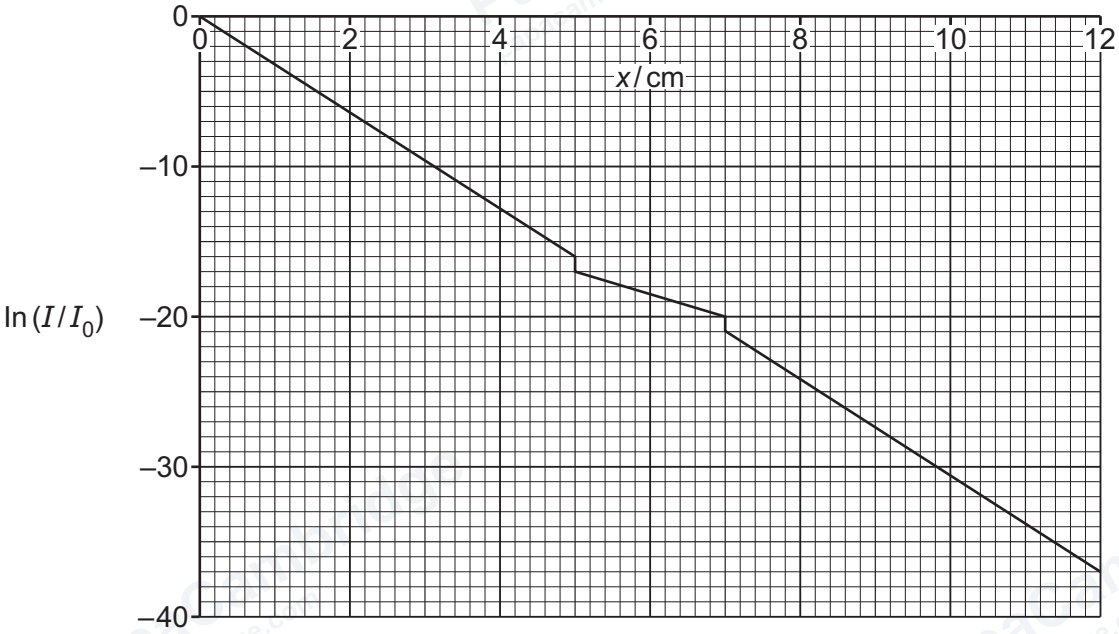


Figure 9.2

Determine **three** conclusions about the structure and the materials from which it is made that may be drawn from Figure 9.2. The conclusions may be qualitative or quantitative. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 8]

25 Astronomy and cosmology

13 questions · Paper 4 · 2024–2026

考纲要求见下页 · 题目按年份排列（2024 → 2025 → 2026）

Syllabus requirements · 25 Astronomy and cosmology

考纲要求（9702 · 2025 – 2027 syllabus）

25.1

25.1.1 understand the term luminosity as the total power of radiation emitted by a star

25.1.2 recall and use the inverse square law for radiant flux intensity F in terms of the luminosity L of the source F = L / (4πd 2)

25.1.3 understand that an object of known luminosity is called a standard candle

25.1.4 understand the use of standard candles to determine distances to galaxies

25.2

25.2.1 recall and use Wien’s displacement law λmax ∝ 1 / T to estimate the peak surface temperature of a star

25.2.2 use the Stefan–Boltzmann law L = 4πσr 2 T 4

25.2.3 use Wien’s displacement law and the Stefan–Boltzmann law to estimate the radius of a star

25.3

25.3.1 understand that the lines in the emission and absorption spectra from distant objects show an increase in wavelength from their known values

25.3.2 use Δλ / λ . Δf / f . v / c for the redshift of electromagnetic radiation from a source moving relative to an observer

25.3.3 explain why redshift leads to the idea that the Universe is expanding

25.3.4 recall and use Hubble’s law v . H0d and explain how this leads to the Big Bang theory (candidates will only be required to use SI units) 4 Details of the assessment Paper 1 Multiple Choice Written paper, 1 hour 15 minutes, 40 marks Forty multiple-choice items of the four-choice type testing assessment objectives AO1 and AO2. Questions are based on the AS Level syllabus content. Paper 2 AS Level Structured Questions Written paper, 1 hour 15 minutes, 60 marks Structured questions testing assessment objectives AO1 and AO2. Questions are based on the AS Level syllabus content. Paper 3 Advanced Practical Skills Practical test, 2 hours, 40 marks This paper tests assessment objective AO3 in a practical context. Two questions assess the AS Level practical skills in the

10 (a) The Sun has a surface temperature of 5780 K. The luminosity of the Sun is 3.85×10^{26} W.

(i) Calculate the radius of the Sun.

radius = ... m [2]

(ii) The Earth is a distance of 1.50×10^{11} m from the Sun.

Calculate the radiant flux intensity F of the radiation from the Sun at a distance of 1.50×10^{11} m. Give a unit with your answer.

F = ... unit ..... [2]

(iii) The variation with wavelength of the intensity of radiation emitted from the Sun is shown in Fig. 10.1.

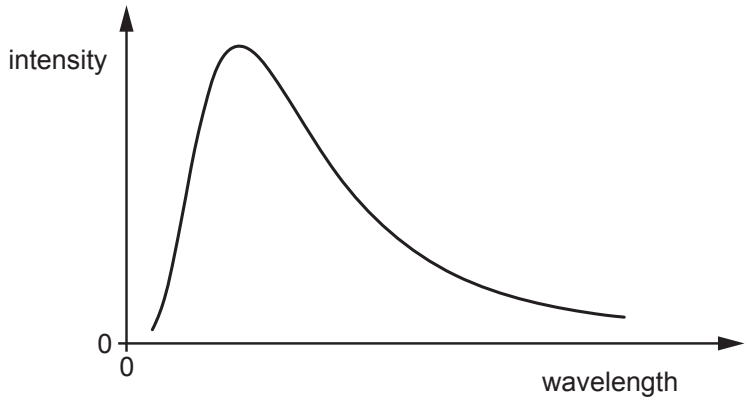


Fig. 10.1

Another star has the same radius as the Sun but has a lower surface temperature.

On Fig. 10.1, sketch a line to show the variation with wavelength of the intensity of the radiation emitted for this star. [2]

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(b) A galaxy in the constellation Corona Borealis is moving away from the Earth.

(i) The visible emission spectrum for the Sun is shown in Fig. 10.2.

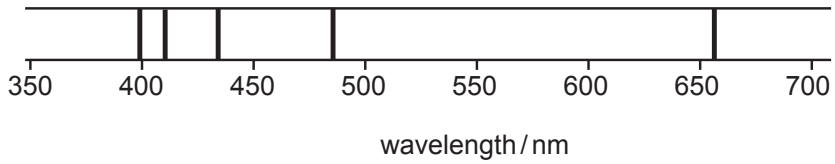


Fig. 10.2

The lines are at wavelengths of 397 nm, 410 nm, 434 nm, 486 nm and 656 nm. The compositions of the Sun and a star in the Corona Borealis galaxy are similar.

On Fig. 10.3, sketch the emission spectrum for the star in the Corona Borealis galaxy as observed from the Earth. No calculations are required.

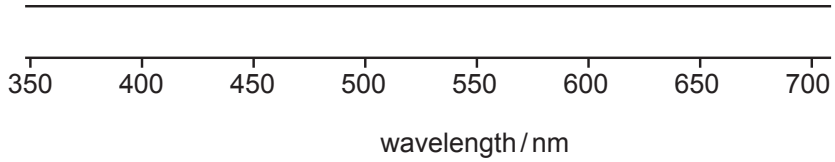


Fig. 10.3

[1]

(ii) The galaxy in Corona Borealis is moving away from the Earth at a speed of 21 400 km s^{-1}.

Use information from **(b)(i)** to calculate, in nm, the observed wavelength of the lowest visible energy emission for the star in the Corona Borealis galaxy.

wavelength = ... nm [2]

(iii) The wavelength in **(b)(ii)** is used to calculate a value for the surface temperature of the star in the Corona Borealis galaxy. The calculation does not give an accurate value.

State and explain whether this value of temperature is too high or too low.

.....

.....

..... [1]

[Total: 10]

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10 (a) (i) State what is meant by the luminosity of a star.

.....

.....

..... [2]

(ii) Explain how a standard candle in a distant galaxy can be used to determine the distance of the galaxy from an observer.

.....

.....

.....

.....

..... [3]

(b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5780 K.
Light from the Sun is observed to have a peak intensity at a wavelength of 501 nm.

(i) Calculate the luminosity of the Sun. Give a unit with your answer.

luminosity = ... unit ..... [2]

(ii) Another star emits radiation that has a peak intensity at a wavelength of 624 nm.

Determine the surface temperature of this star.

surface temperature = ...K [2]

[Total: 9]

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* 0019655342318 *

8 Fig. 8.1 shows part of the emission spectrum of visible radiation emitted by hydrogen gas in a star in a distant galaxy.

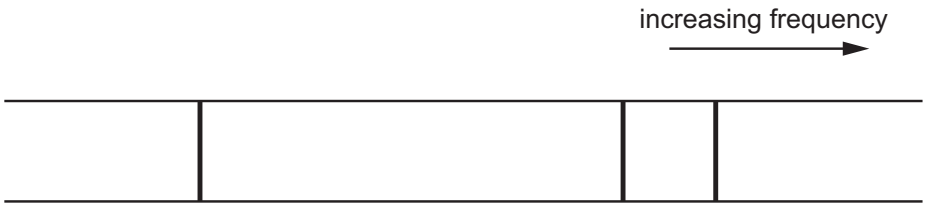


Fig. 8.1

The galaxy is moving away from the Earth at a speed of $6.2 \times 10^6\,\text{m s}^{-1}$.

(a) (i) Explain how the positions of the lines in the emission spectrum seen by an observer on the Earth differ from the positions shown in Fig. 8.1.

.....

.....

..... [2]

(ii) On Fig. 8.1, draw the three lines in possible positions in the spectrum seen by the observer. [2]

(b) The lines in Fig. 8.1 correspond to electron transitions down to the energy level –3.40 eV. One of the lines represents emitted radiation of wavelength 488 nm.

(i) Calculate the energy of a photon of this radiation.

photon energy = ... J [2]

(ii) Determine the energy, in eV, of the energy level from which the electron transition originates to cause the emission of this radiation.

energy level = ... eV [2]

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(iii) Determine the wavelength, in nm, of this radiation as detected by the observer on the Earth.

wavelength = ... nm [2]

(c) A value for the Hubble constant is $2.3 \times 10^{-18}\,s^{-1}$.

Determine the distance of the galaxy from the Earth.

distance = ... m [2]

[Total: 12]

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10 **(a)** Explain how redshift leads to the idea that the Universe is expanding.

.....

.....

.....

.....

..... [3]

(b) Stars in a distant galaxy emit radiation. The total luminosity of the stars in the galaxy is 1.90×10^{36} W.
The emission spectrum of the radiation contains a line X at a wavelength of 658 nm.

Radiation from the galaxy is observed on the Earth. The observed radiation has a radiant flux intensity of 8.42×10^{-16} W m^{-2}. In the observed emission spectrum, line X is at a wavelength of 726 nm.

Determine:

(i) the distance d of the galaxy from the Earth

d = ... m [2]

(ii) the speed v of the galaxy relative to the Earth.

v = ... m s^{-1} [2]

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(c) Observations of many galaxies, such as the one in **(b)**, lead to many pairs of values of d and v. Plotting these values reveals a trend.

(i) On Fig. 10.1, sketch the variation of v with d.

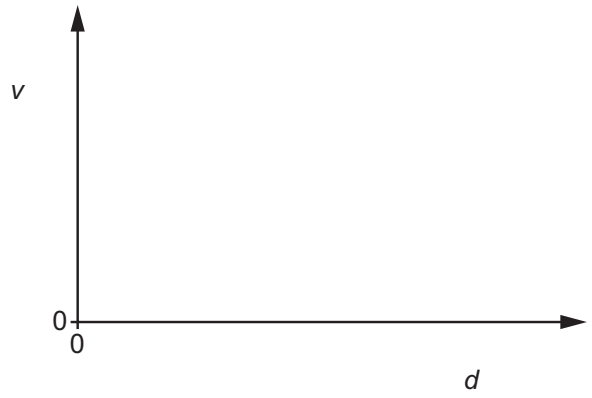


Fig. 10.1

[2]

(ii) State the name of the quantity represented by the gradient of the line in Fig. 10.1.

..... [1]

[Total: 10]

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10 (a) (i) State what is meant by the luminosity of a star.

.....

..... [1]

(ii) Explain how standard candles are used to determine the distance to a galaxy.

.....

.....

.....

.....

.....

..... [3]

(b) The Sun rotates on its axis. Points X, Y and Z are on the equator of the Sun as shown in Fig. 10.1.

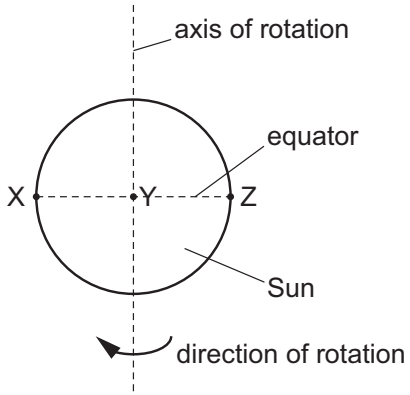


Fig. 10.1

The wavelengths of light from points X and Y are observed and recorded in Table 10.1.

Table 10.1

observed wavelength from X / nm	observed wavelength from Y / nm
656.2877	656.2831

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(i) The Sun rotates with a period of 2.07×10^6 s.

Show that the radius of the Sun is 6.93×10^8 m.

[3]

(ii) State and explain how the expected wavelength of the light observed from Z compares with the emitted wavelength.

.....

.....

..... [2]

(iii) The luminosity of the Sun is 3.8×10^{26} W.

Use the information in **(b)(i)** to calculate the surface temperature of the Sun.

temperature = ... K [2]

[Total: 11]

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10 **(a)** State Hubble's law.

.....

.....

..... [2]

(b) A star in a distant galaxy emits radiation that has a maximum intensity of emission at a wavelength of 4.62×10^{-7} m.

Observations of the galaxy made on the Earth detect the maximum intensity of emission from the star at a wavelength of 4.91×10^{-7} m.

(i) Explain why the observed wavelength and the emitted wavelength have different values.

.....

.....

..... [2]

(ii) Calculate the speed of the star relative to the Earth.

speed = ... $m\,s^{-1}$ [2]

(iii) The wavelength of maximum intensity of emission is used to determine a value for the surface temperature of the star.

Explain how the temperature determined using the observed wavelength compares with the true value of temperature determined using the emitted wavelength.

.....

.....

..... [2]

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(c) A value for the Hubble constant is $2.3 \times 10^{-18}\,s^{-1}$.

Use your answer in **(b)(ii)** to determine the distance of the star in **(b)** from the Earth.

distance = ... m [2]

[Total: 10]

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10 **(a)** State Hubble's law.

.....

.....

..... [2]

(b) A star in a distant galaxy emits radiation that has a maximum intensity of emission at a wavelength of 4.62×10^{-7} m.

Observations of the galaxy made on the Earth detect the maximum intensity of emission from the star at a wavelength of 4.91×10^{-7} m.

(i) Explain why the observed wavelength and the emitted wavelength have different values.

.....

.....

..... [2]

(ii) Calculate the speed of the star relative to the Earth.

speed = ... $m\,s^{-1}$ [2]

(iii) The wavelength of maximum intensity of emission is used to determine a value for the surface temperature of the star.

Explain how the temperature determined using the observed wavelength compares with the true value of temperature determined using the emitted wavelength.

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(c) A value for the Hubble constant is $2.3 \times 10^{-18}\,s^{-1}$.

Use your answer in **(b)(ii)** to determine the distance of the star in **(b)** from the Earth.

distance = ... m [2]

[Total: 10]

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9 **(a)** State Wien's displacement law.

.....

.....

..... [2]

(b) Fig. 9.1 shows the variation with d^{-2} of the radiant flux intensity F observed from a star X, where d is the distance of the observer from the star. Fig. 9.2 shows the variation with wavelength λ of the rates of emission P of radiation by star X and the Sun.

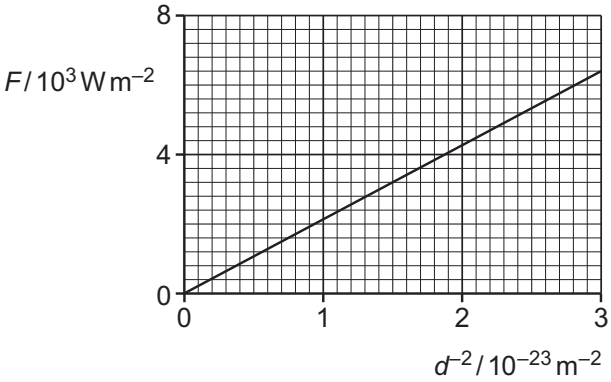


Fig. 9.1

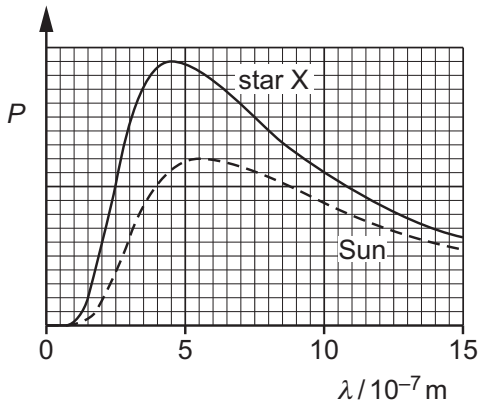


Fig. 9.2

The surface temperature of the Sun is 5770 K.

State **three** conclusions about star X that can be drawn from this data. The conclusions may be qualitative or quantitative. Use the space for any working.

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..... [3]

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(c) Star X is in a galaxy that is moving away from the Earth.

Suggest, with a reason, how the line for star X in Fig. 9.2 would appear differently if it had been obtained from data measured on the Earth.

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..... [2]

[Total: 7]

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9 **(a)** State Wien's displacement law.

.....

.....

..... [2]

(b) Fig. 9.1 shows the variation with d^{-2} of the radiant flux intensity F observed from a star X, where d is the distance of the observer from the star. Fig. 9.2 shows the variation with wavelength λ of the rates of emission P of radiation by star X and the Sun.

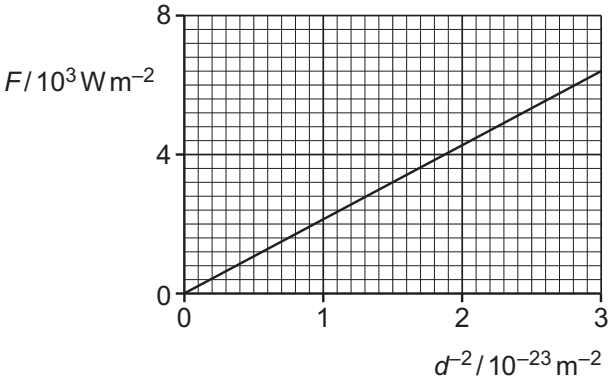


Fig. 9.1

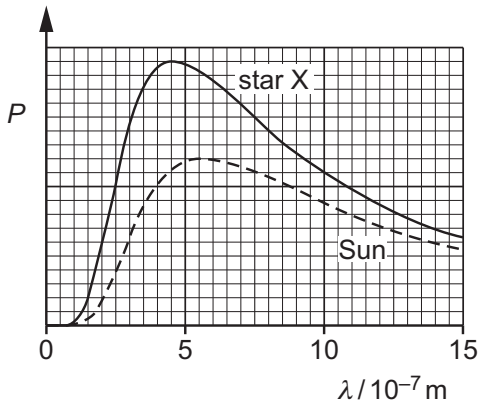


Fig. 9.2

The surface temperature of the Sun is 5770 K.

State **three** conclusions about star X that can be drawn from this data. The conclusions may be qualitative or quantitative. Use the space for any working.

1 .....

.....

2 .....

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..... [3]

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(c) Star X is in a galaxy that is moving away from the Earth.

Suggest, with a reason, how the line for star X in Fig. 9.2 would appear differently if it had been obtained from data measured on the Earth.

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[Total: 7]

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10 **(a)** State what is meant by redshift.

.....

.....

..... [2]

(b) Explain how observations of redshift lead to the idea that the universe is expanding.

.....

.....

..... [2]

(c) Explain how Hubble's law leads to the Big Bang theory of the origin of the universe.

.....

.....

.....

.....

..... [3]

[Total: 7]

10 **(a)** State Wien's displacement law.

.....

.....

..... [2]

(b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5770 K.

The Earth is at a distance of 1.52×10^{11} m from the Sun.

(i) Calculate the luminosity of the Sun.

luminosity = ... W [2]

(ii) Determine the radiant flux intensity at the Earth of radiation emitted by the Sun. Give a unit with your answer.

radiant flux intensity = ... unit ..... [3]

(iii) Star X has a wavelength of maximum emission of $0.581\lambda_s$, where λ_s is the wavelength of maximum emission from the Sun.

Calculate the surface temperature of star X.

surface temperature = ... K [1]

[Total: 8]

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10 **(a)** State Wien's displacement law.

.....

.....

..... [2]

(b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5770 K.

The Earth is at a distance of 1.52×10^{11} m from the Sun.

(i) Calculate the luminosity of the Sun.

luminosity = ... W [2]

(ii) Determine the radiant flux intensity at the Earth of radiation emitted by the Sun. Give a unit with your answer.

radiant flux intensity = ... unit ..... [3]

(iii) Star X has a wavelength of maximum emission of $0.581\lambda_s$, where λ_s is the wavelength of maximum emission from the Sun.

Calculate the surface temperature of star X.

surface temperature = ... K [1]

[Total: 8]

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10 **(a)** **(i)** State what is meant by a standard candle.

.....

..... [1]

(ii) Explain how standard candles may be used to determine the distance of a galaxy from an observer.

.....

.....

..... [2]

(b) A star in a distant galaxy has luminosity L.

The star emits radiation that has maximum intensity at a wavelength that is known to be λ_{star} after correcting for redshift.

The Sun has surface temperature T_0 and emits radiation that has maximum intensity at wavelength λ_0.

(i) Explain why the wavelength of the radiation observed from the star has to be corrected for redshift.

.....

.....

..... [2]

(ii) State an expression, in terms of λ_{star}, λ_0 and T_0, for the surface temperature T_{star} of the star.

T_{star} = ... [1]

(iii) Determine an expression, in terms of L and T_{star}, for the radius of the star. Identify any other symbols that you use.

radius = ... [1]

[Total: 7]